# GOVERNMENT OF ANDHRA PRADESH
# SAMAGRA SHIKSHA
# DEPARTMENT OF EDUCATION

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## TENDER BID DOCUMENT

## VOLUME-I

**Tender Notice No** : **NIT No. 27/ APSS/CWs/Autism Centres/2026-27/2nd call, dt: 22.10.2026 of the Chief Engineer, APSS, Vijayawada.**

**Name of the Work** :**Construction of two rooms/ one room and renovation of existing rooms for SG MPL HS ANKASANI CHERUVU for the establishment of Autism Centers in Konaseema District**

**EST. COST** : **Rs. 40.00 Lakhs**

**DEPARTMENT: SAMAGRA SHIKSHA, DEPARTMENT OF SCHOOL EDUCATION, KBC ZP HIGH SCHOOL (BOYS) PREMISES, PATAMATA, VIJAYAWADA, NTR DISTRICT, PIN - 520010.**



Tenderer

Chief Engineer
APSS, Amaravati.

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Tenderer

Chief Engineer
APSS, Amaravati.

# NOTICE INVITING TENDER
# (NIT)



Tenderer

Chief Engineer
APSS, Amaravati.

# GOVERNMENT OF ANDHRA PRADESH
# SAMAGRA SHIKSHA
# DEPARTMENT OF EDUCATION

****

**NIT No. 27/ APSS/CWs/Autism Centres/2026-27/2nd call, dt: 22.10.2026 of the Chief Engineer, APSS, Vijayawada.**

NAME OF WORK: Construction of two rooms/ one room and renovation of existing rooms for SG MPL HS ANKASANI CHERUVU for the establishment of Autism centres in Konaseema District.

**Est. Cost. Rs. 40.00 Lakhs.**

**OFFICER INVITING BIDS:** Chief Engineer, Samagra Shiksha, Amaravati.

1. Bids are invited on the e-procurement platform for the above -mentioned work from the Contractors/ Contracting firms **registered with Government of Andhra Pradesh**. The details of Tender conditions and terms can be downloaded from the electronic procurement platform of Government of Andhra Pradesh i.e. www.ape-Procurement.gov.in

   Approximate Estimate Contract Value of work: **Rs. 32,12,152.00.**

2. Contractors would be required to register on the e-Procurement Market place "**www.ape-Procurement.gov. in**" and submit their bids online. The department will not accept any bid submitted in the paper form.

   a) Non refundable **Processing fee of Rs.20,000/-** to be paid by way of crossed demand draft/ online in favour of the State Project Director, Samagra Shiksha Payable at Vijayawada.

   b) E.M.D. to be paid by way of Crossed Demand Draft/ online payment for **Rs. 32,122/-** (i.e.,1.00% of ECV) drawn in favour of Chief Engineer, APSSA, Amaravati (or) Bank Guarantee in favour of Chief Engineer, APSSA, Amaravati obtained from Nationalized/ Scheduled Commercial Bank at 1.00% of the estimated contract value of work along with bid documents and the balance EMD @ 1.50% of Estimate Contract Value/ Tender Contract Value whichever is higher is to be paid at the time of concluding agreement.

   c) All the participating bidders shall electronically pay mandatorily a Transaction fee of **Rs. ****/- [@ 0.03% of ECV** (Estimate Contract Value) + GST @ 18.00% on transaction fee] with a cap of Rs.20,000/- to M.D., APTS LTD., to the service provider through Payment Gateway service on e procurement platform by using Credit cards (Any MASTER/ VISA Card) issued by any bank or through net banking accounts with ICICI or HDFC Banks as per G.O.Ms.No.13 IT&C Dept. Dt.07.05.2006 with effect from 02.02.2007.

   d) Scanned E.M.D. and Processing fee may be uploaded with the bids and originals must be submitted to the Chief Engineer, Samagra Shiksha, Amaravati, so as to reach before 04.30 PM on or before the date of opening of Price bid either personally or through courier or by post and the receipt of the same within the stipulated time shall be the responsibility of the tenderer. Failure to furnish the original DD/BG towards EMD and DD for processing fee before 04.30 PM on or before Price bid opening date will entail rejection of bid which is mandatory.

   e) Failure to furnish the original Demand Draft/ BG on or before the date stipulated in the NIT will entail rejection of bid and will be suspended/ disqualified from participating in the tenders on e-procurement platform for a period of 12 months



Tenderer

Chief Engineer
APSS, Amaravati.

from the date of bid submission. The suspension of bidder shall be automatically enforced by the 'e-procurement system'.

f) If the lowest bidder backs out at the time of agreement, penalty of forfeiture of EMD will be imposed and business of the agency will be suspended for one year with all the Departments in A.P in respect of conventional tenders also vide G.O.Ms.No. 259 of T, R&B (R.V.) dept., dt. 06.09.2008.

g) The tenderers shall furnish a declaration in online stating that the soft copies uploaded by them are genuine. Any incorrectness/deviation noticed will be viewed seriously and apart from cancelling the work duly forfeiting the EMD, criminal action will be initiated including suspension of business.

3. **i) Vide ref. G.O.Ms.No.174, I&CAD dept dated:1-9-2008, submission of original hard copies of the uploaded scanned copies of DD/BG towards EMD by participating bidders to the tender inviting authority before the opening of the price bid is dispensed forthwith.**

   **ii) All the bidders shall pay the EMD through online and invariably upload the copy in e-Procurement system and this will be the primary requirement to consider the bid responsive.**

   **iii) The department shall carry out the technical evaluation solely based on the uploaded certificates/documents, EMD in the e-Procurement system and open the price bids of the responsive bidders.**

   **iv) The department will notify the successful bidder for submission of original hardcopies of all the uploaded documents prior to entering into agreement. v) The successful bidder shall invariably furnish the Certificates/Documents of the uploaded scan copies to the Tender Inviting Authority before entering into agreement, either personally or through courier or post and the receipt of the same within the stipulated date shall be the responsibility of the successful bidder. The department will not take any responsibility for any delay in receipt/non-receipt of Certificates/Documents from the successful bidder before the stipulated time. On receipt of documents, the department shall ensure the genuinity of the certificates/documents uploaded by the bidder in e-Procurement system. In support of the qualification criteria before concluding the agreement.**

6. **The GO. Ms. No. 174 -I&CAD dated: 1-9-2008 Deactivation of Bidders If any successful bidder fails to submit the original hard copies of uploaded certificates/documents within stipulated time or if any variation is noticed between the uploaded documents and the hardcopies submitted by the bidder, as the successful bidder will be suspended from participating in the tenders on e-Procurement platform for a period of 3 years. The e-Procurement system would deactivate the user ID of such defaulting bidder based on the trigger/recommendation by the Tender Inviting Authority in the system. Besides this, the department shall invoke all processes of law including criminal prosecution of such defaulting bidder as an act of extreme deterrence to avoid delays in the tender process for execution of the development schemes taken up by the government.**

7. **Period of completion of work : 03 Months.**

8. The bidders can view/ down load the tender documents from the 'e' market place.

9. Form of contract – Lump sum contract.

10. **Class of Contractors eligible is as given below:**

Eligible class of contractor as per G.O. Ms. No:

*1) 132 of T(R&B) Dept., Dt.11-8-1999* ***Class-III and above***



Tenderer

Chief Engineer
APSS, Amaravati.

2) *178 of I & CAD Dt 27.9.1997* ***Class-III and above***

3) *94 of I & CAD (PW-COD) Dept. Dt. 01-07-2003* ***Class- IV and above***

4) *8 of T (R&B) Dept. Dt. 08.01-2003* ***Class-IV and above***

**Downloading of Tender Documents: From 24.10.2026, 10.00 AM to 30.10.2026 up to 4.00 P.M.**

**Date and time for Receipt of tenders: From 24.10.2026, 10.00 AM to 31.10.2026 up to 5.00 P.M.**

**Date and time of opening of Technical bid: 30.10.2026 @ 05.30 PM.**

**Date and time of opening Price Bid : 30.10.2026 @ 05.30 PM.**

**Note**: The dates stipulated above are firm and under no circumstances they will be relaxed unless otherwise extended by an official notification or happen to be Public Holidays.

12. **Procedure for submission of Bids**:

(a) Bidders need to contact **Chief Engineer, Samagra Shiksha, Amaravati** for information on e-Procurement.

(b) Bidders need to register on the electronic procurement market place of Government of Andhra Pradesh i.e., **"www.ape-Procurement.gov.in"**. On registration on the e-Procurement market place they will be provided with a user id and password by the system through which they can submit their bids online.

(c) While registering on the e-procurement market Place, Bidders need to scan and upload the required documents as per the Tender requirements onto their profile.

(d) Such uploaded documents pertaining to Technical Bid need to be attached to the tender while submitting the bids on line. The attested copies of all these uploaded documents of Technical Bid, signed undertaking of tender should be submitted off line to **Chief Engineer, Samagra Shiksha, Amaravati** before the date and time of opening of price Bid.

13. **Qualification Requirements:** To qualify for consideration of award of the contract each tenderer should fulfill the following criteria;

i) Availability (either owned or leased) of the following key and critical equipment **(Mandatory)**.

| | | |
|---|---|---|
| (i) Centering | : | **200 Sqm** |
| (ii) Pin vibrators | : | **2 Nos.** |
| (iii) Pan vibrators | : | **2 Nos.** |
| (iv) Concrete mixers | : | **1 Nos.** |
| (v) Water Tanker | **:** | **1 Nos** |

ii) Availability of the **key technical personal** with adequate experience as per clause 6 of conditions of contract. **1. Graduate Engineer – 1 No ( Civil)**

iii) A declaration regarding critical equipment owned/ leased shall be produced by the tenderer on a non judicial stamp paper of Rs.100/-.

iv) Liquid assets/ credit Facilities/ Latest Solvency certificate from Nationalized Bank/Schedule bank of value not less than **Rs. 32.12 Lakhs (Conditional Solvency will not be accepted)**

v) Non refundable **Processing fee of Rs.20,000/-** to be paid by way of crossed demand draft/ online in favour of the State Project Director, Samagra Shiksha Payable at Vijayawada.

vi) E.M.D. to be paid by way of Crossed Demand Draft for **Rs. 32,122/-** (i.e.,1.00% of ECV) drawn in favour of Chief Engineer, APSSA, Amaravati (or) Bank Guarantee in favour of Chief Engineer, APSSA, Amaravati obtained from Nationalized/ Scheduled Commercial Bank at 1.00% of the estimated contract value of work along with bid documents and the balance EMD @ 1.50% of Estimate Contract Value/ Tender Contract Value whichever is higher is to be paid at the time of concluding agreement.

vii) All the participating bidders shall electronically pay mandatorily a Transaction fee of **Rs. ****/- [@ 0.03% of ECV** (Estimate Contract Value) + GST @ 18.00% on transaction fee] with a cap of Rs.20,000/- to M.D., APTS LTD., to the service provider through Payment Gateway service on e procurement platform by using Credit cards (Any MASTER/ VISA Card) issued by any bank or through net banking accounts with ICICI or HDFC Banks as per G.O.Ms.No.13 IT&C Dept. Dt.07.05.2006 with effect from 02.02.2007.

9. The tenderer is subjected to be black listed and the EMD forfeited if he is found to have misled or furnished false information in the forms/ statements/ certificates submitted in proof of qualification requirements or record of performance such as abandoning of work not properly completed in earlier contracts, inordinate delays in completion of the works, litigation history and/ or financial failures and/ or participated in the previous tendering for the same work and had quoted unreasonable high bid prices

10. Even while execution of the work, if found that the contractor had produced false/fake certificates of experience he will be black listed and the contract will be terminated under clause 60 (a) of PS to APSS.

11. Signed declaration (Page 25).

12. The Contractor should submit signed undertaking of tender (Pages 30 to 33).

**Note** : *The certificates in support of Group of works, Single work, Experience in Water Supply & Sanitary arrangements and Electrical works must be in State/Central Government Departments or State/Central Government undertakings only issued by the concerned Executive Engineer and counter signed by the Chief Engineer or equivalent authority. The sub-contractor's/ GPA holders/ Joint ventures experience shall not be taken into account in determining the tenderers compliance with qualifying criteria.*

**Rates for CC/RCC items are as per Standard Data and minimum quantity of cement is adopted for Design Grade mix. Before grounding the work, concrete mix proportions will be designed and the rates will be approved as per design mix. The accepted Tender percentage will be operated on these rates.**

**The rates are inclusive of Overhead &contractors profit @ 13.615% but excluding GST as per govt. norms.**

**14. Procedure for Bid Submission**

i) The Tenderers who are desirous of participating in e-procurement shall submit their Technical bids, price bids etc., in the Standard formats prescribed in the Tender documents, displayed at e-market place. The tenderers should upload the attested scanned copies of documents in support of their Technical bids. The bidders shall sign on all the statements, documents, certificates, uploaded by them, owning responsibility for their correctness / authenticity.



Tenderer

Chief Engineer
APSS, Amaravati.

ii) All the bidders shall invariably upload the scanned copies of all required documents in e-procurement system and this will be the primary requirement to consider the bid responsive.

iii) The department shall carry out the technical evaluation solely based on the uploaded certificates/ documents, in the e-Procurement system and open the price bids of the responsive bidders.

iv) The department will notify the successful bidder for submission of original hardcopies of all the uploaded documents prior to entering into agreement.

v) The successful bidder shall invariably furnish the original Certificates/ Documents of the uploaded scan copies to the Tender Inviting Authority before entering into agreement, either personally or through courier or post and the receipt of the same within the stipulated date shall be the responsibility of the successful bidder. The department will not take any responsibility for any delay in receipt/non-receipt of original Certificates/Documents from the successful bidder before the stipulated time. On receipt of documents, the department shall ensure the genuinity of the documents uploaded by the bidder in e-Procurement system. In support of the qualification criteria before concluding the agreement.

vi) The Department will not hold any risk and responsibility for the loss in transit during uploading of the scanned document, for the invisibility of the scanned document online, and any other problem(s) encountered by the Tenderers while submitting his bids online.

## 15. General Terms & Conditions

**Officer inviting bids**: Chief Engineer, Samagra Shiksha, Amaravati.

1. Bids are invited on the e-procurement platform for the above-mentioned work from the contractors/ contracting firms eligible as per clause-8 of NIT, registered with State Govt.
2. Approximate estimated contract value of work**: Rs. 32,12,152.00/-**
3. Non refundable **Processing fee of Rs.20,000/-** to be paid by way of crossed demand draft/ online in favour of the State Project Director, Samagra Shiksha Payable at Vijayawada.
4. E.M.D. to be paid by way of Crossed Demand Draft for **Rs. 32,122/-** (i.e.,1.00% of ECV) drawn in favour of Chief Engineer, APSSA, Amaravati (or) Bank Guarantee in favour of Chief Engineer, APSSA, Amaravati obtained from Nationalized/ Scheduled Commercial Bank at 1.00% of the estimated contract value of work along with bid documents and the balance EMD @ 1.50% of Estimate Contract Value/ Tender Contract Value whichever is higher is to be paid at the time of concluding agreement.
5. All the participating bidders shall electronically pay mandatorily a Transaction fee of **Rs. ****/- [@ 0.03% of ECV** (Estimate Contract Value) + GST @ 18.00% on transaction fee] with a cap of Rs.20,000/- to M.D., APTS LTD., to the service provider through Payment Gateway service on e procurement platform by using Credit cards (Any MASTER/ VISA Card) issued by any bank or through net banking accounts with ICICI or HDFC Banks as per G.O.Ms.No.13 IT&C Dept. Dt.07.05.2006 with effect from 02.02.2007.
6. Period of completion of work: **03 Months**
7. Tender Schedules: tender schedules can be downloaded from the Web site “www.ape-Procurement.gov.in”
8. Form of Contract: Lump sum contract.



Tenderer

Chief Engineer
APSS, Amaravati.

9. Class of contractor Eligible**: Class- III and above** as per Go. Ms No.178 Date: 27/9/1997 or **Class- III and above** as per Go. Ms No. 132 Date: 11/08/1999 or **Class- IV and above** as per Go. Ms No. 94 Date: 01/07/2003 **Class- IV and above** G.O.No.8, T (R&B), dt 08.01.2003

The eligible intending bidders would be required to enroll themselves on the e-Procurement market place www.e-Procurement.gov.in and submit their bids online

**Downloading of Tender Documents: From 24.10.2026, 10.00 AM to 30.10.2026 up to 4.00 P.M.**

**Date and time for Receipt of tenders: From 24.10.2026, 10.00 AM to 31.10.2026 up to 5.00 P.M.**

**Date and time of opening of Technical bid: 30.10.2026 @ 05.30 PM.**

**Date and time of opening Price Bid : 30.10.2026 @ 05.30 PM.**

Note: The dates stipulated above are extendable by an official notification by the department or happen to be Public Holidays

10. The documents that are uploaded online on e-market place will only be considered for technical bid evaluation.

**Successful bidder.** The department will not take any responsibility for any delay in receipt/ non-receipt of original DD/BG towards EMD, Certificates/Documents from the successful bidder before the stipulated time. On receipt of documents, the department shall ensure the genuinity of the DD/BG towards EMD and all other certificates/documents uploaded by the bidder in e-Procurement system. In support of the qualification criteria before concluding the agreement.



Tenderer

Chief Engineer
APSS, Amaravati.

## INSTRUCTIONS TO TENDERERS

## A – GENERAL

**NAME OF WORK:** **Construction of two rooms/ one room and renovation of existing rooms for SG MPL HS ANKASANI CHERUVU for the establishment of Autism centres in Konaseema District.**

**Est. Cost. Rs. 40.00 Lakhs.**

**1. Scope of work : Buildings**

**Give brief description and location of work: Construction of two rooms/ one room and renovation of existing rooms for SG MPL HS ANKASANI CHERUVU for the establishment of Autism centres in Konaseema District.**

**Est. Cost. Rs. 40.00 Lakhs.**

a) ECV put to tender. : **Rs. 32,12,152.00/-**

b) Period of completion. : **03 Months**

c) SSR adopted : **Common SSR of 2026-27**

d) Rates adopted for

**a) Cement : Rs. 4220/- M.T.**

**b) Steel HYSD : Rs. 53000/- M.T.**

e) Allowances

a) Municipal Allowance – As per Government. Norms

f) Seigniorage Charges - As per Government. Orders

g) GST - As per Government. Orders

The Chief Engineer, Samagra Shiksha, Amaravati invites tenders for the above works vide **NIT No. 27/ APSS/CWs/Autism Centres/2026-27/2nd call, dt: 22.10.2026 of the Chief Engineer, APSS, Vijayawada.**

**2. Firms Eligible to Tender:**

2.1 The Firms who

i) Possess the valid registration in the class and category mentioned in the NIT and satisfy all the conditions therein are the eligible tenderers.

ii) are not blacklisted or debarred or suspended by the Government for what ever the reason, prohibiting them not to continue in the contracting business are the eligible tenderers.

iii) Have complied with the eligibility criteria specified in the NIT are the eligible tenderers.

2.2 **Firms Ineligible to Tender:**

i) A retired officer of the Govt. of AP or Govt. of India executing works is disqualified from tendering for a period of two years from the date of retirement without the prior permission of the Government.

ii) The Tenderer who has employed any retired officer as mentioned above shall be considered as an ineligible tenderer.

iii) The contractor himself or any of his employees is found to be Gazetted Officer who retired from Government Service and had not obtained permission from

the Government for accepting the contractor's employment within a period of 2 years from the date of his retirement.

iv) The Contractor or any of his employees is found at any time after award of contract, to be such a person who had not obtained the permission of the Government as aforesaid before submission of the tender or engagement in the Contractor's service.

v) v) Contractor shall not be eligible to tender for works in the division/ circle where any of his near relatives are employed in the rank of Assistant Engineer or Assistant Executive Engineers and above on the Engineering side and Divisional Accounts Officer and above on the administrative side. The Contractor shall intimate the names of persons who are working with him in any capacity or are subsequently employed. He shall also furnish a list of Gazetted/ Non-Gazetted, State Government Employees related to him. Failure to furnish such information tenderer is liable to be removed from the list of approved contractors and his contract is liable for cancellation.

**Note: Near relatives include**

1. Sons, step sons, daughters, and stepdaughters.
2. Son-in-law, and daughter-in-law.
3. Brother-in-law, and sister-in-law.
4. Brothers and Sisters.
5. Father and Mother.
6. Wife / Husband.
7. Father-in-law and Mother-in-law
8. Nephews, nieces, uncle and aunts
9. Cousins and
10. Any person residing with or dependent on the contractor.

**3. Qualification data of the Tenderers**

3.1(i). The tenderer shall furnish the following particulars in the formats enclosed, supported by documentary evidence as specified in the formats.

a) Check slip to accompany the tender (in Annexure-I).

b) Attested copies of documents relating to the Registration of the firm, Registration as Civil Contractor, Partnership deed, Articles of Association, Commercial Tax Registration, Permanent Account Number with latest IT returns submitted and proof of receipt.

**Note**: The Partnership firms, which are registered as Contractors shall intimate the change in partnership deed, if any as per G.O.Ms.No.58, I & CAD dated 23-04-2002 within one month of such change. Failure to notify the change to the registration authority in time will entail the firms to forfeit their registration and their tender will be rejected. The intimation of change of partners if any and the acceptance by the Registration authority may be enclosed.

3.1(ii) The contractor should have valid GST registration with Commercial Tax department and latest commercial tax clearance certificate.

3.1(iii) The contractor should furnish copy of permanent account number (PAN) card and latest income tax returns submitted along with proof of receipt. (SARAL FORM)

The bidder should submit the particulars preferably in the format specified in the tender schedule along with necessary certificates.



Tenderer

Chief Engineer
APSS, Amaravati.

3.1(iv) The tenderer should further demonstrate:

Availability (either owned or leased) of the following key and critical equipment. **(Mandatory)**

| | | |
|---|---|---|
| (i) Centering | : | **200 Sqm** |
| (ii) Pin vibrators | : | **2 Nos.** |
| (iii) Pan vibrators | : | **2 Nos.** |
| (iv) Concrete mixers | : | **1 Nos.** |
| (v) Water Tanker | : | **1 Nos.** |

3.1(v) Availability of the **key technical personal** with adequate experience as per clause 6 of conditions of contract. **1. Graduate Engineer – 1 Nos (Civil)**

3.1(vi) Liquid assets/ credit Facilities/ Latest Solvency certificate from Nationalized Bank/ Schedule bank of value not less than **Rs. 32.12 Lakhs (Conditional certificates are not accepted).**

3.1(vii) Copy of Registration as mentioned in Tender Notice.

3.1(viii) Information on Litigation history in which tenderer is involved if any.

3.1(ix) Non refundable **Processing fee of Rs.20,000/-** to be paid by way of crossed demand draft/ online in favour of **State Project Director, Samagra Shiksha**, Payable at Vijayawada.

**3.1(x).** E.M.D. to be paid by way of Crossed Demand Draft for **Rs. 32,122/-** (i.e.,1.00% of ECV) drawn in favour of Chief Engineer, APSSA, Amaravati (or) Bank Guarantee in favour of Chief Engineer, APSSA, Amaravati obtained from Nationalized/ Scheduled Commercial Bank at 1.00% of the estimated contract value of work along with bid documents and the balance EMD @ 1.50% of Estimate Contract Value/ Tender Contract Value whichever is higher is to be paid at the time of concluding agreement.

**3.**1(xi). All the participating bidders shall electronically pay mandatorily a **Transaction fee of Rs. ****/- [@ 0.03% of ECV** (Estimate Contract Value) + GST @ 18.00% on transaction fee] with a cap of Rs.20,000/- to M.D., APTS LTD., to the service provider through Payment Gateway service on e procurement platform by using Credit cards (Any MASTER / VISA Card) issued by any bank or through net banking accounts with ICICI or HDFC Banks as per G.O.Ms.No.13 IT&C Dept. Dt.07.05.2006 with effect from 02.02.2007.

3.1(xii) Declaration (Page 20)

3.1(xiii) Signed undertaking of tender (Pages 22 to 25)

**Note 1** : *The certificates in support of 3.1(i), 3.1(ii), 3.1(iii), 3.3(iv), 3.3(vii) must be in State/Central Government Departments or State/Central Government undertakings only issued by the concerned Executive Engineer and counter signed by the Chief Engineer or equivalent authority. The sub-contractor's/GPA holders/ Joint ventures experience shall not be taken into account in determining the tenderers compliance with qualifying criteria.*

**Note 2** : *@ Estimate Contract Value is of 2017-18 price level. Weightage of 10% per annum should be given for the annual turnover, single work experience, experience of contractor in building electrification works, water supply and sanitary engineering works to bring them to present price level.*



Tenderer

Chief Engineer
APSS, Amaravati.

**Note 3** : *To qualify for a contract made up of this work and other works for which the tenders are invited in the CTN this tenderer must demonstrate having experience and resources sufficient to meet the aggregate of all qualifying criteria except for 3.1(iii) and 3.1(iv) for the individual contracts.*

***Note 4 :*** *Tenders from joint ventures will not be accepted.*

***Note 5:*** ***The bidders shall sign on all the statements, documents, certificates uploaded by him owning responsibility for their correctness/authenticity.***

3.1 Even though the tenderers meet the above qualifying criteria, they are liable to be disqualified/ debarred/ suspended/ blacklisted if they have

- Furnished false/ fabricated particulars in the forms, statements and/ annexures submitted in proof of the qualification requirements and/or
- Not turned up for entering into agreement, when called upon.
- record of poor progress such as abandoning the work, not properly completing the contract, inordinate delays in completion, litigation history or financial failures etc. and/or
- participated in the previous bidding for the same work and had quoted unreasonably high tender percentage and
- even while execution of the work, if found that the work was awarded to the Contractor based on false/ fake certificates of experience, the Contractor will be blacklisted and work will be taken over invoking clause 60(a) of PS to APSS.
- The tender of the contractor will be disqualified along with the forfeiture of E.M.D. and the contractor will be debarred from future tendering for a further period of 2 years in the event of furnishing of false/ fraudulent certificates along with the tender.

3.2 **Tenders with more than the estimated contract value shall summarily be rejected.**

3.3 For tenders up to 15**% less than** the estimated contract value of work, no additional security deposit is required. But for tenders **less than 15%** of the estimated Contract Value of work, the difference between the tendered amount and **85%of the estimated contract value**, shall be paid by the successful tenderer at the time of concluding agreement as an additional security to fulfill the contract through a Bank Guarantee or Demand Draft on a Nationalized Bank/ Scheduled bank in the prescribed format valid till completion of the work in all respects

3.4 a) If the percentage quoted by a tenderer is found to be either abnormally high or with in the permissible ceiling limits prescribed but under collusion or due to unethical practices adopted at the time of tendering process, such tenders shall be rejected.

b) A tenderer submitting a Tender which the tender accepting authority considers excessive and or indicative of insufficient knowledge of current prices or definite attempt of profiteering will render himself liable to be debarred permanently from tendering or for such period as the tender accepting authority may decide. The tenderer overall percentage should be based on the controlled prices for the materials, if any, fixed by the Government or the reasonable prices permissible for the tenderer to charge a private purchaser under the provisions of clause-6 of the hoarding and profiteering prevention ordinance of 1943 as amended from time to time and on similar principle in regard to labour supervision on the construction.

3.5 **CONDITIONAL TENDER**

Conditional tenders are not accepted. Submission of tender would be construed as acceptance to all the terms and conditions of the tender which include conditions of contract, drawings and accompanying specifications.

**4. One Tender per Tenderer:**

4.1 Each Tenderer shall submit only one Tender for the work. A Tenderer who submits more than one Tender will cause disqualification of all the Tenders submitted by the Tenderer.

**5. Cost of Tendering**

5.1 The Tenderer shall bear all costs associated with the preparation and submission of his Tender and the tender inviting authority will in no case be responsible and liable for those costs.

**6. Site Visit.**

6.1 The Tenderer, at the Tenderer's own responsibility and risk is advised to visit and examine the Site of Work and its surroundings and obtain all information that may be necessary for preparing the Tender for entering into a contract, for construction of the work. The costs of visiting the site shall be at the Tenderer's own expense.

## B. TENDER DOCUMENT

**7. Contents of Tender document.**

7.1 One set of Tender document, comprises of the following:

**Technical bid**

1) Notice Inviting Tenders (NIT)
2) Instruction to Tenderer
3) Forms of Tender and qualification information
4) Conditions of Contract.
5) Specifications.
6) Drawings.
7) Forms of Securities. i.e., EMD, Additional Security etc.

**Price bid**

Bill of Quantities and Price bid.

**8. Clarification on Tender Documents**

8.1 A prospective Tenderer requiring any clarification on Tender documents may contact the Tender Inviting Officer at the address indicated in the NIT. The Tender Inviting Officer will also respond to any request for clarification, received through post.

**9. Amendment to Tender Documents**

9.1 Before the last date for submission of Tenders, the Tender Inviting Officer may modify any of the Contents of the Tender Notice, Tender documents by issuing amendment/ Addendum.

9.2 Any addendum/ amendments issued by the Tender Inviting Officer shall be part of the Tender Document and it shall be attached to the Tender Notice on web site (i.e) e-Procurement.gov.in

9.3 To give prospective Tenderers reasonable time to take an addendum into account in preparing their bids, the Tender Inviting Officer may extend if necessary, the last date for submission of tenders.

## C. PREPARATION OF TENDERS.

**10. Language of the Tender.**

All documents relating to the tender shall be in the English Language only.

**11. Documents comprising of the Tender.**

11.1 The bidders who are desirous of participating in e-procurement shall submit their technical bids, price bids etc., in the standard prescribed format in the tender documents, displayed at e market place. The bidders should upload the scanned copies of all the relevant certificates, documents etc., in the e market place in support of their technical bids. The bidders shall sign on all the statements, documents, certificates, uploaded by him, owning responsibility for their correctness/authenticity.

11.2 The department shall carry out the technical evaluation solely based on the uploaded certificates/documents in the e-Procurement system and open the price bids of the responsive bidders.

**12. Bid Offer:**

12.1 Bill of Quantities called Schedule "A" and the bid offer accompanies the tender document as Volume.II. It shall be explicitly understood that the Tender Inviting Officer does not accept any responsibility for the correctness or completeness of this schedule 'A' and this schedule 'A' is liable to alterations by omissions, deductions or additions at the discretion of the Chief Engineer or as set forth in the conditions of the contract. The tenderers will have to state clearly their willingness to execute the work at certain specific percentage of excess or less or at par of the ECV indicated at the space provided therein in Schedule 'A'. The tenderer should however quote his lump sum tender based on this schedule of quantities. He should quote his offer as a overall tender percentage. The overall tender percentage should be written both in words and figures. The bid offers i.e., percentage shall be quoted both in figures and words.

12.2 The Schedule –A (or Price-bid) contains not only the quantities but also the rates worked out by the Department and the amount for each item and total value of the estimated contract. The tenderer should workout his own rates keeping in view the work, site conditions and quote his overall tender percentage with which he intends to execute the work.

12.3 The bid offer shall be for the whole work and not for individual items/ part of the work.

12.4 All duties, taxes, and other levies payable by the contractor as per State/ Central Government rules, shall be included in the tender percentage quoted by the tenderer.

12.5 The tendered contract amount as computed based on overall tender percentage is subject to variation during the performance of the Contract in accordance with variation in quantities etc.

**13. Validity of Tenders:**

13.1 Tenders shall remain valid for a period of not less than three months from the last date for receipt of Tender specified in NIT.

13.2 During the above-mentioned period no plea by the tenderer for any sort of modification of the tender based upon or arising out of any alleged misunderstanding of misconceptions or mistake or for any reason will be entertained.

13.3 In exceptional circumstances, prior to expiry of the original time limit, the Tender Inviting Officer may request the bidders to extend the period of validity for a specified additional period. Such request to the Tenderers shall be made in writing. A Tenderer may refuse the request without forfeiting his E.M.D. A Tenderer agreeing to the request will not be permitted to modify his Tender, but will be required to extend the validity of his E.M.D. for a period of the extension.

**14. Return of E.M.D. to unsuccessful tenderer.**

The earnest money deposit will be refunded to the unsuccessful tenderer by registered post at the expiry of the period of validity of tender or the entrustment of the work to the successful tenderer whichever is earlier.

**15. Return of EMD to successful tenderer**

The EMD paid by the successful tenderer at the time of tendering will be discharged when the tenderer has signed the agreement and furnished the required EMD pursuant to Clause 14.2

**16.** The earnest money deposited by the successful tenderer will not carry any interest and it will be dealt with as provided in the conditions stipulated in the tender.

16.1 The E.M.D. shall be forfeited.

(a) if the Tenderer withdraws the Tender during the validity period of Tender.

(b) in the case of a successful Tenderer, if he fails to sign the Agreement for whatever the reason.

16.2 **In consideration of the Chief Engineer undertaking to investigate and to take into account each tender and in consideration of the work thereby involved, all earnest monies deposited by the tenderer will be forfeited to the department in the event of such tenderer either modifying or with-drawing his tender at his instance within the said validity period of three months.**

## D. SUBMISSION OF TENDERS.

**17. Submission of Tenders:**

17.1 The Tenderers who are desirous of participating in e-procurement shall submit their Technical bids, price bids etc., in the Standard formats prescribed in the Tender documents, displayed at e-market place. The tenderers should upload the scanned copies in support of their Technical bids. The documents are to be uploaded in ZIP format only. The bidders shall sign on all the statements, documents, certificates, uploaded by him, owning responsibility for their correctness/ authenticity. The documents uploaded online will only be considered for evaluation.

17.2 The Department will not hold any risk and responsibility for the loss in transit during uploading of the scanned document, for the invisibility of the scanned document online, and any other problem(s) encountered by the Tenderers while submitting his bids online.

17.3 Related certificates, documents etc., duly attested are to be scanned and uploaded on to the e-procurement platform at **www.ape-Procurement.gov.in** in support of items mentioned in clause 3.

17.4 Any other condition regarding receipt of tenders in conventional method appearing in Tender document may be treated as Non-applicable.

**18. Last date/ time for Submission of the Tenders.**

18.1 Tenders must be submitted online not later than the date and time specified in NIT.

18.2 The Chief Engineer or his nominee may extend the dates for issue and receipt of Tenders by issuing an amendment in which case all rights and obligations of the Chief Engineer and the Tenderers will remain same as previously.

**19. Late Tenders.**

Tenders will not be received after the last date/ time prescribed in NIT.

**20. Modification to the Tender.**

Tenderers can modify their Tender online before the last date/time prescribed in NIT and amendments issued, if any.

**21.** It would not be possible to modify tenders after the last date /time of submission of Tenders.

**22. Submission of Tenders:**

22.1 The tenderer shall invariably ensure that the following are furnished in hard copy to the tender receiving authority, that is, Chief Engineer, Samagra Shiksha, Amaravati before opening of the technical bid besides uploading them on line.

a) Check slip

b) Copy of contractors registration certificate under appropriate class with Government of Andhra Pradesh

c) Copy of permanent account number (PAN) card and latest income tax returns submitted along with proof of receipt. (SARAL FORM 2023-24)

d) Copy of GST Tax Registration certificate with TIN No. and latest commercial tax clearance certificate.

e) The availability of Key/ critical construction/ quality control equipment in Statement V & VIII **(Mandatory)**.

f) The availability of Key personnel in Statement VI

g) The information and litigation history in Statement VII7

h) Proof of liquid assets/ credit Facilities/ Latest Solvency certificate from Nationalized Bank/ Schedule bank for the required amount.

i) EMD by way of online payment/ DD.

j) Signed declaration (Page 25).

k) Signed under taking of tender (Pages 30 to 33).

## E. TENDER OPENING AND EVALUATION

**23. Tender opening**

23.1 The Technical bids will be opened online by the Chief Engineer or his nominee at the time and date as specified in the Notice Inviting Tender. All the Statements, documents, certificates, etc, uploaded by the Tenderers will be verified and downloaded, for technical evaluation. The technical bids will be evaluated against the specified parameters/ criteria same as in the case of conventional tenders and the technically qualified bidders will be identified. The result of Technical bids evaluation will be displayed on the e-market place, which can be seen by all the tenderers who participated in the Tenders.

**24. Clarification on the Technical Bid.**

24.1 The tender opening authority may call upon any tenderer for clarification on the statements, documentary proof relating to the technical bid. The request for clarification and response thereto shall be in writing and it shall be only on the qualification information uploaded online by the tenderer. The clarification called for from the tenderers shall be furnished within the stipulated time, which shall not be more than a week.

24.2 The tenderer if so desirous, shall agree in writing to furnish the clarification called for within the stipulated time and, for disqualification and rejection of his tender in the event of failure to do so.

**25. Examination of technical Bids and Determination of Responsiveness**

25.1 The Chief Engineer or his nominee will evaluate whether each Tenderer is satisfying the eligibility criteria prescribed in the tender document and declares them as a qualified Tenderer.

25.2 If any alteration is made by the tenderer in the tender documents, the conditions of the contract, the drawings, specifications or statements/ formats or quantities the tender will be rejected.

**26. Price Bid Opening:**

26.1 At the specified date and time, the price bids of all the technically qualified bidders will be opened online by the Chief Engineer or his nominee and the result will be displayed on the e-market place which can be seen by all the bidders who participated in the Tenders.

26.2 The Price Bid of the Unqualified Tenderers will not be opened.

26.2.1 Tenders shall be scrutinised in accordance with the conditions stipulated in the Tender document. In case of any discrepancy of non-adherence to conditions, the Tender accepting authority shall communicate the same which will be binding both on the tender Opening authority and the Tenderer. In case of any ambiguity or dispute, the decision taken by the Tender Accepting Authority on tenders shall be final.

**27. Evaluation and Comparison of Price Bids**

27.1 The Chief Engineer or his nominee will evaluate and compare the price bids of all the qualified Tenderers.

27.2 Negotiations at any level are strictly prohibited.

27.3 Selection of Tenderer among the lowest & equally quoted tenderers will be in the following orders:

a) The tenderer whose bid capacity is higher will be selected.

b) In case the bid capacity is also same the tenderer whose annual turnover is more will be preferred.

c) Even if the criteria incidentally become the same, the turnover on similar works and thereafter machinery available for the work and then the clean track record will be considered for selection.

**28. Process to be Confidential.**

28.1 Information relating to the examination, clarification, evaluation and comparison of Tenders and recommendations for the award of a contract shall not be disclosed to Tenderers or any other persons not officially concerned with such process until the award to the successful Tenderer has been announced by the tender accepting

authority. Any effort by a Tenderer to influence the processing of Tenders or award decisions may result in the rejection of his Tender.

28.2 No Tenderer shall contact the Chief Engineer or any authority concerned with finalisation of tenders on any matter relating to its Tender from the time of the Tender opening to the time the Contract is awarded. If the Tenderer wishes to bring additional information to the notice of the Chief Engineer, it should do so in writing.

28.3 Before recommending/ accepting the tender, the tender recommending / accepting authority shall verify the correctness of certificates submitted to meet the eligibility criteria and specifically experience. The authenticated agreements of previous works executed by the lowest tenderer shall be called for.

## F. AWARD OF CONTRACT

### 29. Award Criteria

29.1 The Chief Engineer or his nominee will award the contract on approval of the tender by Competent authority.

29.2 The tender accepting authority reserves the right to accept or reject any Tender or all tenders and to cancel the Tendering process, at any time prior to the award of Contract, without thereby incurring any liability to the affected Tenderer or Tenderers or any obligation to inform the affected Tenderer or Tenderers of the reasons for such action.

### 30. Notification of Award and Signing of Agreement.

30.1 The Tenderer whose Tender has been accepted will be notified of the award of the work prior to expiration of the Tender validity period by registered letter. This letter (hereinafter and in the Conditions of Contract called “Letter of Acceptance”) will indicate the sum that the Government will pay the Contractor in consideration of the execution, completion, and maintenance of the Works by the Contractor as prescribed by the Contract (hereinafter and in the Contract called the “Contract Amount”).

30.2 When a tender is to be accepted the concerned tenderer shall attend the office of the Chief Engineer concerned on the date fixed in the Letter of acceptance. Upon intimation being given by the Chief Engineer, of acceptance of his tender, the tenderers shall make payment of the E.M.D., and additional security deposit wherever needed by way of Demand Draft obtained from a Nationalized / Scheduled Bank with a validity period of duration of Contract period plus defects liability period of 2 years and sign an agreement in the form prescribed by the department for the due fulfillment of the contract. Failure to attend the Chief Engineer’s office on the date fixed, in the written intimation, to enter into the required agreement shall entail forfeiture of the Earnest Money deposited. The written agreement to be entered into between the contractor and the Department shall be the foundation of the rights and obligations of both the parties and the contract shall not be deemed to be complete until the agreement has first been signed by the contractor and then by the proper officer authorized to enter into contract.

30.3 The successful tenderer has to sign an agreement within a period of 15 days from the date of receipt of communication of acceptance of his tender. On failure to do so his tender will be cancelled duly forfeiting the E.M.D., paid by him without issuing any further notice and action will be initiated for black listing the tenderer.

### 31. Corrupt or Fraudulent Practices

31.1 The Government requires that the bidders/ suppliers/ contractors under Government financed contracts, observe the highest standard of ethics during the procurement and execution of such contracts. In pursuance of this policy, the Government

(a) define for the purposes of the provision, the terms set forth below as follows:

(i) "corrupt practices" means the offering, giving, receiving or soliciting of any thing of value to influence the action of a Government official in procurement process or in contract execution: and

(ii) "fraudulent practice" means a misrepresentation of facts in order to influence a procurement process or the execution of a contract to the detriment of the Government and includes collusive practice among Tenderers (prior to or after Tender submission) designed to establish in Tender prices at artificial non-competitive levels and to deprive the Government of the benefits of free and open competition.

(b) Will reject a proposal for award if it determines that the Tenderer recommended for award has engaged in corrupt or fraudulent practices in competing for the contract in question.

(c) Will blacklist/ or debar a firm, either indefinitely or for a stated period of time, if at any time determines that the firm has engaged in corrupt or fraudulent practices in competing for, or in executing a Government Contract.

(d) Further more, Tenderers shall be aware of the provisions stated in the General Conditions of Contract.

**33 Rights of the Department**

33.1 The Chief Engineer, Samagra Shiksha reserves the right to reject any or all of the tenders, without assigning any reason whatsoever.

33.2 In the event of any dispute regarding any of the tender conditions, the decision of the Chief Engineer shall be final.



Tenderer

Chief Engineer
APSS, Amaravati.

**FORMS OF TENDER**
**QUALIFICATION INFORMATION**
**AND**
**UNDER TAKING OF TENDERER**



Tenderer

Chief Engineer
APSS, Amaravati.

## QUALIFICATION INFORMATION

**Annexure –I**

## CHECKLIST TO ACOMPANY THE TENDER

**Check slip**

**Documents to be submitted by tenderer on the E-procurement plot form on line**

| Sl. No. | Document to be uploaded to profile | Description to be given | Scanned attested documents along with electronic bid (on line) (needs to be attached to Tech. Bid) |
|---|---|---|---|
| 1.a)<br><br>b) | Contractor's registration under appropriate class with Govt. of A.P. Partnership deed in case of firms & Article of Association in case of companies | Registration | Yes |
| 2. | Copy of Permanent Account Number (PAN) card along with latest IT returns submitted and proof of receipt. (SARAL FORM AY 2024-25) | ITCC | Yes |
| 3. | Availability of key and Critical equipment on non Judicial stamp paper worth Rs.100/- (Format at Statement V) | Key and critical Equipment | Yes |
| 4. | Availability of key personal as in Statement VI | Key personnel | Yes |
| 5 | Declaration of lab equipment on a non judicial stamp paper of Rs.100/- as in the statement - VIII | Lab equipment | Yes |
| 6 | Proof of liquid assets/ Credit facilities/ Latest solvency certificates from banks in the format given in schedules | Credit facilities | Yes |
| 7. | Information of litigation history as in Statement VII | Litigation history | Yes |
| 8. | Transaction fee | Transaction fee | Yes |
| 9. | E.M.D | EMD | Yes |
| 10. | GST Registration and latest valid commercial tax clearance certificate | GST Registration | Yes |
| 11. | Declaration | Declaration | Yes |
| 12. | Undertaking of Tender | Undertaking | Yes |

Note: 1. Please upload documents in ZIP format with suitable description as defined above.

2. Only attested copies of the certificates shall be scanned and scanned documents shall be legible failing which they will not be considered.

3. Shall sign on all statements, documents, certificates uploaded owning responsibility for their correctness/ authenticity.

4. All experience certificates including that of the contractor and certificates in support of existing commitments issued by an Officer not below the rank of Executive Engineer and should be countersigned by the Chief Engineer or equivalent authority.



Tenderer

Chief Engineer
APSS, Amaravati.

**DECLARATION**
**[Also to be submitted along with the Hard copies of work done certificates.]**

I / WE ……. have gone through carefully all the Tender conditions and solemnly declare that I / we will abide by any penal action such as disqualification or black listing or determination of contract or any other action deemed fit, taken by, the Department against us, if it is found that the statements, documents, certificates produced by us are false / fabricated.

I / WE hereby declare that, I / WE have not been blacklisted / debarred / Suspended / demoted in any department in Andhra Pradesh or in any State due to any reasons.

**Signature of the Tenderer**

**STATEMENT - V**

**Availability of Critical Equipment (Mandatory)**

The tenderer should furnish the information required below, regarding the availability of the equipment, required for construction/ quality control.

| Sl. No | Details of Equipment | Number required | Number | | |
|---|---|---|---|---|---|
| | | | Owned | Leased | To be procured |
| 1 | 2 | 3 | 4 | 5 | 6 |
| | | | | | |

**Signature of the Tenderer**

A declaration regarding the equipment owned shall be produced by the Tenderer on a non-judicial stamp paper of Rs. 100/- as below;

**DECLARATION**

**"I ……. do hereby solemnly affirm and declare that I /we own the following equipment for using on the subject work and also declare that I / We will abide by any action such as disqualification or determination of Contract or blacklisting or any action deemed fit, if the department detects at any stage that I/we do not possess the equipment listed below.**

| Sl. No. | Details of each Equipment | No. | Year of purchase | Regn. Number | Capacity | Any other data. | Is it in working condition |
|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |



Tenderer

Chief Engineer
APSS, Amaravati.

| |
|---|

## STATEMENT – VI.

**Availability of Key Personnel**

Qualification and experience of Key Personnel proposed to be deployed for execution of the Contract.

| Sl. No | Name | Designation | Qualification | Total Experience | Working with the Tenderer since. |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | | | | | |

**Signature of the Tenderer**

## STATEMENT - VII

Information on litigation history in which Tenderer is the Petitioner.

| S. No | Case No. / Year | Court where filed. | Subject Matter / Prayer in the case. | Respondents i.e., SE / CE | Present Stage. |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| | | | | | |

**Signature of the Tenderer**

## STATEMENT - VIII

### UNDER TAKING OF TENDER

**[Also to be submitted along with the Hard copies of work done certificates.]**

Date: ……

To
The Chief Engineer,
Samagra Shiksha,
Amaravati.
Sir,

I / We do hereby tender and if this tender be accepted, under take to execute the following work Construction of two rooms/ one room and renovation of existing rooms for SG MPL HS ANKASANI CHERUVU for the establishment of Autism centres in Konaseema District as shown in the drawings and described in the specifications available on online and also in the office of the Chief Engineer, Samagra Shiksha, Amaravati. with such variations by way of alterations or additions to, and omissions from the said works and method of payment as provided for in the "conditions of the contract" for the sum as quoted online or such other sum as may be arrived under the clause of the standard preliminary specifications relating to "Payment on lump-sum basis or by final measurement at unit rates"

I/WE have also quoted percentage excess or less on E.C.V., in Schedule 'A', annexed (in words and figures) for which I/We agree to execute the work when the lump sum payment under the terms of the agreement is varied by payment on measurement quantities.

I/WE have quoted Percentage excess or less on E.C.V., in Schedule 'A' both in words & figures. In case of any discrepancy between the Percentage excess or less on E.C.V., in words and figures, the rates quoted words only shall prevail.

I/WE agreed to keep the offer in this tender valid a period of Three month(s) mentioned in the tender notice and not to modify the whole or any part of it for any reason within above period. If the tender is withdrawn by me/us for any reasons whatsoever, the earnest money paid by me/us will be forfeited to Department.

I/WE hereby distinctly and expressly, declare and acknowledge that, before the submission of my/our tender I/We have carefully followed the instructions in the tender notice and have read the A.P.S.S. and the preliminary specifications therein and the A.P.S.S. addenda volume and that I/We have made such examination of the contract documents and the plans, specifications and quantities and of the location where the said work is to be done, and such investigation of the work required to be done, and in regard to the material required to be furnished as to enable me/us to thoroughly understand the intention of same and the requirements, covenants, agreements, stipulations and restrictions contained in the contract, and in the said plans and specifications and distinctly agree that I/We will not hereafter make any claim or demand upon the Government based upon or arising out of any alleged misunderstanding or misconception /or mistake on my/or our part of the said requirement, covenants, agreements, stipulations, restrictions and conditions.

I / WE enclosed to my/our tender a crossed demand draft (No…….. ……..…….dated:……..) for Rs…….towards earnest money deposit not to bear any interest.

I/WE enclosed to my/our tender a crossed demand drafts (No……..…….dated:……..) for Rs…….towards transaction fees which is non-refundable.



Tenderer

Chief Engineer
APSS, Amaravati.

I/WE enclosed to my/our tender a crossed demand draft (No……..……..dated:……..) for Rs…….towards Transaction fee which is non-refundable

I/WE shall not assign the contract or sublet any portion of the same. In case if it becomes necessary such subletting with the permission of the Engineer-in-Charge shall be limited to (1) Labour contract (2) Material contract (3) Transport contract (4) Engaging specialists for special items of work enjoined in A.P.S.S.

IF MY/OUR tender is not accepted the sum shall be returned to me/us on application when intimation is sent to me/us of rejection or at the expiration of three months from last date of receipt of this tender, whichever is earlier. If my/our tender is accepted the earnest money shall be retained by the Department as security for the due fulfillment of this contract. If upon written intimation to me/us by the Chief Engineer Office, I/We fail to attend the said office on the date herein fixed or if upon intimation being given to me/us by the Chief Engineer or acceptance of my/our tender, and if I/We fail to make the additional security deposit or to enter into the required agreement as defined in condition- 14.2 of the tender conditions, then I/We agree the forfeiture of the earnest money. Any notice required to be served on me/us here under shall be sufficiently served on me/us if delivered to me/us personally or forwarded to me/us by post to (registered or ordinary) or left at my/our address given herein. Such notice shall if sent by post be deemed to have been served on me/us at the time wherein due course of post it would be delivered at the address to which it is sent.

I/WE fully understand that the written agreement to be entered into between me/us and Department shall be the foundation of the rights of the both the parties and the contract shall not be deemed to be complete until the agreement has first been signed by me/us and then by the proper officer authorised to enter into contract on behalf of Government.

I/WE agree to pay the Transaction fee at 0.04% or as amended from time to time on Estimated contract value of the work through a Demand draft/ online in favour of M/s A.P.T.S. Hyderabad at the time of conclusion of Agreement.

I AM/WE ARE professionally qualified an my/our qualifications are given below:

| Name | Qualified |
|---|---|
| | |

I/WE will employ the following technical staff for supervising the work and will see that one of them is always at site during working hours, personally checking all items of works and pay extra attention to such works as required special attention (eg) Reinforced cement concrete work.

| Name of technical staff proposed to be employed | Qualification. |
|---|---|
| | |



Tenderer

Chief Engineer
APSS, Amaravati.

I / WE declare that I/WE agree to recover the salaries of the technical staff actually engaged on the work by the department, from the work bills, if I/We fail to employ technical staff as per the tender condition.

**TENDERERS / CONTRACTOR'S CERTIFICATE.**

(1) I/WE hereby declare that I/We have perused in detail and examined closely the Andhra Pradesh Standard Specifications, all clauses of the preliminary specifications with all amendments and have either examined all the standards specifications or will examine all the standard specifications for items for which I/We tender, before I/We submit such tender and agree to be bound and comply with all such specifications for this agreement which I/We execute in the Department.

(2) I/WE certify that I/We have inspected the site of the work before quoting my Percentage excess or less on ECV, I /We have satisfied about the quality, availability and transport facilities for all the materials.

(3) I/WE am/are prepared to furnish detailed data in support of all my quoted rates, if and when called upon to do so without any reservations.

(4) I/WE hereby declare that I/We will pay an additional security deposit in terms of conditions, the difference between **85% of ECV** and my/our tender amount, in case if my / our offer is less than **15%** as per clause 3.6

(5) Tenders with an excess of above the estimated contract value shall summarily be rejected as per G.O.Ms.No.230 T.R&B (RI) Department, dt.13.8.2007

(6) I/WE hereby declare that I am/We are accepting for the defect liability period as 24 months instead of 6 months under clause 28 of APSS.

(7) a) I/WE declare that I/WE will procure the required construction materials including earth and use for the work after approval of the Engineer-in-Charge. The responsibility for arranging and obtaining the land for borrowing or exploitation in any other way shall rest with me/us for the materials for construction, I/WE shall ensure smooth and un-interrupted supply of materials.

b) I/WE declare that the responsibility for arranging and obtaining the land for disposal of spoil/soil not useful for construction purposes shall rest with me/us.

c) I/WE declare that I/WE shall not claim any compensation or any payment for the land so arranged for disposal of soil and the land for borrow area. My/our quoted percentage excess or less on ECV., are inclusive of the land so arranged and I/We will hand over the land so arranged for disposal of soil to the Department after completion of work.

d) I/WE declare that I/WE will not claim any extra amount towards any material used for the work other than the quoted work for respective schedule 'A' items.

(8) I/WE declare that I/WE will execute the work as per the mile stone programme, and if I/WE fail to complete the work as per the mile stone programme I abide by the condition to recover liquidated damages as per the tender conditions.

(9) I/WE declare that I/WE will abide for settlement of disputes as per the tender conditions.



Tenderer

Chief Engineer
APSS, Amaravati.

## DECLARATION OF THE TENDERER.

1) I/WE have not been black listed in any department / Department of State / Central Govt due to any reasons.

2) I/WE have not been demoted to the next lower category for not filing the tenders after buying the tender schedules in a whole year and my/our registration has not been cancelled for a similar default in two consecutive years.

3) I/WE agree to disqualify me/us for any wrong declaration in respect of the above and to summarily reject my/our tender.

Address of the Tenderer :

Phone No.:

Fax No.:

**Signature of the tenderer.**

**Note: If the tender is made by an individual, it shall be signed with his full name and his address shall be given. If it is made by a firm, it shall be signed with the co-partnership name by a member of the firm, who shall also sign his own name, and the name and address of each member of the firm shall be given, if the tender is made by a Corporation it shall be signed by a duly authorised officer who shall produce with his tender satisfactory evidence of his authorisation. Such tendering Corporation may be required before the contract is executed, to furnish evidence of its corporate existence. Tenders signed on behalf of G.P.A. holder will be rejected**



Tenderer

Chief Engineer
APSS, Amaravati.

## CONDITIONS OF CONTRACT

### A. GENERAL

**1. Interpretation:**

1.1 In interpreting these Conditions of Contract, singular also means plural, male also means female, and vice-versa. Headings have no significance. Works have their normal meaning under the language of the contract unless specifically defined. The Engineers-in-charge will provide instructions clarifying queries about the conditions of Contract.

1.2 The documents forming the Contract shall be interpreted in the following order of priority:

1) Agreement
2) Letter of Acceptance, notice to proceed with the works
3) Contractor's Tender (Technical bid)
4) Conditions of contract
5) Specifications
6) Drawings
7) Bill of quantities (Price-bid)
8) Any other document listed as forming part of the Contract.

**2. Engineer-in-Charge's Decisions:**

Except where otherwise specifically stated, the Engineer-in-charge will decide the contractual matters between the Department and the Contractor in the role representing the Department.

**3. Delegation:**

The Engineer-in-charge may delegate any of his duties and responsibilities to other officers and may cancel any delegation by an official order issued.

**4. Communications:**

Communications between parties, which are referred to in the conditions, are effective only when in writing. A notice shall be effective only when it is delivered (in terms of Indian Contract Act)

**5. Other Contractors:**

The Contractor shall cooperate and share the Site with other contractors, Public authorities, utilities, and the Department. The Contractor shall also provide facilities and services for them as directed by the Engineer-in-charge.

**6. Personnel:**

6.1 The Contractor shall employ the required Key Personnel named in the Schedule of Key Personnel to carry out the functions stated in the Schedule or other personnel approved by the Engineer-in-charge. The Engineer-in-charge will approve any proposed replacement of Key Personnel only if their qualifications, abilities, and relevant experience are substantially equal to or better than those of the personnel listed in the Schedule.

**6.2 Schedule of Key Personnel:**

The successful tenderer shall have to employ the following technical staff on full time basis to be available at site.

| Cost of work (Technical sanction amount) | Qualification of Technical Staff |
|---|---|
| 1 | 2 |
| From Rs.50,000/- to Rs.1 Lakh | One I.T.I. candidate |
| Above Rs.1.00 lakh up to Rs.15.00 lakhs | One diploma holder |
| Above Rs.15.00 lakh up to Rs. 50.00 Lakhs. | One Graduate Engineer |
| Above Rs.35.00 lakh upto Rs.100.00 lakhs | One Graduate Engineer and One Diploma holder |
| Above Rs. 100.00 Lakhs upto Rs.1000.00 lakhs | Two Graduate Engineers |
| Above Rs. 1000.00 Lakhs and above | Four Graduate Engineers |

6.3 Employment of technical personnel shall be with reference to the estimate cost of work put to tender.

6.4 In case the contractor himself is a Diploma holder/ Graduate Engineer no agent need to be appointed to supervise works costing upto Rs.5.00 lakhs. Even if the contractor is himself a technically qualified person, he shall employ technical staff on the scale prescribed for supervising works when more than one work is undertaken and if they are beyond a radius of 5 Kms. One or more works within a radius of say 5 Km. shall be treated as a single work for the purpose of employment of technical staff.

6.5 The appointment of technical staff shall be on full time basis and they shall be available at work site for supervising the work including quality checking of all items from time to time. Failure to employ the required technical personnel by the contractor the following amounts will be recovered from the contractor.

| **Cost of work** | **Rate of penalty** |
|---|---|
| From Rs. 50,000/- to Rs.1,00,000/- | Rs.4350 per month |
| Above Rs.1.00 lakh and up to Rs.15.00 lakhs | Rs.4770 per month |
| Above Rs.15.00 lakh and up to Rs.35.00 lakhs | Rs.7589 per month |
| Above Rs. 50.00 Lakhs. and up to Rs.100.00lakhs | Rs.12,090 per month |
| Above Rs. 100.00 lakhs and upto Rs.1000 lakhs | Rs.14,640 per month |
| Above Rs. 1000.00 lakhs and above | Rs.29,280 per month |

6.6 The Engineer-in-charge is the sole judge (a) to decide whether qualified technical staff is actually supervising the work and (b) to decide the actual period of absence of such staff which requires the above recovery to be enforced and his decision is final and binding on the contractor.

6.7 The technical agents appointed by the contractor shall have to maintain properly all the records required by the department under safe custody at site, like checklists, calibration registers/records, Quality Test Registers, Test reports file, site order

book, etc. and make signatures at appropriate places towards proof of verifications, conduction of tests, compliance to instructions etc.

6.8 The technical personnel should be on full time and available at site whenever required by Engineer in Charge to take instructions.

6.9 The names of the technical personnel to be employed by the contractor should be furnished in the statement enclosed separately.

6.10 In case the contractor is already having more than one work on hand and has undertaken more than one work at the same time, he should employ separate technical personnel on each work.

6.11 If the contractor fails to employ technical personnel the work will be suspended or department will engage a technical personnel and recover the cost thereof from the contractor

6.12 If the Engineer-in-charge asks the Contractor to remove a person who is a member of Contractor's staff or his work force stating the reasons the Contractor shall ensure that the person leaves the site forthwith and has no further connection with the work in the contract

**7 Contractor's Risks:**

All risks of loss of or damage to physical property and of personnel injury and death, which arise during and in consequence of the performance of the Contract are the responsibility of the Contractor.

**8. Insurance: Deleted**

**9.** The Contractor must submit a copy of Registration with Provident Fund authority before entering into agreement.

**10. Site Inspections:**

10.1 The contractor should inspect the site and also proposed quarries of choice for materials source of water and quote his percentage including quarrying, conveyance and all other charges etc.

10.2 The responsibility for arranging the land for borrow area rests with the Contractor and no separate payment will be made for procurement or otherwise. The contractor's quoted percentage will be inclusive of land cost.

**11. Contractor to Construct the Works:**

The Contractor shall construct and Commission the Work in accordance with the specifications and Drawings.

**12. Diversion of streams/ Vagus/ Drains.**

12.1 The contractor shall at all times carry out construction of cross drainage works in a manner creating least interference to the natural flow of water while consistent with the satisfactory execution of work. A temporary diversion shall be formed by the contractor at his cost where necessary. No extra payment shall be made for this work.

12.2 No separate payment for bailing out sub-soils, water drainage or locked up rain water for diversion, shoring, foundations, bailing of pumping water either from excavation of soils from foundations or such other incidental will be paid. The percentage to be quoted by the contractor are for the finished item of work in situ and including all the incidental charges. The borrow pits are also to be de-watered by the contractor himself at his expense, if that should be found necessary.

12.3 The work of diversion arrangements should be carefully planned and prepared by the contractor and forwarded to the Executive Engineer technically substantiating the proposals and approval of the Executive Engineer obtained for execution.

12.4 The contractor has to arrange for bailing out water, protection to the work in progress and the portion of works already completed and safety measures for men and materials and all necessary arrangements to complete the work.

12.5 All the arrangements so required should be carried out and maintained at the cost of the contractor and no separate or additional payments is admissible..

12.6 **Coffer Dams.**

Necessary Coffer Dams and ring bunds have to be constructed at the cost of contractor and same are to be removed after the completion of the work. The contractor has to quote his percentage keeping the above in view.

**13. Power Supply.**

13.1 The contractor shall make his own arrangements for obtaining power from the Electricity dept., at his own cost. The contractor will pay the bills of Electricity Department for the cost of power consumed by him.

13.2 The contractor shall satisfy all the conditions and rules required as per Indian Electricity Act 1910 and under Rule-45(I) of the Indian Electricity Rules, 1956 as amended from time to time and other pertinent rules.

13.3 The power shall be used for bonafide Department work only.

13.4 The contractor shall at all times during the currency of the contract, comply fully with all existing Acts, regulations and bylaws including all statutory amendments and reenactment's of state or central govt., and other local authorities and any other enactment's, notification and acts that may be passed in future either by the state or the central government or local authority including Indian workmen's compensation Act- 1923, Control labour (Regulation and Abolition) Act- 1970, The child labour prohibition and regulation Act-1986 and equal remuneration Act- 1976, Factories Act, minimum wage Act- 1948, provident fund regulations, Employees provident fund Act- 1952 schedules made under the same Act. The buildings and other construction workers (Regulation of employment and condition of service ) Act- 1996, The Cess Act- 1996 and also applicable labour regulations, health and sanitary arrangement for workmen, insurance and other benefit and shall keep department indemnified in case any action is commenced by competent authorities for contravention by the contractor.

13.5 The electrical contractor has to keep his licence in currency till the work is completed. If the licence is suspended during the period in which the work is in progress the contract will be terminated and awarded to some other agency recovering the extra cost if any.

13.6 The materials used in the work should be as per the list of materials enclosed. The department reserves the right to insist upon using any of the materials from this list of approved materials.

13.7 The work shall be carried out strictly in conformity with (i) code of practice for Electrical wiring and fittings in Government Buildings, (ii) The Indian standard specification (iii) The Department specification. If the work carried out does not comply with the code of practice and the Departmental specifications and if the workmanship is unsatisfactory it will be binding on the contractor to redo the job without any extra cost and pay penalty as decided by the Department towards inconvenience caused if any.

13.8 The work should be carried out under the direct supervision of persons holding a certificate of competency for the type of work involved.

13.9 After completion of work a plan of building installation should be prepared and furnished indicating the location of various main and sub boards and also the fittings together with a circuit diagram duly numbered (in the diagram). The final bill will not be paid till the above plan and the diagram is submitted and approved after verification. Such completion drawings shall be signed by the licenced electrical contractor through whom the work is executed.

13.10 Lugs should be provided for all earth connections.

13.11 The contractor himself should arrange for the transportation of men and materials to the work spot.

13.12 Concreting to the pole and providing independent earthing should be done in presence of Departmental staff.

13.13 On completion of the Electrical Installation a certificate shall be furnished by the Contractor countersigned by a licensed supervisor, that under direct supervision the installation was carried out. This certificate shall be in the prescribed from as required by the local supply authority. The contractor shall be responsible for getting the Electrical Installation inspected and approved by the local authority concerned.

13.14 The contractor shall pay for any inspection fees and for permits required for the installation of the work wherever necessary. The Samagra Shiksha shall arrange only for payment of service connection charges and any other security deposit for getting electrical supply. On completion of the work, the contractor shall obtain and deliver to the Samagra Shiksha, certificates of final inspection and approval by the local Electric Authority as may require. The Samagra Shiksha shall have full powers to require the materials or work to be tested by an independent agency at the Electrical contractor's expense in order to prove their soundness and adequacy.

13.15 Contractor shall provide everything necessary for the proper execution of works according to the intent and meaning of the drawings, specifications, schedule of quantities. Any discrepancy in the documents shall be brought to the notice of the Samagra Shiksha and got clarified prior to taking up the installation.

13.16 Materials and Workmanship:

All materials and workmanship shall confirm to the specifications, relevant IS standards and code of practice and comply with APSEB/CEIG requirements as the case may be. Any work that is not up to the standards shall be dismantled and reconstructed by the contractor to the satisfaction of the Samagra Shiksha.

13.17 Liaison Work:

The Contractor shall be responsible for all liaison work with Electricity Board/CEIG

13.18 "The Civil Contractor shall get all the electrical works done only through the licensed electrical contractors/ permit holders as registered with or licensed by A.P.S. Electrical licensing Board. A copy of such valid license/ permit should be furnished by Civil Contractor, along with the tender, for each work separately. The commencement and completion certificate and drawings for each work issued by the electrical licensed Contractor/ Permit holder should be furnished by Civil Contractor, for releasing payment of electrical work".

**Ramps:**

Ramps required during execution may be formed wherever necessary and same are to be removed after completion of the work. No separate payment will be made for this purpose.

**14. Monsoon Damages:**

Damages due to rain or flood either in cutting or in banks shall have to be made good by the contractor till the work is handed over to the Department. The responsibility of de-silting and making good the damages due to rain or flood rests with the contractor. No extra payment is payable for such operations and the contractor shall therefore, have to take all necessary precautions to protect the work done during the construction period.

**15. The works to be Completed by the Intended Completion Date:**

The Contractor may commence execution of the Works on the Start Date and shall carry out the Works in accordance with the programme submitted by the Contractor, as updated with the approval of the Engineer-in-Charge, and complete the work by the Intended Completion Date.

**16. Safety:**

The Contractor shall be responsible for the safety of all activities on the Site.

**17. Discoveries:**

Anything of historical or other interest or of significant value unexpectedly discovered on the Site is the property of the Government. The Contractor is to notify the Engineer-in-charge of such discoveries and carry out the Engineer-in-Charge's instructions for dealing with them.

**18. Possession of the Site.**

The Department shall give possession of the site to the Contractor. If possession of a part site is given, the Department will ensure that the part site so handed over is amenable to carryout the work at site by the Contractor.

**19. Access to the Site:**

The Contractor shall provide the Engineer-in-Charge and any person authorised by the Engineer-in-Charge, access to the site and to any place where the work, in connection with the Contract, is being carried out or is intended to be carried out.

**20. Instructions:**

The Contractor shall carry out all instructions of the Engineer-in-charge and comply with all the applicable local laws where the Site is located.

**21. Settlement of disputes:**

21.1 If any dispute or difference of any kind whatsoever arises between the Department and the Contractor in connection with, or arising out of the Contract, whether during the progress of the works or after their completion and whether before or after the termination, abandonment or breach of the Contract, it shall in the first place, be referred to and settled by the Engineer-in-charge who shall, within a period of thirty days after being requested by the Contractor to do so, give written notice of his decision to the Contractor. Upon receipt of the written notice of the decision of the Engineer-in-Charge the Contractor shall promptly proceed without delay to comply with such notice of decision.

21.2 If the Engineer-in-Charge fails to give notice of his decision in writing within a period of thirty days after being requested or if the Contractor is dissatisfied with the notice of the decision of the Engineer-in-Charge, the Contractor may within thirty days after receiving the notice of decision appeal to the Department which

shall offer an opportunity to the contractor to be heard and to offer evidence in support of his appeal, the Department shall give notice of his decision within a period of thirty days after the Contractor has given the said evidence in support of his appeal, subject to arbitration, as hereinafter provided. Such decision of the Department in respect of every matter so referred shall be final and binding upon the Contractor and shall forthwith be given effect to by the Contractor, who shall proceed with the execution of the works with all due diligence whether he requires arbitration as hereinafter provided, or not. If the Department has given written notice of his decision to the Contractor and no claim to arbitration has been communicated to him by the Contractor within a period of thirty days from receipt of such notice, the said decision shall remain final and binding upon the Contractor. If the Department fail to give notice of his decision, as aforesaid within a period of thirty days after being requested as aforesaid, or if the Contractor be dissatisfied with any such decision, then and in any such case the contractor within thirty days after the expiration of the first named period of thirty days as the case may be, require that the matter or matters in dispute be referred to arbitration as detailed below:-

**SETTLEMENT OF CLAIMS:**

Except as otherwise provided in the contract, any disputes and differences arising out of or relating to the contract shall be referred to adjudication as follows.

**ARBITRATION FOR SETTLEMENT OF ALL CLAIMS**

The procedure for arbitration will be as follows:

(a) In case of Dispute or difference arising between the Department and a contractor relating to any matter arising out of or connected with this agreement, such disputes or difference shall be settled in accordance with the Arbitration and Conciliation Act, 1996. The arbitral tribunal shall consist of 3 arbitrators one each to be appointed by the Chief Engineer and the Contractor. The third Arbitrator shall be chosen by the two Arbitrators so appointed by the Parties and shall act as presiding arbitrator. In case of failure of the two arbitrators appointed by the parties to reach upon a consensus within a period of 30 days from the appointment of the arbitrator appointed subsequently, the Presiding Arbitrator shall be appointed by the President of the Institution of Engineers (India).

(b) If one of the parties fails to appoint its arbitrator in pursuance of sub-clause (a) above within 30 days after receipt of the notice of the appointment of its arbitrator by the other party then the President of the Institution of Engineers (India) both in cases of the Foreign Contractor as well as Indian Contractor, shall appoint the arbitrator. A certified copy of the order of the President of the Institution of Engineers (India) making such appointment shall be furnished to each of the parties.

(c) Arbitration proceedings shall be held at Hyderabad, India, and the language of the arbitration proceedings and that of all documents and communications between the parties shall be English.

(d) The decision of the majority of arbitrators shall be final and binding upon both parties. The cost and expenses of Arbitration proceedings will be paid as determined by the arbitral tribunal. However, the expenses incurred by each party in connection with the preparation, presentation, etc. of its proceedings as also the fees and expenses paid to the arbitrator appointed by such party or on its behalf shall be borne by each party itself.

(e) In the event of value of the contract is Rs.50 millions, the disputes or differences arising shall be referred to the Sole Arbitrator. The Sole Arbitrator

should be appointed by agreement between the parties; failing such agreement, by the appointing authority, namely the President of the Institution of Engineers (India).

(g) Performance under the contract shall continue during the arbitration proceedings and payments due to the contractor shall not be withheld, unless they are the subject matter of the arbitration proceedings.

**Note :** Claims means all claims in that contract.

## B. TIME FOR COMPLETION

**22. Program:**

22.1 The total period of completion is **(as specified in the NIT) 03 Months** from the date of entering with agreement to proceed including rainy season. Keeping in view, the schedule for handing over of site given in condition 23.6 below, the work should be programmed such as to achieve the mile-stones as in "Rate of progress statement" enclosed.

22.2 The attention of the tenderer is directed to the contract requirement at the time of beginning of the work, the rate of progress and the dates for the whole work and its several parts as per milestones. Time is the essence of the contract. The following rate of progress and proportionate value of work done from time to time as will be indicated by the Engineer-in-charge's Certificate for the value of work done and completion of mile-stones will be required.

22.3 The following rate of progress will be required to be maintained by the contractor as a minimum. The start date of this work is the date of signing the agreement but not the date of handing over the site. Contractor may give a separate time schedule for the completion of the whole work and the consideration will be given for accelerated programme. It is imperative that the work progress shall be ahead of the rate of progress given below.

23.4 Detailed programme in terms of collection of necessary materials and labour and in terms of finished items of work, to confirmation of the above rate of progress shall be prepared by the contractor and got approved by the Engineer-in-charge concerned and which shall be strictly adhered to. This programme of work shall be given based on PERT /CPM charts for works, where ECV exceeds Rs.100 lakhs, in the approved format.

23.5 After signing the agreement, the contractor shall forthwith begin the work, shall regularly and continuously proceed with them. Work programme of achieving of milestones (statement) should be submitted from time to time.

23.6 The contractor shall commence the works on site as specified under condition 23.1 to 23.3 above after the receipt by him of a written order to this effect from the Chief Engineer and shall proceed with the same with due expedition and without delay, except as may be expressly sanctioned or ordered by the Chief Engineer or his nominee, or be wholly beyond the contractor's control.

23.7 Save in so far as the contractor may prescribe, the extent of portions of the site of which the contractor is to be given possession from time to time and the order in which such portions shall be made available to him and, Subject to any requirement in the contract as to the order in which the works shall be executed, the Chief Engineer or his nominee will, with the Engineer-in-charge's written order to commence the works, give to the contractor possession of so much of the site as may be required to enable the contractor to commence proceed with the execution of the works in accordance with the programme if any, and otherwise in accordance with such reasonable proposals of the contractor as he shall by written

notice to the Chief Engineer or his nominee, may and will from time to time as the works proceed, give to the contractor possession of such further portions of the site as may be required to enable the contractor to proceed with the execution of the works with due dispatch in accordance with the said programme or proposals as the case maybe; if the contractor suffers delay or incurs cost from failure on the part of the Chief Engineer or his nominee to give possession in accordance with the terms of this clause, the Chief Engineer or his nominee shall grant an extension of time for the completion of works.

23.8 The contractor shall bear all costs and charges for special or temporary way leases required by him in connection with access to the site. The contractor shall also provide at his own cost any additional accommodation outside the site required by him for the purposes of the work.

Subject to any requirement in the contract as to completion of any section of the works before completion of the whole of the works shall be completed, in accordance with provisions of clauses in the Schedule within the time stated in the contract calculated from the last day of the period named in the statement to the tender as that within which the works are to be commenced or such extended time as may be allowed.

23.9 **Delays and extension of time:**

No claim for compensation on account of delays or hindrances to the work from any cause whatever shall lie, except as hereafter defined. Reasonable extension of time will be allowed by the Engineer-in-charge or by the office competent to sanction the extension, for unavoidable delays, such as may result from causes, which in the opinion of the Engineer-in-charge, are undoubtedly beyond the control of the contractor. The Engineer-in-charge shall assess the period of delay or hindrance caused by any written instructions issued by him, at twenty five per cent in excess or the actual working period so lost.

In the event of the Engineer-in-charge failing to issue necessary instructions and thereby causing delay and hindrance to the contractor, the latter shall have the right to claim an assessment of such delay by the Chief Engineer whose decision will be final and binding. The contractor shall lodge in writing with the Engineer-in-charge, a statement of claim for any delay or hindrance referred to above, within fourteen days from its commencement, otherwise no extension of time will be allowed.

Whenever authorised alterations or additions made during the progress of the work are of such a nature in the opinion of the Engineer-in-charge as to justify an extension of time in consequence thereof, such extension will be granted in writing by the Engineer-in-charge or other competent authority when ordering such alterations or additions.

**24 Construction Programme:**

24.1 The Contractor shall furnish within one month of the order of the work a programme showing the sequence in which he proposed to carry out the work, monthly progress expected to be achieved, also indicating date of procurement of materials plant and machinery. The schedule should be such that it is practicable to achieve completion of the whole work within the time limit fixed and in keeping with the Mile stone programme specified and shall obtain the approval of the Engineer-in-charge. Further rate of the progress as in the program shall be kept up to date. In case it is subsequently found necessary to alter this program, the contractor shall submit sufficiently in advance the revised program incorporating necessary modifications and get the same approved by the Engineer-in-charge. No revised program shall be operative with out approval of Engineer-in-charge.

24.2 The Chief Engineer shall have all times the right, without any way violating this contract, or forming grounds for any claim, to alter the order of progress of the works or any part thereof and the contractor shall after receiving such directions proceed in the order directed. The contractor shall also report the progress to the Chief Engineer within 7 days of the Engineer-in-charge's direction to alter the order of progress of works.

24.3 The Contractor shall give written notice to the Engineer-in-Charge whenever planning or progress of the works is likely to be delayed or disrupted unless any further drawings or order including a direction, instruction or approval is issued by the Engineer-in-Charge within a reasonable time. The notice shall include details of the drawing or order required and of why and by when it is required and of any delay or disruption likely to be suffered if it is late.

**25 Speed of Work:**

25.1 The Contractor shall at all times maintain the progress of work to conform to the latest operative progress schedule approved by the Engineer-in-Charge. The contractor should furnish progress report indicating the programme and progress once in a month. The Engineer-in-Charge may at any time in writing direct the contractor to slow down any part or whole of the work for any reason (which shall not be questioned) whatsoever, and the contractor shall comply with such orders of the Engineer-in-Charge. The compliance of such orders shall not entitle the contractor to any claim of compensation. Such orders of the Engineer-in-Charge for slowing down the work will however be duly taken into account while granting extension of time if asked by the contractor for which no extra payment will be entertained.

25.2 Delays in Commencement or progress or neglect of work and forfeiture of earnest money, Security deposit and withheld amounts:

If, at any time, the Engineer-in-Charge shall be of the opinion that the Contractor is delaying Commencement of the work or violating any of the provisions, the Contractor is neglecting or delaying the progress of the work as defined by the. "Rate of progress" in the Articles of Agreement, he shall so advise the Contractors in writing and at the same time demand compliance in accordance with conditions of Tender notice. If the Contractor neglects to comply with such demand within seven days after receipt of such notice, it shall then or at any time there after, be lawful for the Engineer-in-Charge to take suitable action in accordance with Clause.60 of APSS.

**26 Suspension of works by the Contractor:**

26.1 If the Contractor shall suspend the works, or sublet the work without sanction of the Engineer-in-Charge, or in the opinion of the Engineer-in-Charge shall neglect or fail to proceed with due diligence in the performance of his part of the Contract as laid down in the Schedule rate of progress, or if he shall continue to default or repeat such default in the respects mentioned in clause 27 of the APSS, the Engineer-in-Charge shall take action in accordance with Clause 61 of APSS.

26.2 If the Contractor stops work for 28 days and the Stoppage has not been authorised by the Engineer-in-Charge the Contract will be terminated under Clause 61 of APSS.

26.3 If the Contractor has delayed the completion of works the Contract will be Terminated under Clause.61 of APSS.

**27 Extension of the Intended Completion Date:**

27.1 The Engineer-in-Charge shall extend or recommend for extension, in accordance with the delegation of powers in force, the Intended Completion Date if a Variation

is issued which makes it impossible for Completion to be achieved by the Intended Completion Date.

27.2 The Engineer-in-Charge shall decide whether and by how much to extend the Intended Completion Date within 21 days of the Contractor asking the Engineer for a decision upon the effect of a Variation and submitting full supporting information. If the Contractor has failed to give early warning of a delay or has failed to cooperate in dealing with a delay, the delay by this failure shall not be considered in assessing the new Intended Completion Date.

**28 Delays Ordered by the Engineer-in-Charge:**

The Engineer-in-Charge may instruct the Contractor to delay the start or progress of any activity within the Work.

**29 Early Warning:**

29.1 The contractor is to warn the Engineer-in-Charge at the earliest opportunity of specific likely future events or circumstances that may adversely affect the Execution of Works.

29.2 The Contractor shall co-operate with the Engineer-in-Charge in making and considering proposals for how the effect of such an event or circumstance can be avoided or reduced by anyone involved in the work and in carrying out any resulting instruction of the Engineer-in-Charge.

**30 Management Meetings:**

The Engineer-in-Charge may require the Contractor to attend a management meeting. The business of a management meeting shall be to review the programme for remaining work and to deal with matters raised in accordance with the early warning procedure.

**31 Quality Control**

31 A) Quality Control

i) Establishment of Quality Control Laboratory: The contractor shall establish a quality control laboratory, at the site of work, equipped with calibrated equipment (as per list given below) to perform field tests, batch wise, for various materials, then and there itself, as per quality plan and standards.

ii) Calibration of Equipment: All the equipment maintained by the contractor at site shall be calibrated from time to time according to the calibration frequency mentioned hereunder, with calibrations traceable to National Standards. Records for proof of such calibrations done for each instrument, with instrument number shall be maintained by the contractor and shall be made available for verification / counter signature by the Engineer-in-charge. Proper storage, handling and use of these instruments shall be ensured so that their calibration does not get disturbed due to weather factors etc., Frequency of the calibration shall be as decided by the Engineer-in-charge.

iii) List of equipment which should be made available at site by the contractor for testing of materials and` cubes etc., is given below **(Mandatory)**:

| Sl. No. | Description of Item | Recommended Calibration Frequency |
|---|---|---|
| 1. | Vernier Calipers 0-150 mm | 1 year |
| 2. | Screw gauge 0-25 mm | 1 year |

| | | |
|---|---|---|
| 3. a<br><br>3. b | Measurement tapes both Steel (3.0 m, 5.0 m) & Fibre (15.0 m)<br>30 cm steel scale | At the time of purchase and the tapes to be changed after 6 months, if any error is observed |
| 4. | Weighing Machine 5.0 kg capacity | 6 months |
| 5. | Sieves for Coarse aggregate | - |
| 6. | Sieves for fine aggregate | - |
| 7. | Cube moulds ISI marked 150 x150 x 150 mm (6 nos.) | - |
| 8. | Compression testing machine 100 MT, hand operated at every work site whose Estimated contract value (ECV) is Rs. 100 lakhs and above. For less than Rs. 100 lakhs, arrangement for testing at approved lab. | 6 months |
| 9. | Slump cone | - |
| 10. | Carpenter's square 150mm with graduations | |
| 11. | Electrical Meggar 1100 V | At the time of purchase |
| 12. | Spirit level | - |
| 13. | Plumbob | - |
| 14. | Measuring Jars (2) 250ml | - |
| 15. | Magnetic compass | - |
| 16. | Batching and mixing plant for concrete mixing (15-20 cum capacity) | - |

iv) Quality plan for Raw Materials: The contractor shall collect various raw materials well in advance before its use and shall get them tested as per the quality plan supplied by the Engineer-in-charge based on Quality Management System of ISO: 9001-2000 He shall ensure that no raw material is used unless it passes all the checks/ tests as per the acceptance criteria given and a record of all checks/ tests/ verifications shall be maintained at site. Tests which are required to be conducted at outside labs shall be done at those labs which have availability of required instruments traceable to national standards and which are approved by the EXECUTIVE ENGINEER. Reports obtained from such labs should indicate the calibration status and traceability to national standards of their equipments for accepting the results.

v) If the Executive Engineer instructs the contractor to carryout a test not specified in the specification to check whether any work has a defect, the contractor shall conduct the test at his own cost.

vi) Quality control inspections: In addition to the normal inspection by the regular in charge of the construction, the work will also be inspected by the staff of Quality Control Wings or by the State or District level Vigilance Cell Unit and any authorized external agency and if any sub-standard work or excess payments are noticed with reference to measurement books etc., during inspection, recovery will be ordered based on their observations and these will be effected by the EXECUTIVE ENGINEER of the execution of the work.

In addition, the Department may engage external agencies for conducting quality audit in which case the following methodology would be adopted:

a) The external agency shall conduct quality control tests as per the standard procedures in the presence of Construction and Quality Control Engineers and the Contractor who is executing the work.

b) The observations of the external agencies on the quality of work should be recorded then and there and signatures of all the concerned obtained as a token of acceptance of the observations.

The recovery for any sub-standard work or excess payments noticed if any on account of the external quality also audit would be made as mentioned in the Para above.

The final bill will be released after a certificate furnished by the EXECUTIVE ENGINEER that the work has been executed as per the standard specifications and then only the final bill can be released.

Records to be maintained at site

vii) Drawings: One copy of the drawings furnished to the contractor shall be kept by the contractor on the site and same shall be available for inspection and use by the Departmental officers.

viii) Variations by way of modification, omissions or additions:

For all modifications omissions from or additions to the drawings and specifications, the EXECUTIVE ENGINEER will issue revised plans, or written instructions, or both and modification, omission or additions shall be made unless authorised and directed by the EXECUTIVE ENGINEER in writing.

ix) The EXECUTIVE ENGINEER shall have the privilege of ordering modifications, omission or additions at any time before the completion of the work and such orders shall not operate to annual those portions of the specifications with which said changes do not conflict.

x) Site Order Book: The site order book shall be maintained by the contractors at the site of the work. As far as possible all orders regarding the work are to be entered in this book. All entries shall be signed and dated by the Departmental Officer who issues such orders and signed by the contractor or by his representative towards compliance. The order book shall not be removed from the work spot except with the written permission of the EXECUTIVE ENGINEER.

xi) Quality Test Register (QTR): The contractor shall maintain the QTR at the site in the format specified and record therein the results of all the tests conducted by him. The relevant reports of the tests conducted shall be maintained in a separate file.

xii) Other Documents: Other prescribed documents like register on calibration of equipments maintained at site and other checklists shall be maintained at site and produced for verification of inspecting officers.

xiii) Return of site documents: All the site records/ documents mentioned therein must be returned to the engineer-in-charge in full shape after the satisfactory completion of work

**31 b) Identifying Defects:**

i) The Engineer-in-Charge shall check the Contractor's work and notify the Contractor of any Defects that are found. Such checking shall not affect the Contractor's responsibilities. The Engineer-in-Charge may instruct the Contractor

to verify the Defect and to uncover and test any work that the Engineer considers may be a Defect.

**32 Tests:**

If the Engineer-in-Charge instructs the Contractor to carry out a test not specified in the Specification to check whether any work has a Defect and the Contractor shall pay for the test and any samples.

**33 Correction of Defects:**

33.1 The Engineer-in-Charge shall give notice to the Contractor of any Defects before the end of the Defects Liability Period, which begins on Completion. The defect liability period shall be extended for as long as defects remain to be corrected by the Contractor.

33.2 Every time notice of a Defect is given, the Contractor shall correct the notified defect within the length of time specified by the Engineer-in-Charge's notice.

**34 Uncorrected Defects:**

34.1 If the contractor has not corrected the defect within the time specified in the Engineer-in-Charge's notice, the Engineer-in-Charge will assess the cost of having the defect corrected and the contractor will pay this amount.

34.2 The Engineer-in-Charge shall introduce O.K. cards and prescribed the formats there of. O.K. cards shall relate to all major components of the work. The contractor/ his authorised representative shall be required to initiate and fill in and present the O.K. card to the construction staff who would check the respective items and send to the quality control staff for final check and clearance/ O.K. Any defects pointed out by the construction supervision staff or by the Quality Control staff shall promptly be attended to by the contractors and the fact of doing so be duly recorded on the back of O.K. card.

34.3 The Engineer-in-Charge may also introduce checklists, which shall be kept in Bound registers by the construction supervision staff. The contractor may be required to fill up these lists in the first instance and shall be subsequently checked by the Construction/ Quality Control engineers.

**35 Deleted**

## C. COST CONTROL

**36 Bill of Quantities:**

36.1 The Bill of Quantities shall contain items for the construction work to be done by the Contractor.

36.2 The Contractor is paid for the quantity of the work done at the estimate rate in the Bill of Quantities for each item plus or minus Tender percentage.

**37 Supplemental items:**

37.1 The contractor is bound to execute all supplemental works that are found essential, incidental and inevitable during execution of main work.

37.2 The payment of rates for such supplemental items of work will be regulated as under;

Supplemental items directly deducible from similar items in the original agreement.

37.2.1 The rates shall be derived by adding to or subtracting from the agreement rate of such similar item the cost of the difference in the quantity of materials labour between the new items and similar items in the agreement worked out with

reference to the Standard Schedule of Rates adopted in the sanctioned estimate with which the tenders are accepted plus or minus over all tender percentage.

37.2.2 (a) Similar items but the rates of which cannot be directly deduced from the original agreement.

(c) Purely new items which do not correspond to any item in the agreement.

37.2.3 The rates of all such items shall be Estimated Rates plus or minus overall Tender premium.

**38 Extra Items:**

38.1 Extra items of work shall not vitiate the contract. The contractor shall be bound to execute extra items of work as directed by the Engineer-in-Charge. The rates for extra items shall be worked out by the Engineer-in-Charge as per the conditions of the Contract and the same are binding on the Contractor.

38.2 The contractor shall before the 15th day of each month, submit in writing to the Engineer-in-charge a statement of extra items if any that they have executed during the preceding month failing which the contractor shall not be entitled to claim any.

38.3 **Entrustment of additional items**:

38.3.1 Where ever additional items not contingent on the main work and outside the scope of original agreement are to be entrusted to the original contractor dispensing with bids and if the value of such items exceeds the limits upto which the officer is empowered to entrust works initially to contractor without calling for tenders, approval of **competent authority** shall be obtained. Entrustment of such items on nomination shall be the estimated rates or agreement rates (-) Tender Percentage whichever is less.

38.3.2 Entrustment of the additional items contingent on the main work will be authorised by the officers upto the monetary limits upto which they themselves are competent to accept items in the original agreement so long as the rates for such items shall be worked out in accordance with the procedure - For all items of work in excess of the quantities shown in the Bill of Quantities of the Tenders, the rate payable for such items shall be estimate rates for the items (+) or (-) over all tender percentage accepted by the competent authority.

38.3.3 Entrustment of either the additional or supplemental items shall be subject to the provisions of the agreement entered into by a Competent Authority after the tender is accepted. The Chief Engineer being the authority next higher to the Chief Engineer, who entered into the agreement approves the rate for the items/ variation in quantity in the current agreement. The items shall not be ordered by an officer on his own responsibility if the revised estimate or deviation statement providing for the same requires the sanction of competent authority.

**Note:** It may be noted that the term Estimate Rate used above means the rate in the sanctioned estimate with which the tenders are accepted, or if no such rates is available in the estimate, the rate derived will be with reference to the Standard Schedule of Rates adopted in the sanctioned estimate with which tenders are accepted.

**39 Cash flow forecasts:**

39.1 When the program is updated, the contractor is to provide the Engineer-in-charge with an updated cash flow forecast.

**40 Payment Certificates:**

40.1 The Contractor shall submit to the Engineer-in-charge monthly statements of the estimated value of the work completed less the cumulative amount certified previously.

40.2 The Engineer-in-charge shall check the Contractor's monthly statement within 14 days.

40.3 The value of work executed shall be determined by the Engineer-in-charge.

40.4 The value of work executed shall comprise the value of the quantities of the items in the Bill of Quantities completed.

40.5 The Engineer-in-charge may exclude any item certified in a previous certificate or reduce the proportion of any item previously certified in any certificate in the light of later information.

**41 Payments:**

41.1 Payment for the work done by the contractor will be made for the finished work based on the measurements recorded in measurement books by concerned officer of the Department not lower in rank than a Assistant Engineer and check measured by concerned officer not lower in rank than a Deputy Executive Engineer. The measurement shall be recorded at various stages of the work done and also after work is completed. The contractor shall be present at the time of recording of each set of measurement and their check measurement and accept them then and there so as to avoid disputes at a later stage. If the contractor is not available at the workspot at the time of recording measurements or check measurements, the particulars of measurements shall be signed by the authorised agent of contractor based on which the contractor shall accept the set of measurements without any further dispute. If for any reason the contractor's authorised agent is also not available at site when the department decides to suspend the work recording of measurements in the absence of the contractor or his authorised representative, the department shall not entertain any claim from the contractor for any loss incurred by him on this account. The Contractor shall however note that the Department cannot indefinitely wait for recording the measurement due to the absence of the Contractor and his authorised agent and check measure them even in the absence of the contractor.

Unless otherwise directed, measurements shall not be taken until sufficient materials for use on work have been collected and stacked. Immediately after measurement, the stack shall be marked by white wash or other means as directed by the Engineer-in-charge.

**41.2 Payments and Certificates:**

41.2.1 Payments shall be adjusted for recovery of advance payments, liquidated damages in terms of tender conditions and security deposit for the due fulfillment of the contract. Payment will be made to the Contractor under the certificate to be issued at reasonably frequent intervals by the Engineer-in-Charge, and intermediate payment will be the sum equal to 92½% of the value of work done as so certified and balance of 7½% will be withheld and retained as security for the due fulfillment of the contract under the certificate to be issued by the Engineer-in-Charge. On completion of the entire works the contractor will receive the final payment of all the moneys due or payable to him under or by virtue of the contract except earnest money deposit retained as security and a sum equal to 2 ½ percent of the total value of the work done. The amount withheld from the final bill will be retained under deposits and paid to the contractor together with the earnest money deposit retained as security after a period of 24 months as all defects shall have been made good according to the true intent and meaning there of.

41.2.2 In case of over payments or wrong payment if any made to the contractor due to wrong interpretation of the provisions of the contract, APSS or Contract conditions etc., such unauthorised payment will be deducted in the subsequent bills or final bill for the work or from the bills under any other contracts with the Government or at any time there after from the deposits available with the Department.

41.2.3 Any recovery or recoveries advised by the Government Department either state or central, due to non-fulfillment of any contract entered into with them by the contractor shall be recovered from any bill or deposits of the contractor.

41.2.4 No claim shall be entertained, if the same is not represented in writing to the Engineer-in-Charge within 15 days of its occurrence.

41.2.5 The contractor is not eligible for any compensation for inevitable delay in handing over the site or for any other reason. In such case, suitable extensions of time will be granted after considering the merits of the case.

41.3 Intermediate Payments:

41.3.1 For intermediate Stage of work, only part rates as fixed by the Engineer-in-Charge will be paid.

41.3.2 Part rates shall be worked out for the work done portion based on the actual operations involved keeping inview the value of the balance work to be done, to avoid unintended benefit to the Contractor in initial Stage.

41.3.3 Full rate shall be paid when the work is completed to the full profile as noted in the drawings.

41.3.4 Where payment is intended for aggregates by Bill of Quantities item based on stack measurements, 10% of the quantity measured will be withheld. No payment or advance will be made for unfixed materials when the rates are for finished work in site

41.3.5 The contractor shall supply as built drawings drawn to scale in 6 sets along with original tracings within 28 days of the issue of certificate of completion of work failing which an amount of **Rs.1,00,000/-** will be withheld from the amounts due to the contractor.

**43 Interest on Money due to the Contractor:**

Any omission by the Engineer-in-Charge or the sub-divisional officer to pay the amount due upon certificates shall vitiate or make void the contract, nor shall the contractor be entitled to interest upon any guarantee fund or payments in arrear, nor upon any balance which may, on the final settlement of his accounts, found to be due to him.

**44 Certificate of Completion of works:**

44.1 Certificate of Completion of works:

44.1.1 When the whole of the work has been completed and has satisfactory passed any final test that may be prescribed by the Contract, the Contractor may give a notice to that effect to the Engineer-in-Charge accompanied by an undertaking to carryout any rectification work during the period of maintenance, such notice and undertaking shall be in writing and shall be deemed to be request by the Contractor for the Engineer-in-Charge to issue a Certificate of completion in respect of the Works. The Engineer-in-Charge shall, within twenty one days of the date of delivery of such notice either issue to the Contractor, a certificate of completion stating the date on which, in his opinion, the works were completed in accordance with the Contract or give instructions in writing to the Contractor specifying all the Works which, in the Engineer-in-Charge'' opinion, required to be done by the Contractor before the issue of such Certificate. The Engineer-in-Charge shall also

notify the Contractor of any defects in the Works affecting completion that may appear after such instructions and before completion of the Works specified there in. The Contractor shall be entitled to receive such Certificate of the Completion within twenty one days of completion to the satisfaction of the Engineer-in-Charge of the Works so specified and making good of any defects so notified.

44.1.2 Similarly, the Contractor may request and the Engineer-in-Charge shall issue a Certificate of Completion in respect of:

a) Any section of the Permanent works in respect of which a separate time for completion is provided in the Contract, and

b) Any substantial part of the Permanent Works which has been both completed to the satisfaction of the Engineer-in-Charge and occupied or used by the Department.

44.1.3 If any part of the Permanent Works shall have been completed and shall have satisfactorily passed any final test that may be prescribed by the Contract, the Engineer-in-Charge may issue such certificate, and the Contractor shall be deemed to have undertaken to complete any outstanding work in that part of the Works during the period of Maintenance.

**45 Taxes included in the bid:**

The percentage quoted by the contractor shall be deemed to be inclusive of the sales Tax and other taxes on all materials that the contractor will have to purchase for performance of this contract.

## Price Adjustment:

45.1 Price adjustment shall be applicable as per G.O. Ms.No. 62, Water Resources (Reforms) department, dated.30.11.2021 **and other related latest instructions/ orders of the government from time to time will be implemented in respect of price variation for all works irrespective of cost and agreement period subject to following conditions**.

i) Price adjustment shall be both for increase and decrease in the prices for the works completed within original agreement period.

ii) Price adjustment scheme will be applied in all cases where the variation between estimated rates and Board of Chief Engineers approved rates (increase or decrease) is more than 5 % for the month.

iii) **Any changes in sand policy by the Government will be applicable from time to time**.

**46 Retention:**

**46.1 The department shall retain from each payment due to the contractor @ the rate of 7.5% of bill amount until completion of the whole of the works.**

**46.2 On completion of the whole of the works 5% of the total amount retained is re-paid to the Contractor and 2½% when the Defects Liability Period has passed and the Engineer-in-Charge has certified that all the Defects notified by the Engineer-in-Charge to the Contractor before the end of this period have been corrected.**

**47 Liquidated Damages:**

47.1 If for any reason, which does not entitle the contractor to an extension of item, the rate of progress of works, or any section is at any time, in the opinion of the Chief Engineer too slow to ensure completion by the prescribed time or extended time for completion Chief Engineer shall so notify the contractor in writing and the contractor shall there upon take such steps as are necessary and the Chief

Engineer may approve to expedite progress so as to complete the works or such section by the prescribed time or extended time. The contractor shall not be entitled to any additional payment for taking such steps. If as a result of any notice given by the Chief Engineer under this clause the contractor shall seek the Chief Engineer's permission to do any work at night or on Sundays, if locally recognised as days of rest, or their locally recognised equivalent, such permission shall not be unreasonably refused.

47.2 If the contractor fails to complete whole of the works or any part thereof or section of the works within the stipulated periods of individual mile stones (including any bonafide extensions allowed by the competent authority without levying liquidated damages), the Chief Engineer may without prejudice to any other method of recovery will deduct for the period of delays subject to a maximum of 10% of the contract value from any monies in his hands due or which may become due to the contractor. The payment or deductions of such damages shall not relieve the contractor from his obligation to complete the works, or from any other of his obligations and liabilities under the contract.

47.3 The liquidated damages for the whole of the work are Rs.100/- per day per lakh of mile stone amount/contract value

The maximum amount of liquidated damages for the whole of the works is ten percent of final contract price.

**48 Mobilisation Advance:**

48.1 The contractors for works costing more than Rs.1.00 crore of estimated cost value are permitted to avail the facility of mobilization advance upto 10% of the value of the contract against an unconditional, irrevocable B.G. Interest will be charged on the advance payment at the rate of 18% per annum and will be recovered from the bills of contractor.

**48.2 Repayment of advance payment for mobilization.**

The advance loan shall be repaid with percentage deductions from the interim payments certified by the Engineer-in-Charge under the contract. Deductions shall commence in the next Interim Payment Certificate following that in which the total of all such payments to the contractor has reached 15 percent of the contract price or 2 months from the date of payment of first installment of advance which ever period concludes earlier and shall be made at the rate of 25 percent of the amounts of each of the following interim cash payments certificates, until such time as the loan has been repaid, always provided that the loan shall be completely repaid prior to the expiry of the 85% of the original time for completion. (pursuant to Clause 18).

On recovery of mobilization advance to that extent, the Contractor is eligible for release of Bank Guarantee referred in clause 49.1

**49 Securities:**

The Earnest Money Deposit and Additional Security (for discount tender percentage beyond 10%) shall be provided to the Department not later than the date specified in the Letter of Acceptance and shall be issued in an amount and form and by a bank acceptable to the Department. The Earnest Money shall be valid until a date 28 days from the date of expiry of Defects Liability Period and the additional security shall be valid until a date 28 days from the date of issue of the certificate of completion.

**50 Cost of Repairs:**

Loss or damage to the Works or materials to the Works between the Start Date and the end of the Defects Correction Periods shall be remedied by the Contractor

at the Contractor's cost if the loss or damage arises from the Contractor's acts or omissions.

## E. FINISHING THE CONTRACT

**51 Completion:**

The Contractor shall request the Engineer-in-Charge to issue a Certificate of completion of the Works and the Engineer-in-Charge will do so upon deciding that the work is completed.

**52 Taking Over:**

The Department shall takes over the Site and the Works within seven days of the Engineer-in-Charge issuing a certificate of Completion.

**53 Final Account:**

The Contractor shall supply to the Engineer-in-Charge a detailed account of the total amount that the Contractor considers payable under the Contract before the end of the Defects Liability Period. The Engineer-in-Charge shall issue a Defects Liability Certificate and certify any final payment that is due to the Contractor within 56 days of receiving the Contractor's account if it is correct and complete. If it is not, the Engineer-in-Charge shall issue within 56 days a schedule that states the scope of the corrections or additions that are necessary. If the final Account is still unsatisfactory after it has been resubmitted, the Engineer-in-Charge shall decide on the amount payable to the Contractor and issue a payment certificate with in 56 days of receiving the Contractor's revised account.

**54 Termination:**

54.1 The Department may terminate the Contract if the contractor causes a fundamental breach of the Contract.

54.2 Fundamental breaches of Contract include, but shall not be limited to the following.

a) The Contractor stops work for 28 days when no stoppage of work is shown on the current program and the stoppage has not been authorised by the Engineer-in-Charge.

b) The Contractor is made bankrupt or goes into liquidation other than for a reconstruction or amalgamation.

c) The Engineer-in-Charge gives Notice that failure to correct a particular Defect is a fundamental breach of Contract and the Contractor fails to correct it within a reasonable period of time determined by the Engineer-in-Charge; and

d) The Contractor does not maintain a security which is required and

e) The Contractor has delayed the completion of works by the number of days for which the maximum amount of liquidated damages can be paid as defined.

f) If the contractor, in the judgement of the Department has engaged in corrupt or fraudulent practices in competing for or in the executing the contract.

g) The Contractor has contravened Sub-Clause 6 of Conditions of Contract and sublet the work.

h) The Contractor does not adhere to the agreed construction program (Clause 23.1, 23.2, 23.3 of Conditions of Contract) and also fails to take satisfactory remedial action as per agreements reached in the management meetings (Clause 31) for a period of 15 days.

i) The Contractor fails to carry out the instructions of Engineer-in-Charge within a reasonable time determined by the Engineer-in-Charge.

For the purpose of this paragraph: “corrupt practice” means the offering, giving, receiving or soliciting of any thing of value to influence the action of a public official in the procurement process or in contract execution. “Fraudulent practice” means a misrepresentation of facts in order to influence a procurement process or the execution of a contract to the detriment to the Government and includes collusive practice among Tenderers (prior to or after Tender submission) designed to establish Tender prices at artificial non-competitive levels and to deprive the Government of the benefits of free and open competition.

54.3 Notwithstanding the above the Department may terminate the contract for convenience.

54.4 If the Contract is terminated, the Contractor shall stop work immediately, make the Site safe and secured, leave the Site as soon as reasonably possible.

**55 Payment upon Termination:**

If the Contract is terminated because of a fundamental breach of Contract by the Contractor, the Engineer-in-Charge shall issue a certificate for the value of the work done less advance payments received upon the date of the issue of the certificate, less other recoveries due in terms of the Contract, less taxes due to be deducted at source as per applicable law and the percentage to apply to the value of the work not completed representing additional cost for completing the works at the rate of 20 percent of balance work. Additional Liquidated Damages shall not apply. If the total amount due to the Department exceeds any payment due to the Contractor the difference shall be a debt payable to the Department. In case of default for payment within 28 days from the date of issue of notice to the above effect, the contractor shall be liable to pay interest at 12% per annum for the period of delay.

**56 Property:**

All materials on the Site, Plant, Equipment, Temporary Works and Works are deemed to be the property of the Department if the Contract is terminated because of Contractor’s default.

**57 Release from Performance:**

If the Contract is frustrated by the outbreak of war or by any other event entirely outside the control of either the Department or the Contractor the Engineer-in-Charge shall certify that the contract has been frustrated. The Contractor shall make the site safe and stop work as quickly as possible after receiving this certificate and shall be paid for all works carried out before receiving it and for any work carried out after wards to which commitment was made.

## F. SPECIAL CONDITIONS

**58 Water Supply:**

The Contractor has to make his own arrangements for water required for the work and to the colonies and work sites, which are to be established by the Contractor.

**59 Land:**

59.1 Land for Contractor’s use:

The contractor will be permitted to use Government land for execution of work. The contractor shall have to make his own arrangements for acquiring and clearing the site, leveling, providing drainage and other facilities for labour staff colonies, site office, work-shop or stores and for related activities. The Contractor shall apply to the Department within a reasonable time after the award of the contract and atleast 30 days in advance of its use, the details of land required by him for the work at site and the land required for his camp and should any private land which

has not been acquired, be required by the contractor for his use. The same may be acquired by the contractor at his own cost by private negotiations and no claim shall be admissible to him on this account.

The Engineer-in-Charge reserves the right to refuse permission for use of any government land for which no claim or compensation shall be admissible to the contractor. The contractor shall, however, not be required to pay cost or any rent for the Government land given to him.

59.2 Surrender of Occupied Land:

a) The Government land as here in before mentioned shall be surrendered to the Engineer-in-Charge within seven days, after issue of completion certificate. Also no land shall be held by the contractor longer than the Engineer-in-Charge shall deem necessary and the contractor shall on the receipt of due notice from the Engineer-in-Charge, vacate and surrender the land which the Engineer-in-Charge may certify as no longer required by the Contractor for the purpose of the work.

b) The contractor shall make good to the satisfaction of the Engineer-in-Charge any damage to areas, which he has to return or to other property or land handed over to him for purpose of this work. Temporary structures may be erected by the contractor for storage sheds, offices, residences etc., for non-commercial use, with the permission of the Engineer-in-charge on the land handed over to him at his own cost. At the completion of the work these structures shall be dismantled site cleared and handed over to the Engineer-in-charge. The land required for providing amenities will be given free of cost from Government lands if available otherwise the contractor shall have to make his own arrangements.

59.3 Contractor not to dispose off Spoil etc. :-

The contractor shall not dispose off or remove except for the purpose of fulfillment of this contract, sand, stone, clay ballast, earth, trees and shrubs or other materials obtained in the excavation made or lying on the site of the work, and all such materials and produce shall remain property of the Government. The Department may upon request from the contractor, or if so stipulated in the conditions of the contract allow the contractor to use any of the above materials for the works either free of cost or after payment as may be specifically mentioned or considered necessary during the execution of the work.

**60 Roads:**

In addition to existing public roads and roads Constructed by Government, if any, in work area all additional approach roads inside work area and camp required by the Contractor shall be constructed and maintained by him at his own cost. The layout design, construction and maintenance etc. of the roads shall be subject to the approval of the Engineer-in-Charge. The contractor shall permit the use of these roads by the Government free of charge.

It is possible that work at, or in the vicinity of the work site will be performed by the Government or by other contractors engaged in work for the Government during the contract period. The contractor shall without charge permit the government and such other contractor and other workmen to use the access facilities including roads and other facilities, constructed and acquired by the contractor for use in the performance of the works.

The contractor's heavy construction traffic or tracked equipment shall not traverse any public roads or bridges unless the contractor has made arrangement with the authority concerned. In case contractor's heavy construction traffic or tracked equipment is not allowed to traverse any public roads or bridges and the contractor is required to make some alternative arrangements, no claim on this account shall be entertained.

The contractor is cautioned to take necessary precautions in transportation of construction materials to avoid accidents.

**61 Payment for Camp Construction:**

No payment will be made to the contractor for construction, operation and maintenance of camp and other camp facilities and the entire cost of such work shall be deemed to have been included in the tendered rate for the various items of work in the schedule of quantities and bids.

**62 Explosive And Fuel Storage Tanks:**

No explosive shall be stored within ½ (half) KM of the limit of the camp sites. The storage of gasoline and other fuel oils or of Butane, Propane and other liquefied petroleum gases, shall confirm to the regulations of Andhra Pradesh State Government and Government of India. The tanks, above ground and having capacity in excess of 2000 litres, shall not be located within the camp area, nor within 200m, of any building.

**63 Labour:**

The contractor shall, make his own arrangements for the engagement of all staff and labour, local or other, and for their payment, housing, feeding and transport.

Labour importation and amenities to labour and contractor's staff shall be to the contractor's account. His quoted percentage shall include the expenditure towards importation of labour amenities to labour and staff;

The contractor shall, if required by the Engineer-in-Charge, deliver to the Engineer-in-Charge a written in detail, in such form and at such intervals as the Engineer-in-Charge may prescribe, showing the staff and the numbers of the several classes of labour from time to time employed by the contractor on the Site and such information pertaining to Contractor's Equipment as the Engineer-in-Charge may require.

**63.1 Transportation of Labour:**

The contractor shall make his own arrangement for the daily transportation of the labour and staff from labour camps colonies to the work spot and no labour or staff of the contractor shall stay at the work spot. No extra payment will be made to the contractor for the above transportation of the labour and his quoted percentage to the work shall include the transportation charges of labour from colonies to work spot and back.

The contractor will at all times duly observe the provisions of employment of children Act XXVI of 1938 and any enactment or modification of the same and will not employ or permit any person to do any work for the purpose under the provisions of this agreement in contravention of said Act. The contractor here by agrees to indemnify the department from and against all claims, penalties which may be suffered by the department or any person employed by the department by any default on the part of the contractor in the observance and performance of the provisions of the employment of children Act. XXVI of 1938 or any enactment or modification of the same.

As per Govt. memo No.721/Gr.(1)/81-35, dt:17.11.87. The contractor shall obtain the insurance at his own cost to cover the risk on the works to labour engaged by him during period of execution against fire and other usual risks and produce the same to the Engineer-in-charge concerned before commencement of work.

**64 Safety Measures:**

1. The contractor shall take necessary precautions for safety of the workers and preserving their health while working in such jobs, which require special

protection and precautions. The following are some of the measures listed but they are not exhaustive and contractor shall add to and augment these precautions on his own initiative where necessary and shall comply with directions issued by the Engineer-in-charge or on his behalf from time to time and at all times.

2. Providing protective foot wear to workers situations like mixing and placing of mortar or concrete, sand in quarries and places where the work is done under much wet conditions.
3. Providing protective head wear to workers at places like under ground excavations to protect them against rock falls.
4. Providing masks to workers at granulates or at other locations where too much fine dust is floating about and sprinkling water at frequent intervals by water hoses on all stone crushing area and storage bins abate to dust.
5. Getting the workers in such jobs periodically examined for chest trouble due to too much breathing in to fine dust.
6. Taking such normal precautions like fencing and lighting in excavation of trenches, not allowing rolls and metal parts of useless timber spread around, marking danger areas for blasting providing whistles etc.
7. Supply work men with proper belts, ropes etc., when working in precarious slopes and heights etc.
8. Avoiding un-insulated electrical wire etc., as they would electrocute the workers.
9. Taking necessary steps towards training the workers concerned on the machinery before they are allowed to handle them independently and taking all necessary precautions in and around the areas where machines hoists and similar units are working.

**65 Fair Wage Clause:**

1. The contractor shall pay not less than fair wages to labourers engaged by him on the work.
2. “Fair” wages means wages whether for time of piecework notified by the Government from time in the area in which the work is situated.
3. The contractor shall not with-standing the revisions of any contract to the contrary cause to be paid to the labour, in directly engaged on the work including any labour engaged by the sub-contractor in connection with the said work, as if the labourers had been directly employed by him.
4. In respect of labour directly or indirectly employed in the works for the purpose of the contractors part of the agreement the contractor shall comply with the rules and regulations on the maintenance of suitable records prescribed for this purpose from time to time by the Government. He shall maintain his accounts and vouchers on the payment of wages to the labourers to the satisfaction of the Engineer-in-charge.
5. The Engineer-in-charge shall have the right to call for such record as required to satisfy himself on the payment of fair wages to the labourers and shall have the right to deduct from the contract amount a suitable amount for making good the loss suffered by the worker or workers by reason of the “fair wages” clause to the workers.

6. The contractor shall be primarily liable for all payments to be made and for the observance of the regulations framed by the Govt., from time to time without prejudice to his right to claim indemnity from his sub-contractors.
7. As per contract labour (Regulation and abolition) Act. 1970 the contractor has to produce the license obtained from the licensing officers of the labour department along with the tender or at the time of agreement.
8. Any violation of the conditions above shall be deemed to be a breach of his contract.
9. Equal wages are to be paid for both men and women if the nature of work is same and similar.
10. The contractor shall arrange for the recruitment of skilled and unskilled labour local and imported to the extent necessary to complete the work within the agreed period as directed by the Engineer-in-charge in writing.

**66 Indemnity Bond:**

The tenderers should submit Indemnity Bond at the time of the Agreement as specified below.

Name of work:…….

I _____contractor S/o._____

aged _____ Resident of _____

do hereby bind myself to pay all the claims may come (a) under Workmen's Compensation Act. 1933 with any statutory modification there of and rules there under or otherwise for or in respect of any damage or compensation payable in connection with any accident or injury sustained (b) under Minimum wages Act 1948 (c) under payment of wages Act.1936 (d) under the Contractor labour (Regulation and Abolition) Act. 1970 by workmen engaged for the performance of the business relating to the above contract ie., Failing such payment of claims of workmen engaged in the above work, I abide in accepting for the recovery of such claims, effected from any of my assets with the Department and with other Government Departments / Departments.

**67 Compliance With Labour Regulations:**

During continuance of the contract, the contractor and his sub contractors shall abide at all times by all existing labour enactments and rules made there under, regulations, notifications and bye laws of the State or Central Government or local authority and any other labour law (including rules), regulations, bye laws that may be passed or notifications that may be issued under any labour law in future either by the State or the Central Government or the local authority and also applicable labour regulations, health and sanitary arrangements for workmen, insurance and other benefits. Salient features of some of the major labour laws that are applicable

to construction industry are given below. The contractor shall keep the Department indemnified in case any action is taken against Department by the competent authority on account of contravention of any of the provisions of any Act or rules made thereunder, regulations or notifications including amendments. If the Department is caused to pay or reimburse, such amounts as may be necessary to cause or observe, or for non-observance of the provision stipulated in the notifications/bye laws/Acts/Rules/regulations including amendments, if any, on the part of the contractor, the Department shall have the right to deduct any money due to the contractor including his amount of performance security. The Department shall also have right to recover from the contractor any sum required or estimated to be required for making good the loss or damage suffered by the Department.

The employees of the Contractor and the Sub-contractor in no case shall be treated as the employees of the Department at any point of time.

**68 Salient features of some major labour laws applicable to establishment engaged in buildings and other construction work:**

(a) Workmen compensation Act 1923: The Act provides for compensation in case if injury by accident arising out of and during the course of employment.

(b) Payment of Gratuity Act 1972: Gratuity is payable to an employee under the Act on satisfaction of certain conditions on separation if any employee has completed 5 years service or more, or on death, the rate of 15 days wages for every completed year of service. The Act is applicable to all establishments, employing 10 or more employees.

(c) Employees P.F. and Miscellaneous provision Act 1952: The Act provides for monthly contributions by the Department plus workers @ 10% or 8.33%. The benefits payable under the Act are:

(i) Pension or family pension on retirement or death, as the case may be.

(ii) Deposit linked insurance on the death in harness of the worker.

(iii) Payment of P.F. accumulation on retirement/death etc.,

(d) Maternity Benefit Act 1951: The Act provides for leave and some other benefits to women employees in case of confinements or miscarriage etc.

(e) Contract Labour (Regulation & Abolition) Act 1970: The Act provides for certain welfare measures to be provided by the contractor to contract labour and in case the Contractor fails to provide, the same are required to be provided by the Principal Department by Law. The Principal Department is required to take certificate of Registration and the contractor is required to take license from the designated Officer before concluding agreement. The Act is applicable to the establishments or Contractor of Principal Department if they employ 20 or more contract labour.

(f) Minimum wages Act 1948: The Department is supposed to pay not less than the Minimum wages fixed by appropriate Government as per provisions of the Act if the employment is a scheduled employment construction of Buildings, Roads, Runways are scheduled employments.

(g) Payment of wages Act 1936: It lays down as to by what date the wages are to be paid, when it will be paid and what deductions can be made form the wages of the workers.

(h) Equal Remuneration Act 1979: The Act provides for payment of equal wages for work of equal nature to Male or Female workers and for not making

discrimination against Female employee in the matters of transfers, training and promotions etc.

(i) Industrial Disputes Act 1947: The Act lays down the machinery and procedure for resolution of Industrial disputes, in what situations a strike or lock- out becomes illegal and what are the requirements for laying off or retrenching the employees or closing down the establishment.

(j) Industrial Employment (Standing Orders) Act 1946: It is applicable to all establishments employing 100 or more workmen (employment size reduced by some of the State and Central Government to 50). The Act provides for laying down rules governing the conditions of employment by the Department on matters provided in the Act and get the same certified by the designated Authority.

(k) Trade Unions Act 1926: The Act lays down the procedure for registration of trade unions of workmen and Departments. The Trade Unions registered under the act have been given certain immunities from civil and criminal liabilities.

(l) Child Labour (Prohibition & Regulation) Act 1986: The Act prohibits employment of children below 14 years of age in certain occupations and processes and provides for regulation of employment of children in all other occupations and processes, Employment Child Labour is prohibited in Building and Construction Industry.

(m) Inter-State Migrant workmen's (Regulation of Employment & Conditions of service) Act 1979. The Act applicable to an establishment, which employs 5 or more inter-state migrant workmen through an intermediary (who has recruited workmen in one state for employment in the establishment situated in another State). The inter State migrant workmen, in an establishment to which this Act becomes applicable, are required to be provided certain facilities such as housing, medical aid, travelling expenses from home upto the establishment and back, etc.

(n) The Building and Other Construction workers (regulation of Employment and conditions of service) Act 1996 and the Cess Act of 1996: All the establishments who carryon any building or other construction work and employs 10 or more workers are covered under this Act. All such establishments are required to pay cess at the rate not exceeding 2% of the cost of construction as may be modified by the Government. The Department of the establishment is required to provide safety measures at the Building or construction work and other welfare measures, such as Canteens, First-aid facilities, Ambulance, Housing accommodations for workers near the work place etc. The Department to whom the Act applies has to obtain a registration certificate from the Registering Officer appointed by the Government.

(o) Factories Act 1948: The Act lays down the procedure for approval of plans before setting up a factory, health and safety provisions, welfare provisions, working hours, annual earned leave and rendering information regarding accidents or dangerous occurrences to designated authorities. It is applicable to premises employing 10 person or more with aid of power or 20 or more persons without the aid of power engaged in manufacturing process.

(p) Payment of bonus act 1965: The Act is applicable to all establishments employing 20 or more employees. The Act provides for payment of annual bonus subject to a minimum of 8.33% of wages and maximum of 20% of wages to employees drawing Rs. 3500/- per month or less. The bonus to be paid to employees getting Rs.2500/- per months or above and upto Rs.3500/- per month shall be worked out by taking wages as Rs.2500/- per monthly only.

The Act does not apply to certain establishments. The newly set-up establishments are exempted for five years in certain circumstances. Some of the State Governments have reduced the employment size from 20 to 10 for the purpose of applicability of this Act.

**69 Liabilities of the Contractor:**

69.1 Accident Relief and workmen compensation:

The contractor should make all necessary arrangements for the safety of workmen on the occurrence of the accident, which results in the injury or death of any of the workmen employed by the contractor, the contractor shall within 24 hours of the happenings of the accident and such accidents should intimate in writing to the concerned Asst. Engineer/ Asst. Executive Engineer of the Department the act of such accident. The contractor shall indemnify Department against all loss or damage sustained by the Department resulting directly or indirectly from his failure to give intimation in the manner aforesaid including the penalties or fines if any payable by Department as a consequence of Department failure to give notice under workmen's compensation Act or otherwise conform to the provisions of the said Act. in regard to such accident.

69.2 In the event of an accident in respect of which compensation may become payable under the workmen's compensation Act VIII 23 whether by the contractor, by the Government it shall be lawful for the Engineer-in-charge to retain such sum of money which may in the opinion of the Engineer-in-charge be sufficient to meet such liability. The opinion of the Engineer-in-charge shall be final in regard to all matters arising under this clause.

69.3 The contractor shall at all times indemnify the Department against all claims which may be made under the workmen's compensation act or any statutory modification thereafter or rules thereunder or otherwise consequent of any damage or compensation payable in consequent of any accident or injuries sustained or death of any workmen engaged in the performance of the business relating to the contractor.

**70 Contractor's Staff, Representatives and Labour:**

(a) The contractor shall, at all times, maintain on the works, staff of qualified Engineers, and Supervisors of sufficient experience of similar other jobs to assure that the quality of work turned out shall be as intended in the specifications. The contractor shall also maintain at the works, a Work Manager or sufficient status, experience and office and duly authorise him to deal with all aspects of the day-today work. All communications to any commitments by the Work Manager shall be considered as binding on the Contractor.

(b) The Contractor shall at all times submit details of skilled and unskilled labour and equipment employed to the Engineer-in-Charge in prescribed proforma as he may require to assess and ensure the proper progress of work.

**71 Accommodation and food:**

The contractor should arrange accommodation he needs, at his own cost. The contractor shall make his own arrangements for supply of food grains, fuel and other provision to his staff and labourers including controlled commodities.

**72 Relationship :**

Contractor shall have to furnish information along with tender, about the relationship he is having with any officer of the Department.

**73 Protection of adjoining premises:**

The contractor shall protect adjoining sites against structural, decorative and other damages that could be caused by the execution of these works and make good at his cost any such damages.

**74 Work during night or on Sundays and holidays:**

The works can be allowed to be carried out during night, Sundays or authorised holidays in order to enable him to meet the schedule targets and the work shall require almost round the clock working keeping inview:

(i) The provisions of relevant labour laws being adhered to:

(ii) Adequate lighting, supervision and safety measures are established to the satisfaction of the Engineer-in-Charge and

(iii) The construction programme given by the Contractor and agreed upon by the Engineer-in-Charge envisages such night working or working during Sundays or authorised holidays.

**75 Layout of materials stacks:**

The contractor shall deposit materials for the purpose of the work on such parts only of the ground as may be approved by the Engineer-in-Charge before starting work. A detailed survey, clearly indicating position and areas where materials shall be stacked and sheds built is to be conducted by the contractor at his own cost and only after obtaining necessary approval of the plan for use of sites by the Engineer-in-Charge, the Contractor can use the sites accordingly.

**76 Use of blasting materials:**

Procurement of blasting materials and its storage is the responsibility of the contractor. The contractor shall engage licensed blaster for blasting operation. The contractor is to act in accordance with Indian Explosive Act and other rules prevailing, during the execution of work. It is the responsibility of the contractor to see, that works by other agencies in the vicinity are not hampered, in such cases if any claim is made by other agencies that should be borne by the contractor. Carriage of blasting materials, from the magazine to the work site, is the responsibility of the contractor.

**77 Plant and Equipment:**

77.1 The contractor shall have sufficient plant, equipment and labour and shall work such hours and shifts as may be necessary to maintain the progress on the work as per the approval progress schedule. The working and shifts hours shall comply with the Govt. Regulations in force.

77.2 It is to expressly and clearly understood that contractor shall make his own arrangements to equip himself with all machinery and special tools and plant for the speedy and proper execution of the work and the Department does not undertake responsibility towards their supply.

77.3 The Department shall supply such of the machinery that may be available on hire basis but their supply cannot be demanded as matter of right and no delay in progress can be attributed to such non-supply of the plant by the Department and the Department cannot be made liable for any damage to the contractor. The Contractor shall be responsible for safe custody of the Department machinery supplied to him (which will be delivered to contractor at the machinery yard at site of work) and he has to make good all damages and losses if any other than fire, wear and tear to bring it to the conditions that existed at the time of issue to the contractor before handing over the same to the Department. The hire charges for the machinery handed over to the contractor will be recovered at the rate prevalent

at the time of supply. The contractor will have to execute supplemental agreement with Engineer-in-charge at the time of supply of the machinery.

77.4 The acceptance of Department machinery on hire is optional to the contractor.

**78 Steel forms:**

Steel forms should be used for all items involving and use of centering and shuttering shall be single plane without any dents and undulations.

**79 Inconvenience to public:**

The contractor shall not deposit materials at any site, which will cause inconvenience to public. The Engineer-in-Charge may direct the contractor to remove such materials or may undertake the job at the cost of the contractor.

**80 Conflict of interest:**

Any bribe, commission, gift or advantage given, promised or offered by on behalf of contractor or his partner, agent or servant or any one on his behalf to any officer, servant, representatives, agents of Engineer-in-Charge, or any persons on their behalf, in relation to the obtaining or to execution of this, or any other contract with Engineer-in-Charge shall in addition to any criminal liability, which it may occur, subject to the cancellation of this or all other contracts and also to payment of any loss or damage resulting from any such cancellation. Engineer-in-Charge shall then be entitled to deduct the amount, so payable from any money, otherwise due to the contractor under this or any other contract.

**81 Contract documents and materials to be treated as confidential:**

All documents, correspondences, decisions and orders, concerning the contract shall be considered as confidential and/or restricted in nature by the contractor and he shall not divulge or allow access to them by any unauthorised person.

**82 General obligations of Contractor:**

82.1 The contractor shall, subject to the provision of the contract and with due care and diligence, execute and maintain the works in accordance with specifications and drawings.

82.2 The contractor shall promptly inform the Department and the Engineer-in-Charge of any error, omission, fault, defect in the design of or specifications for the works which are discovered when reviewing the contract documents or in the process of execution of the works.

**83 Disputes**

If Contractor believes that a decision taken by the Engineer-in-Charge was either outside the authority given to the Engineer-in-Charge by the Contract or that the decision was wrongly taken, the decision shall be referred to the technical expert within 14 days of the notification of the Engineer-in-Charge's decisions on occurrence of such event.

**84 Settlement of disputes:**

84.1 If any dispute of difference of any kind whatsoever arises between the department and the Contractor in connection with, or arising out of the Contract, whether during the progress of the works or after their completion and whether before or after the termination, abandonment or breach of the Contract, it shall in the first place, be referred to and settled by the Engineer-in-charge who shall, within a period of thirty days after being requested by the Contractor to do so, give written notice of his decision to the Contractor. Upon receipt of the written notice of the decision of the Engineer-in-Charge the Contractor shall promptly proceed without delay to comply with such notice of decision.

If the Engineer-in-Charge fails to give notice of his decision in writing within a period of thirty days after being requested or if the Contractor is dissatisfied with the notice of the decision of the Engineer-in-Charge, the Contractor may within thirty days after receiving the notice of decision appeal to the Department who shall offer an opportunity to the contractor to be heard and to offer evidence in support of his appeal, the Department shall give notice of his decision within a period of thirty days after the Contractor has given the said evidence in support of his appeal, subject to arbitration, as hereinafter provided. Such decision of the Department in respect of every matter so referred shall be final and binding upon the Contractor and shall forthwith be given effect to by the Contractor, who shall proceed with the execution of the works with all due diligence whether he requires arbitration as hereinafter provided, or not. If the Department has given written notice of his decision to the Contractor and no claim to arbitration, has been communicated to him by the Contractor within a period of thirty days from receipt of such notice the said decision shall remain final and binding upon the Contractor. If the Department fail to give notice of his decision, as aforesaid within a period of thirty days after being requested as aforesaid, or if the Contractor be dissatisfied with any such decision, then and in any such case the contractor within thirty days after the expiration of the first named period of thirty days as the case may be, require that the matter or matters in dispute be referred to arbitration as detailed below:

SETTLEMENT OF CLAIMS:

Settlement of claims for Rs. 50,000/- and below by Arbitration.

All disputes or difference arising of or relating to the contract shall be referred to the adjudication as follows:

a) Claims upto a value of rupees 10,000/-: Chief Engineer, APSS, Amaravati.

b) Claims above value of rupees 10,000/- and up to rupees 50,000/- : Tenders Committee.

The arbitration shall be conducted in accordance with the provisions of Indian Arbitration and Conciliation Act 1996 or any statutory modification thereof.

The arbitrator shall state his reasons in passing the award.

Claims above Rs.50,000/-.

All claims of above Rs.50,000/- are to be settled by a Civil Court of competent jurisdiction by way of Civil suit and not by arbitration.

A reference for adjudication under this clauses shall be made by the contractor within six months from the date of intimating the contractor of the preparation of final bill or his having accepted payment whichever is earlier.

84.2 Pending finalisation of disputes, the contractor shall proceed with execution of work with all due diligence.

**85 Security measures:**

a) Security requirements for the work shall be in accordance with the Department/ Government's general requirements including provisions of this clause and the Contractor shall conform to such requirements and shall be held responsible for the actions of all his staff, employees and the staff and employees of his sub-contractors.

b) All contractors' employees, representatives and sub-contractor's employees shall wear identifications badges provided by the contractor. Badges shall identify the contractor, showing employee's number and shall be worn at all

times while at the site. Individual labour will not be required to wear identification badges.

c) All vehicles used by the contractor shall be clearly marked with contractor's name.

d) The contractor shall be responsible for the security of the works for the duration of the contract and shall provide and maintain continuously adequate security personnel to fulfill these obligations. The requirements of security measures shall include, but not limited to maintenance of order on the site, provision of all lighting, fencing, guard flagmen and all other measures necessary for the protection of the works within the colonies, camps and elsewhere on the site, all materials delivered to the site, all persons employed in connection with the works continuously throughout working and non working period including nights, Sundays and holidays for duration of the contract.

e) Other contractors working on the site concurrently with the contractor will provide security for their own plant and materials. However, their security provisions shall in no way relieve the contractor of his responsibilities in this respect

f) Separate payment for provision of security services will not be made and its cost shall be deemed to have been included in the offer of the tenderer.

**86 Fire fighting measures:**

a) The contractor shall provide and maintain adequate fire fighting equipment and take adequate fire precaution measures for the safety of all personnel and temporary and permanent works and shall take action to prevent damage to destruction by fire of trees shrubs and grasses.

b) Separate payment will not be made for the provision of fire prevention measures.

**87 Provisions of Health and Sanitation:**

The contractor shall implement the sanitary and watch and ward rules and regulations for all forces employed under this contract and if the Contractor fails to enforce these rules, the Engineer-in-Charge may enforce them at the expenses of the Contractor.

The contractors special attention is invited to clause 37, 38, 39 and 51 of the preliminary specification to the A.P.S.S. and he is requested to provide at his own expenses the following amenities to the satisfaction of Engineer-in-charge concerned.

87.1 **First Aid:** At the work site there shall be maintained in a readily accessible place, first aid appliances and medicine including adequate supply of sterilized dressing and sterilized cotton wool. The appliance shall be kept in good order. They shall be placed under the charge of a responsible person, who shall be readily available during working hours.

87.2 **Drinking water:**

Water of good quality for drinking purpose shall be provided for the worker on a scale of not less than 2 gallons per head per day.

a) Where drinking water is obtained from an intermittent public water supply each work site shall be provided with a storage tank, where such drinking water shall be stored.

b) Every water supply storage shall be at a distance of not less than 10 M. from any latrine drain or other source of pollution where water has to be drained.

Any existing well, which is within such proximity of any latrine, drain or other source of pollution, the well shall be properly chlorinated before water is drawn from it for drinking. All such wells shall be dust and water proof.

c) A reliable pump shall be fitted to each inner well. The trap door shall be keep locked and opened only for inspection or cleaning which shall be done at least once a month.

87.3 **Washing and bathing place:**

Adequate washing and bathing places shall be provided separately for men and women. Such place shall be keep clean and well drained, bathing or washing should not be allowed in or near any drinking water well.

87.4 **Latrine and Urinals:**

There shall be provided within the area of every work site latrines and urinals in an accessible place to men and women separately. For each of them shall be on the following scales or the scale as directed by Engineer-in-charge in any particular case.

| | | |
|---|---|---|
| 1. | Where the number of persons employed does not exceed 50 | 2 |
| 2. | Where the number of persons employed exceeds 50 but does not exceed 100 | 3 |
| 3. | For every additional 100 | 3 |

If women are employed, separate latrines and urinals separated from those for men shall be provided on the same scale.

Except in work site provided with water flushed latrines connected with a water borne sewage systems all latrine shall be cleaned at least four times daily and at least twice during working hours and kept in a strict sanitary condition. The receipt scales shall be tarred inside and outside at least once a year.

The excrete from the latrines shall be disposed off at the contractors expenses in a way approved by the local public health authority. The contractor shall also employ adequate number of scavengers and conservancy shall to keep the latrines and urinals in a clean condition.

87.5 **Shelters during Rest:**

At he work site there shall be provided free of cost two suitable sheds, one for meals and other for rest for the use of workers.

**87.6 Creches:**

At every work site at which 50 or more women workers are ordinarily employed there shall be provided two huts of suitable size for use of children under the age of 6 years. One hut shall be used for infants games and other as a bed room. The hut shall be constructed on a standard not lower than the following.

1. Thatched roots
2. Mud floors and wall
3. Planks spread over the mud floor and covered with matting. The use of huts shall be restricted to children their attendants and mothers of the children.

**86.7 Canteens**:

A cook for canteen on a moderate scale shall be provided for the benefit of works if it is considered essential.

**86.8 Sheds for the workers:**

The contractor should provide at his own expense sheds for housing the workers. The sheds shall be on a standard not less than the cheap shelter type to have in which the workers in the locality are accustomed. The sheds are to be in rows with 1.5 Mts., clear space between sheds and 2.5 Mts. clear space between roofs. If conditions permit, the workers camp shall be laid out in units of 400 persons each unit to have a clear space of 4' each side.

86.9 Land should be acquired temporarily for Storing Contractor's materials or for housing their staff.

The contractor should make his own arrangements for temporary acquisition of land required for storing his materials and for the housing of his staff at his own expenses

**87 Ecological balance:**

a) The contractor shall maintain ecological balance by preventing de-forestation, water pollution and defacing of natural landscape. The contractor shall so conduct his construction operation as to prevent any unnecessary destruction, scarring, or defacing of the natural surrounding in the vicinity of the work. In respect of the ecological balance, Contractor shall observe the following instructions.

i) Where unnecessary destruction, scarring, damage or defacing may occur, as result of the operation, the same shall be repaired replanted or otherwise corrected at the contractor's expense. The contractor shall adopt precautions when using explosives, which will prevent scattering of rocks or other debris outside the work area. All work area including borrow areas shall be smoothened and graded in a manner to conform to the natural appearances of the landscape as directed by the Engineer-in-Charge.

ii) All trees and shrubbery which are not specifically required to be cleared or removed for construction purposes shall be preserved and shall be protected from any damage that may be caused by the contractor's construction operation and equipment. The removal of trees and shrubs will be permitted only after prior approval by the Engineer-in-Charge. Special care shall be exercised where trees or shrubs are exposed to injuries by construction equipment, blasting, excavating, dumping, chemical damage or other operation and the contractor shall adequately protect such trees by use of protective barriers or other methods approval by the Engineer-in-Charge. Trees shall not be used for anchorages. The contractor shall be responsible for injuries to trees and shrubs caused by his operations. The term "injury" shall include, without limitation bruising, scarring, tearing and breaking of roots, trunks or branches. All injured trees and shrubs be restored as nearly as practicable without delay to their original condition at the contractor's expense.

(iii) The contractor's construction activities shall be performed by methods that will prevent entrance or accidental spillage of solid matter contaminants, debris and other objectionable pollutants and wastage into river. Such pollutant and waste include earth and earth products, garbage, cement concrete, sewage effluent, industrial wastes, radio-active substances, mercury, oil and other petroleum products, aggregate processing, mineral salts and thermal pollution. Pollutants and wastes shall be disposed off in a manner and at sites approved by the Engineer-in-Charge.

(iv) In conduct of construction activities and operation of equipments the contractor shall utilise such practicable methods and devices as are reasonably available to control, prevent and otherwise minimise the air pollution. The excessive omission of dust in to the atmosphere will not be permitted during the manufacture, handling and storage of concrete aggregates and the contractor shall use such methods and equipment as a necessary for collection and disposal or prevention of dust during these operations. The contractor's methods of storing and handling cement shall also include means of eliminating atmospheric discharges of dust, equipment and vehicles that give objectionable omission of exhaust gases shall not be operated. Burning of materials resulting from clearing of trees, bushes, combustible construction materials and rubbish may be permitted only when atmospheric conditions for burning are considered favourable.

b) Separate payment will not be made for complying with the provisions of this clause and all cost shall be deemed to have been included in the unit rates and prices included in the contract if any provision is not complied with within a reasonable time even after issue of a notice in this respect, the necessary operations would be carried out by the Engineer-in-Charge at the cost of the Contractor, Orders of the Engineer-in-Charge in this respect would be final and binding on the contractor.

**88 Preservation of existing vegetation:**

a) The contractor will preserve and protect all existing vegetation such as trees, on or adjacent to the site which do not unreasonably interfere with the construction as may be determined by the Engineer-in-Charge. The contractor will be held responsible for all unauthorised cutting or damage of trees, including damage due to careless operation of equipment, stockpiling of materials or trecking of grass areas by equipment. Care shall be taken by the Contractor in felling tress authorised for removal to avoid any unnecessary damages to vegetation and tress that are to remain in place and to structures under construction or in existence and to workmen.

b) All the produce from such cutting of trees by the contractor shall remain the property of Government and shall be properly stacked at site, approved by the Engineer-in-Charge. No payment whatsoever, shall be made for such cutting and its stacking by the Contractor. If any produce from such cutting is not handed over to the Government by the contractor, he shall be charged for the same at the rates to be decided by the Engineer-in-Charge. The recovery of this amount shall be made in full from the intermediate bill that follows.

c) The contractor shall also make arrangements of fuel deposits for supply of required fuel for the labourer to be employed for cooking purpose at his own cost in order to prevent destruction of vegetation growth in the surrounding area of the work site.

**89 Possession prior to completion:**

The Engineer-in-charge shall have the right to take possession of or use any completed part of work or works or any part there of under construction either temporarily or permanently. Such possession or use shall not be deemed as an acceptance of any work either completed or not completed in accordance with the contract with in the interest of Clause 28 of APSS except where expressly otherwise specified by the Engineer-in-charge.

**90 Payment upon termination:**

If the Contract is terminated because of a fundamental breach of Contract by the Contractor, the Engineer-in-Charge shall issue a certificate for the value of the

work done less advance payments received upon the date of the issue of the certificate, less other recoveries due in terms of the Contract, less taxes due to be deducted at source as per applicable law and the percentage to apply to the value of the work not completed representing additional cost for completing the works at the rate of 20 percent of balance work. Additional Liquidated Damages shall not apply. If the total amount due to the Department exceeds any payment due to the Contractor the difference shall be a debt payable to the Department. In case of default for payment within 28 days from the date of issue of notice to the above effect, the contractor shall be liable to pay interest at 12% per annum for the period of delay.

**91 Access to the contractor's books:**

Whenever it is considered necessary by the Engineer-in-Charge to ascertain the actual cost of execution of any particular extra item of work or supply of the plant or material on which advance is to be made or of extra items or claims, he shall direct the contractor to produce the relevant documents such as payrolls, records of personnel, invoices of materials and any or all data relevant to the item or necessary to determine its cost etc. and the contractor shall when so required furnish all information pertaining to the aforesaid items in the mode and manner that may be specified by the Engineer-in-Charge.

**92 Drawing to be kept at site:**

One copy of the drawings furnished to the contractor shall be kept by the contractor on the site and the same shall at all reasonable time be available for inspection and use by the Engineer-in-Charge and the Engineer-in-Charge's representative and by any other persons authorised by the Engineer-in-Charge in writing.

**93 B.I.S. [I.S.I.] books and APSS to be kept at site:**

A complete set of Indian Standard specification referred to in "Technical Specifications" and A.P.S.S. shall be kept at site for reference.

**94 Variations by way of modification, omissions or additions:**

For all modifications, omissions from or additions to the drawings and specifications, the Engineer-in-charge will issue revised plans, or written instructions, or both and no modification, omission or addition shall be made unless so authorised and directed by the Executive Engineer in writing.

The Executive Engineer shall have the privilege of ordering modifications, omission or additions at any time before the completion of the work and such orders shall not operate to annual those portions of the specifications with which said changes do not conflict.

Engineer-in-Charge's Decision:

It shall be accepted as in separable part of the contract that in matters regarding materials, workmanship, removal of improper work, interpretation of the contract drawings and contract specification, mode of the procedure and the carrying out of the work, the decision of the Engineer-in-Charge, which shall be given in writing shall be binding on the contractor.

**95 Order Book**: An order book shall be kept at the Department office on the site of the work. As far as possible all orders regarding the work are to be entered in this book. All entries shall be signed and dated by the Departmental officer who issues such orders and by the contractor or by his representative. The order book shall not be removed from the workspot except with the written permission of the Executive Engineer.

**96 Care and diversion of river/stream:**

The contractor shall submit details regarding the diversion and care of river or stream during construction of the work along with a separate print-out of the time table showing earliest and latest start and finish dates of various activities. He should submit a detailed layout plan with drawings for the diversion and care of river of stream during construction of work. The above arrangements shall be at contractor's cost.

**97 Income tax:**

a) During the currency of the contract deduction of income tax at 2.24% or amended from time to time shall be made from the gross value of each bill of the contract, the contract value of which is in excess of Rs.20,000/- for deduction of tax at rates stipulated under section 194-C(4) of Income Tax Act, 1961 shall be followed.

b) Attested copy of Permanent Account Number (PAN) card and latest Income tax returns certificate submitted along with proof of receipt should be furnished, failing which the final payment will be with held (SARAL FORM).

c) The contractor's staff, personnel and labour will be liable to pay personnel income taxes in respect of their salaries and wages as are chargeable under the laws and regulations for the time being in force, and the contractor shall perform such duties in regard to such deductions thereof as may be imposed on him by such laws and regulations.

**98 Seigniorage charges:**

The Seigniorage charge will be recovered from the contractor's bills as per the rates prescribed by the State Government/ Mines and Geology Department from time to time and as per the government orders in vogue at that time for the materials used on the work.

**99 Sales tax:**

99.1 For tax on works contract, tax deduction at the rate prescribed by the State Government from time to time at source shall be made on the total value of the contract.

99.2 The contractor has to pay the sales tax on items wherever payable and no separate payment on this account shall be made by the department. The rates for various items of works shall remain unaffected by any change in the sales tax that may be made from time to time.

**100 Supply of construction materials:**

i) The contractor has to make his own arrangements for procurements, supply and use of construction materials.

ii) All materials so procured should confirm to the relevant specifications indicated in the bidding documents.

iii) The contractor shall follow all regulations of the Department/ Government of India in respect of import licenses etc., of the procurement of the materials is through imports and he shall be responsible for the payment of applicable duties and taxes, port clearances, inland transportation etc.

iv) The contractor shall make his own arrangements for adequate storage of the materials.

**101 CONSTRUCTION MATERIALS**

101.1 The contractor has to make his own arrangement for procurement, supply and use of all construction materials including cement, steel and blasting materials etc., and compliance of following should be ensured.

a) All materials so procured should confirm to the relevant specifications indicated in the tender documents or to alternative standards or specifications which are equal or higher in quality than those specified subject to Engineer-in-Charge's prior review and written approval difference between the standards specified and the proposed alternatives must be fully described by the contractor and submitted to Engineer-in-Charge's at least 30 days prior to the date when the contractor desires Engineer-in-Charge's approval. In the event Engineer-in-Charge determines that the alternative do not ensure equal or higher quality the same will be rejected and the contractor shall comply with the standards set forth within the documents.

b) All materials to be supplied should fully confirm to provisions of A.P.S.S. / I.S. Specifications as applicable.

101.2 Materials, workmanship, period and certificate of maintenance and defect liability, quality:-

All materials and workmanship shall be of the respective kinds described in the contract and in accordance with Engineer-in-Charge's instructions and shall be subjected from time to time to such tests as the Engineer-in-Charge may direct at the place of manufacture or fabrication or on the site or at such other place or places as may be specified in the contract, or at all or any of such places. The contractors shall provide such assistance, instruments, machines, labour and materials as are normally required for examining measuring and testing the work and the quality weight or quantity of any materials used and shall supply samples of materials before in Department in the works for testing as may be selected and required by the Engineer-in-Charge.

101.3 Tests, inspection of defective materials:

The contractor shall without extra cost provide samples and co-operate in the testing of materials. The Engineer-in-Charge shall have access at all times to the places of storage and where materials are being manufactured and proceeded for use in the works under contract to determine whether their manufacture and process are proceeding in accordance with the drawings and specifications. The Engineer-in-Charge shall during the progress of the works have power to order in writing from time to time in respect of the following.

a) The removal from the site, within such time or times as may be specified in the order of any materials which in opinion of the Engineer-in-Charge, are not in accordance with contract.

b) The substitution of proper and suitable materials and

c) The removal and proper re-execution, not with standing of any work which in respect of materials or workmanship is not, in the opinion of the Engineer-in-Charge, in accordance with contract.

102. The contractor shall carry out such order at no extra cost to the Engineer-in-Charge.

In case of default on the part of the contractor in carrying out such order, the

Engineer-in-Charge shall be entitled to employ and pay other persons to carry out the same and all expenses consequent thereon or incidental thereto shall be recoverable from the contractor by Engineer-in-Charge or may be deducted by the Engineer-in-Charge from any monies due to or which may become due to the contractor.

In lieu of removing the work or materials not in accordance with the contract the Engineer-in-Charge may order such work or materials to remain and in that case such may be paid at the reduced rates as may be decided by Engineer-in-Charge. However any action by the Engineer-in-Charge under this para shall not any way absolve the contractor from his responsibility and liabilities as per conditions of contract.

103. **CEMENT**

The contractor has to make his own arrangements for the procurement of cement of required specification for works subject to the following:

a) The contractor shall procure bulk cement required for the works, only from cement factories (Main producers) of approved make and brand only as approved by the Engineer-in-charge. The contractor shall make own arrangements for adequate storage of cement.

b) The contractor shall procure cement in standard packing (50 Kg per bag) from the authorised manufacturers. The contractor shall make necessary arrangement at his own cost to the satisfaction of Engineer-in-charge for actual weighment of random sample from the available stock and shall confirm with the specification laid down by the Bureau of Indian standards or other standard institutions as the case may be. Cement shall be got tested for all the tests as directed by the Engineer-in-charge at least once in a month in advance before the use of cement bags brought and kept at site godown.

   Cement bags required for testing shall be supplied by the contractor free of cost.

c) The contractor should store the cement of 60 days requirement at least one month in advance to ensure the quality of cement so brought to site and shall not remove the same without the written permission of the Engineer-in-charge.

The contractor shall forthwith remove from the works area any cement that the Engineer-in-charge may disallow for use on account of failure to meet with required quality and standard.

d) The contractor will have to construct sheds for storing cement having capacity not less than the cement required for 90 days use at appropriate locations at the work site. The Engineer-in-charge or the representatives shall have free access to such stores at all times.

e) The contractor shall further at all times satisfy the Engineer-in-charge on demand by production of records and books or by submission of returns and other proofs as directed that the cement is being used as tested and approved by Engineer-in-charge for the purpose and the contractor shall at all times keep his records up to date to enable the Engineer-in-charge to apply such checks as he may desire.

f) Cement which has been unduly long in storage with the contractor or alternatively has deteriorated due to inadequate storage and thus become unfit for use on the work shall be rejected by the department and no claims will be entertained. The contractor shall forthwith remove from the work area any cement the Engineer-in-charge may disallow for use on work and replace it by cement complying with the relevant Indian Standards.

**104 STEEL**

The contractors shall procured mild steel (M S) reinforcement bars, High yield strength deformed bars (HYSD) bars, rods and structural steel etc., required for the works, only from the main or secondary producers, manufacturing steel to the prescribed specification of Bureau of Indian Standards or equivalent and licensed

to affix ISI or other equivalent certifications marks and acceptable to the Engineer-in-charge. Necessary test certificates for each consignment are to be produced to Engineer-in-charge before use on works. The original bills of procurement should be submitted to the Engineer-in-charge for making payment of the item.

104.1 The diameter and weight of steel should be as follows:

| S.No. | Diameter of rod | Sectional weight in Kg/RM both for Plain and HYSD steel |
|---|---|---|
| 1. | 6 MM | 0.22 |
| 2. | 8 MM | 0.39 |
| 3. | 10 MM | 0.62 |
| 4. | 12MM | 0.89 |
| 5 | 14 MM | 1.21 |
| 6. | 16 MM | 1.56 |
| 7. | 18 MM | 2.00 |
| 8. | 20 MM | 2.47 |
| 9. | 22 MM | 2.98 |
| 10. | 25 MM | 3.85 |
| 11. | 28 MM | 4.83 |
| 12. | 32 MM | 6.31 |
| 13. | 33 MM | 6.71 |
| 14. | 36 MM | 7.99 |
| 15. | 40 MM | 9.86 |
| 16. | 42 MM | 10.88 |

Note: If any rods other than those diameters specified above are procured the weights shall be as per standard steel tables.

## 105 I.S. 383/1970 TABLE - I

### 105.1 COARSE AGGREGATE

| I.S. Sieve Designation | Percent passing for single sized aggregate of Metal Size | | | | | | Percentage passing for graded-aggregate of Nominal Size | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 63 mm | 40 mm | 20 mm | 16 mm | 12.50 mm | 10 mm | 40 mm | 20 mm | 16 mm | 12.50 mm |
| 80 mm | 100 | _ | _ | _ | _ | _ | 100 | _ | _ | _ |
| 63 mm | 85-100 | 100 | _ | _ | _ | _ | _ | _ | _ | _ |
| 40 mm | 0-30 | 85-100 | 100 | _ | _ | _ | 95-100 | 100 | _ | _ |
| 20 mm | 0-5 | 0-20 | 85-100 | 100 | _ | _ | 30-70 | 95-100 | 100 | 100 |
| 16 mm | _ | _ | _ | 85-100 | 100 | _ | _ | _ | 90-100 | _ |
| 12.5 mm | _ | _ | _ | _ | 85-100 | 100 | _ | _ | _ | 90-100 |
| 10 mm | 0-5 | 0-5 | 0-20 | 0-30 | 0-45 | 85-100 | 10-35 | 25-55 | 30-70 | 40-85 |
| 4.75 mm | _ | _ | 0-5 | 0-5 | 0-10 | 0-20 | 0-5 | 0-10 | 0-10 | 0-10 |
| 2.36 mm | _ | _ | _ | _ | _ | _ | 0-5 | _ | _ | _ |



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**TABLE - II**

**105.2 FINE AGGREGATE**

| I.S. Sieve Designation | Grading Zone - I | Grading Zone - II | Grading Zone - III | Grading Zone - IV |
|---|---|---|---|---|
| 10 mm | 100 | 100 | 100 | 100 |
| 4.75 mm | 90-100 | 90-100 | 90-100 | 95-100 |
| 2.36 mm | 60-95 | 75-100 | 85-100 | 95-100 |
| 1.18 mm | 13-70 | 55-90 | 75-100 | 90-100 |
| 600.00 microns | 15-34 | 35-59 | 60-79 | 80-100 |
| 300.00 microns | 5-20 | 8-30 | 12-40 | 15-50 |
| 15.00 microns | 0-100 | 0-10 | 0-10 | 0-15 |



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**TABLE - III**

**105.3 ALL-IN AGGREGATE GRADING**

| I.S. Sieve Designation | 40 mm Nominal | 20 mm Nominal |
|---|---|---|
| 80.00 mm | 100 | _ |
| 40.00 mm | 95-100 | 95-100 |
| 20.00 mm | 45-75 | 30-50 |
| 4.75 mm | 24-75 | 10-35 |
| 600.00 microns | 8-30 | 0-6 |
| 150-00 microns | 0-60 | _ |



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**TABLE - IV**

105.4 For vibrated Reinforced concrete items (V.R.C.C.)

**Characteristic Strength of Cube at the age of 28 days of curing**

| | | | | |
|---|---|---|---|---|
| M-25 | 1:1:2 | 25N/mm2 | = | 250 kgs/cm2 |
| M-20 | 1:1½:3 | 20 N/mm2 | = | 200 kgs/cm2 |
| M-15 | 1:2:4 | 15 N/mm2 | = | 150 kgs/ cm2 |



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## 106 STORAGE OF CEMENT

106.1 It is often necessary to store Portland cement, some times for a period months. This is particularly true when transportation facilities must be need to their capacity and deliveries.

106.2 Portland cement readily absorbs moisture not only in the form of free water but also moisture from the atmosphere or from damp material in contact with it and becomes hydrated and loses strength. It is necessary therefore that it should be protected from absorption of moisture before it is used if it is to fulfill its function. An absorption of one or two percent of water has not appreciable effect but further amounts of absorption, results in hardening of the cement and reduced the strength. If the absorption exceeds 5% the cement is for all ordinary purposes ruined. Finally ground cement stored in stacks lend to deteriorate more than coarse cements. In this respects normal hardening Portland cement and high aluminum cements at least affected than rapid hardening Portland cement.

106.3 American, Spanish and German experiments have shown that on average the strength of cement stress in bags is reduced.

After 3 months by 15 to 20 percent.

After 6 months by 20 to 30 percent.

After 12 months by 30 to 50 percent.

After 2 years by 10 to 50 percent.

After 4 1/2 years by 50 to 60 percent.

106.4 These figures prove that special attention should be paid to the storage of cement, even when its strength is equal to or suspense's the specified normal strength.

106.5 With an extensive range of climate conditions it is difficult to lay down universal rules for the storage of cement by the general principle should always be kept in mind that it must be protected as far as possible from any form of moisture prior to mixing concrete mortar.

106.6 During the dry weather in main parts of the country where the relative humidity of the atmosphere even in nights is low (that is to say when there is very little moisture in the air) little or no protection may be necessary and the cement in its stock may require no more than a tarpaulin through for the stack. But there are parts of the country particularly near the coast where the atmosphere is always damp at any time of day or night and then greater precautions are necessary. In such place such as the west coast and the Niligiries and period when heavy rain falls are encountered such greater care has to be taken of the cement and proper strength provided it from the damp.

106.7 Whenever there is any possibility of the cement exposed to moisture either in the atmosphere or actual ratio it should be stored in a well constructed dry godown or shed. The cement store should be whether right construction preferable with terraced roofing with a sound wooden or ground to ensure that it is damp proof building with plant roofing are prohibited because of their tendency to leak. Corrugated sheets roofing has tendency to the condense moisture and should be protected by field to prevent wind and rain driving through cement should not be placed directly on cement plaster flooring and other types of flooring commonly meant with which are not damp proof. A wooden platform or false floor a sheet of water proof paper should be provided.

If none of these is possible than floor should be covered with straw, hay, cinder or ash or such other material densely and uniformly packed to a thickness of at least one inch and over a laid worth tarpaulin of old cement large windows and ventilators if any should be slightly shut to prevent from circulation of air inside the stones drainage should be provided if necessary to prevent accumulation of water in the vicinity of the store.

106.8 Cement should be stored in piles arranged parallel to the walls. It is advisable to pile bags against the walls and an allowance of at least 0.3M all round should be made between the exterior walls piles at least 0.6M wide should be left for each access and delivery. The outside stacks deterior a similar pile. Successive consignments covered with some water proof cover as a both measure of protection and prevent the free circulation of air as each lot of proper fresh air will bring in more moisture. Once the cement has been properly stored should not be disturbed until it is to be used. There is no advantage in moving and stacking the bags to reduce where house set as this practice only exposes fresh cement to the air resulting in loss due to the shifting of cement through the cloth mesh and in damage to the stacks.

106.9 Cement required for use immediately after delivery to the site may be stored in the open on a raised damp proof floor so long as it is fully protected by tarpaulin or either weather resisting covers. Storage under these conditions should be limited to 48 hours. The tarpaulin should be raised well above the top most Tie of bags and must be sloped for rapid drainage in case of showers.

106.10 The storage place required for a given quantity of cement can be calculated from the following date. If spread losses over the floor of a store to a depth of 1st floor a ton of cement required about 2.50 Sqm. If stored in paper bags laid on their side the area required is 5.00 Sqm to 6.00 Sqm. per ton if laid in a single tier and proportionately less if laid on more than one tier. If the bags are stacked in any other manner feet portion the minimum area provided should be increased to allow space for passenger etc., to avoid house set in any case not more than 15 bags. If stacked higher than this the pressure on the bottom bags is liable to burst at or form clad in damp water apart from handling difficulties because of their height.

106.11 Consignments should be used in the same sequences as they are delivered. To ensure this the date of arrival of each consignment should be clearly indicated. This is best done by tying a piece of country twins or cord to the end bags in the bottom most tier of the days pile, tacking the two places of card up the sides and along the top of pile an tying the main the center. The date of receipt in the store being clearly written on a bin card high from the card. Dead storage where the cement remains in place for a long time which other consignments of cement come in and out should be avoided.

106.12 In issuing cement from a store the cement bags should be removed in vertical column of the pile and not horizontal so as to avoid dead stoppage space.

106.13 As a rule cement should not be stored longer than three months and if time is exceeded the material should be retested being needed. Especially in the rainy season prolonged storage should be avoided. If stuck is likely to be held over for more than three months anticipatory measures should be taken to use it on the works.

106.14 Cement that has become supply due to storage in damp positions due to exposure to the weather is generally useless for making concrete and should be removed from the site. Air set lumps that can be broken down to floor with the pieces. If such lumps are innumerous it is easier to screen them out and discord

them if the proportion of air cement is considerable. The fine material after screening should be tested to determine whether it has become defective.

106.15 The cement in bags is stored in high piles for long periods. There is often a slight tendency in the lower layers to harden, caused by the pressure above this is known as warehouse set. Cement in this condition, an every wet not for service and can be reconditioned by letting each drop on a solid surface for using the cement contained.

106.16 All cement concrete shall be machine mixed and machine vibrated.

106.17 The proportions of cement concrete specified in the above schedule are nominal and are indication of approximate proportion of cement, fine aggregate and coarse aggregate which may have to be altered suitably at site to obtain desired strength and workability. However, the quantity of cement shall not be less than specified below:

| Nominal Mix | Cement in bags of 50 Kgs per one Cubic metre (net) of cement concrete |
|---|---|
| a) 1 : 1.5 : 3 | 8.84 bags of 50 Kgs. |
| b) 1 : 2 : 4 | 6.62 bags of 50 Kgs. |
| c) 1 : 2.5 : 5 | 5.30 bags of 50 Kgs. |
| d) 1 : 3 : 6 | 4.42 bags of 50 Kgs. |
| e) 1 : 4 : 8 | 2.65 bags of 50 Kgs. |
| g) 1 : 6 : 12 | 2.21 bags of 50 Kgs. |
| h) 1 : 8 : 16 | 1.66 bags of 50 Kgs. |

**106.18 Theoretical requirement of cement for**

| | |
|---|---|
| a) C.R.S. Masonry in C.M. (1 : 6) | 1.54 bags per Cum |
| b) C.R.S. Masonry in C.M. (1 : 8) | 1.15 bags per Cum |
| c) Brick Masonry in C.M. (1 : 4) | 1.44 bags per Cum |
| d) Brick Masonry in C.M. (1 : 6) | 0.96 bags per Cum |
| e) Brick Masonry in C.M. (1 : 8) | 0.72 bags per Cum |
| f) 12mm plastering in C.M. (1:5) & (1:3) | 1.02 bags per 10 Sqm. |
| g) 12mm plastering in C.M. (1:6) & (1:4) | 1.02 bags per 10 Sqm. |
| h) 20mm plastering in C.M. (1:6) & (1:4) | 1.02 bags per 10 Sqm. |
| i) 12mm plastering in C.M. (1:4) | 1.08 bags per 10 Sqm. |
| j) 12mm plastering in C.M. (1:6) | 0.72 bags per 10 Sqm. |

**107 Conditions on Roof Slabs and Stripping time**

107.1 The R.C.C. slab laid should be leak proof. After observing for two rainy seasons if the roof or floor is found to be perfectly leak proof and no moisture or dampness is seen underneath at ceiling of the slab, the contractor can ask for refund of E.M.D. or F.S.D. from the department. If there are any defects noticed after laying of roof they must be attended to by the contractor at his own cost. Further the contractor must arrange to get the structure treated as per clause 21 of ISI code No.456/2000 at his own cost on the instructions of the department.

When R.C.C. slab is laid the following tests may be carried out by the contractor at his own cost to prove that the slab is impervious.

a) After the centering is removed and curing period is over the slab shall be put to test by pouring water to 15 cms. depth and watched carefully for period not less than a week.

b) If leakage is observed immediate action should be taken to rectify it by the contractor at his own cost and again tested to see that there are no leakages.

c) The officer observing the leakage test shall issue a certificate to this effect before final bill is made.

d) The variation thickness of R.C.C. roof slab due to varying spans, or special covering materials should not effect the general roof bed which should be uniform unless otherwise shown in drawings or instructed.

e) For roof slab to be laid MS hooks to be provided as directed by the department for fixing fans and lighting G.I. pipes of 12mm or 20mm dia meter supplied by the department at site has to be provided in the masonry walls of a concrete at the specified places as directed by the department for making electrical wiring. No payment will be made to the contractor for these sundry items of work.

f) For roof slabs water has to be stagnated for 15 cms depth for one week to test the leakages if any. If there are any leakages the contractor has to rectify the same as directed by the department at the cost of the contractor. No payment will be made to the contractor on this account either for testing or for rectifications thus stagnated.

107.2 Stripping Time: Forms shall not be struck until the concrete has reached a strength at least twice the stress to which the concrete may be subjected at the time of removal of form work. The strength referred to shall be that of concrete using the same cement and aggregates, with the same proportions and cured under conditions of temperature and moisture similar to these existing on the work. Where possible, the form work shall be left longer as it would assist the curing.

Note 1:- In normal circumstances and where ordinary Portland cement is used form work may generally be removed after the expiry of the following period.

| | | |
|---|---|---|
| a) | Walls, Columns and Vertical faces of all structural members | 24 to 48 hours as may be decided by the Engineer-in-charge |
| b) | Slabs (props left under) | 3 days |
| c) | Beam soffits (props left under) | 7 days |
| d) | Removal of props under slabs | |
| | i) Spanning upto 4.5 Mts. | 7 days |
| | ii) Spanning over 4.5 Mts. | 14 days |
| e) | Removal of props under Beams & Arches | |
| | i) Spanning upto 6 Mts. | 14 days |
| | ii) Spanning over 6 Mts. | 21 days |

For other cements the stripping time recommended for ordinary Portland cement may be suitably modified.

Note2:- The number of props left under the concrete element, their sizes and description, shall be such that they shall be able to safely carry the full dead load of the slab, beam or arch as the case may be together with any live load likely to occur during curing or further construction.

## TECHNICAL SPECIFICATIONS

**[INCORPORATED AS PER REQUIREMENT OF THE WORK PUT TO TENDER WITH THE RELEVANT SPECIFICATION AND NUMBER OF AP STANDARD SPECIFICATION / SPECIAL SPECIFICATION]**



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## SPECIFICATIONS

### 1.0 PREAMBLE

The technical specifications for various items of work contained here in shall be read in conjunction with the specifications mentioned for each item of work in bill of quantities (Schedule – A) and also plans and drawings.

### 2.0 PART I - GENERAL TECHNICAL SPECIFICATIONS

The following are the general technical specifications normally adopted for construction of buildings. Each item of work shall be executed according to the relevant standard specification number as described in the "Andhra Pradesh Standard Specification" (APSS) and Indian Standard (I.S) Specifications, including Water supply, Sanitary and Electrical Installations. In the absence of any definite provisions on any particular item of work in the aforesaid specifications in A.P.S.S., reference may be made to the latest codes and specifications of Indian Standards or Indian Roads congress (IRC in case of Roads). Where even these are silent, the construction and completion of works shall conform to sound engineering practice as approved by Engineer-in-charge and in case of dispute arising out of the interpretation of the above, the decision of Engineer-in-charge shall be final and binding on the contractor.

### 3.0 GENERAL INSTRUCTIONS

### 3.1 Site Clearance and Demolition

The site shall be cleared of all trees, stumps, roots, brush wood, bushes and other objectionable materials. Useful and saleable material shall be the property of the Owner and shall be stacked properly as directed by the Engineer. The areas to be covered with embankments shall be stripped of top soil to required depths to expose acceptable founding strata. Top soil unsuitable for use in embankment construction and other fills shall be disposed off as directed. All combustible materials shall be stacked and burnt in locations sufficiently remote to eliminate all danger of fire hazards. All old concrete, brick works and drains which interfere with construction works shall be dismantled with the approval of the Engineer duly taking all necessary precautions prescribed in safety specification given below. Top soil which is suitable for use in construction work shall be stockpiled for later use. Other objectionable materials such as trash, debris, stones, brick, broken concrete, scrap metal etc., shall be disposed off as directed by the Engineer. Payment for cutting and removal of trees, stumps, dismantling existing structures and stripping shall be regulated by the description in the Schedule of Items or Section 2 of A.P.S.S.

### 3.2 Safety Specification:

**3.2.1** On every demolition job, danger signs shall be conspicuously posted all-round the structure and all door openings giving access to structure shall be barricaded or marked except during the movement of actual workmen or equipment. However provision shall be made for at least two independent exits for escape of workmen during any emergency.

**3.2.2** During night, red lights shall be placed on or about all the barricades.

**3.2.3** Where in any work of demolition it is imperative, because of danger existing to ensure that no unauthorized person shall enter the site of demolition outside working hours, a watchman should be employed. In addition to watching the site, he shall also be responsible for maintaining all notices, lights and barricades.

**3.2.4** All the necessary safety appliances as per IS: 4130 shall be issued to the workers and their use explained. It shall be ensured that the workers are using all the safety appliances while at work.

**3.2.5** The removal of a member may weaken the side wall of an adjoining structure and to prevent possible damage, these walls shall be supported until such time as permanent protection is provided. In case any danger is anticipated to the adjoining structure the same shall be got vacated to avoid any danger to human life.

**3.2.6** The power on all electrical service lines shall be shut off and all such lines cut or disconnected at or outside the property line, before the demolition work is started. Prior to cutting of such lines the necessary approval shall be obtained from the electrical authorities concerned for demolition work itself.

**3.2.7** All gas, water, steam and other service lines shall be shut off and capped or otherwise controlled at or outside the building line, before demolition work is started.

**3.2.8** All the mains and meters of the building shall be removed or protected from damage.

**3.2.9** If a structure to be demolished has been partially wrecked by fire, explosion or other catastrophe, the walls and damaged roofs shall be shored or braced suitably.

**3.2.10** Walkways and passageways shall be provided for the use of the workman who shall be instructed to use them and all such walkways and passageways shall be kept adequately lighted, free from debris and other materials.

**3.2.11** All nails in any kind of lumber shall be withdrawn, hammered or bent over as soon as such lumber is removed from the structure being demolished, and placed in pipes for future cleaning or burning.

**3.2.12** All the roads and open area adjacent to the work site shall either be closed or suitably protected.

**3.2.13** No electric cable or apparatus which is liable to be a source of danger or a cable or apparatus used by the operator shall remain electricity charged.

**3.2.14** All practical steps shall be taken to prevent danger to persons employed from risk of fire or explosion or flooding. No floor, roof or other part of the building shall be so overloaded with debris or materials as to render it unsafe.

**3.3 Drawings, Instructions, Measurements**

All works shall be done according to the detailed drawings and specifications. Figured dimensions shall be followed. Measurement shall be taken of the actual work done but shall not exceed those marked on the drawings for payments.

**3.4 Quality of work**

The materials, Equipment, tools and plants and workmanship should be of high and acceptable quality conforming to the standard specifications.

**3.5 Testing of works and materials**

**3.5.1** All materials used and works done shall be subject to approval of the Engineer.

**3.5.2** The contractor shall arrange sufficiently in advance to test materials and portions of works in order to prove their soundness and efficiency if required, including samples and supporting test results from the approved laboratory and other documentary evidence from the manufacturer, wherever

applicable, and indicate the types of materials and their respective sources. The delivery of materials at site shall commence only after the approval of the quality, grading and sources of the materials by the Engineer.

**3.5.3** The quality of all materials approved shall be maintained throughout the period of construction and periodical tests shall be carried out to ensure that it is maintained. The contractor shall conduct tests at work site/approved laboratories and shall maintain test reports at site for cement, coarse aggregates, fine aggregates, water, steel, bricks and concrete at the following frequency :-

| Sl. No. | Description of material | Frequency of test | Allowable limits |
|---|---|---|---|
| 1. | CEMENT : (IS : 8112-1989) | | |
| | a) Fineness | One for each source of supply in a month | Shall not be less than 3500 sqcm / gm |
| | b) Setting time | -do- | Initial setting time shall not be less than 30 minutes and final setting time shall not be more than 60 minutes. |
| | c) Soundness | -do- | Expansion (unaerated) shall be not more than 10mm by "Le Chatelier" method; if it fails, expansion of aerated sample shall be not more than 5 mm. |
| | d) Compressive strength of cement mortar cubes 1:3 (1 cement :3 standard sand) by mass | -do- | Compressive strength for 7 days shall not be less than 330 $kg/cm^2$ and compressive strength for 28 days shall not be less than 430 $kg/cm^2$ |
| 2. | Coarse aggregate : (IS383-1970) | | |
| | a) Gradation | One test for 15 Cum or atleast on the day of concrete if concrete quantity is less than 15 cum. | 40mm Metal : a)Seive analysis : -63mm – 100% 40mm-85 to 100% 20mm-0-2-%; 10mm-0.5% b) Flakeness Index : shall be less than 30% by weight |
| | b) Aggregate impact value | Once for each source of supply or when ever change in texture is noticed. | 20mm Metal : a)Sieve analysis : -Limits : 40mm – 100%; 20mm-95 to 100%; 10mm-25 to 55%; 4.75-0 to 10% b) Flakiness Index : less than 25% c) Aggregate impact value: 20-40(IS 2386-1963) |
| 3. | FINE AGGREGATE (IS383 –m1970) | | |
| | a) Gradation for concrete | One test for every 15 cum. | Fineness modules : Fine sand limit 2.2 to 2.6 |
| | b) Gradation for masonry | Atleast once on the day of work | Medium sand limit 2.6 to 2.9 |
| | c) Gradation for finishing | -do- | Coarse sand limit 2.9 to 3.2 |
| | d) Bulkage | Three for each day of work i.e. morning noon | b) Silt Content : shall be less than 4% by weight |

| Sl. No. | Description of material | Frequency of test | Allowable limits |
|---|---|---|---|
| | | and evening | |
| | e) Silt content | Atleast once on the day of work | |
| 4. | WATER : Chemical test | One test for each source | The water quantity shall be as per clause 5.4 of ISI 456-2000. The PH value of water shall not be less than 6. |
| 5. | STEEL : (F2415 (IS1786-1985)) | | |
| | a) 0.2% proof stress | One for each source of supply and once in six months for fresh supply | 4150 kg/cm$^2$ (Minimum) |
| | b) Elongation | -do- | Percentage of elongation 14.5% minimum |
| | c) Tensile strength | -do- | Ultimate tensile strength 4900 kg/cm$^2$ (Minimum) |
| 6. | BRICKS : (IS:1077-1976) | | |
| | a) Compressive strength | One for each source of supply and once in two months when change in texture is noticed | Shall not be less than 40 Kg/cm$^2$ |
| | b) Water absorption | -do- | Shall not be greater than 20% by weight |
| 7. | CONCRETE : (IS456:2000) | | a) Compressive strength (7 days) |
| | a) Cube strength | Frequency of testing as per clause 15.2 of IS 456-2000 for example 6 cube specimens, 3 each for 7 days & 28 days strength for every 15 cum. Cube shall be prepared, cured and tested in accordance with the requirement of IS 516. | M15-100 Kg/cm$^2$ (Minimum)<br>M20-135 KG.cm$^2$ (Minimum)<br>b) Compressive strength (28 days)<br>M15-150 Kg/cm$^2$ (Minimum)<br>M20-200 Kg/cm$^2$ (Minimum) |
| | b) Slump | Thrice in a day of concrete in morning, noon and evening | a) Foundation footing – 10mm to 25mm<br>b) Column beams and slabs – 25mm to 40mm (with normal reinforcement)<br>c) Beams, slabs – 40mm to 50mm (with congested reinforcement) |

***A Register of record of material testing and Register of daily events showing materials received, labour engaged, out turn of work etc. shall be maintained at site and shall be signed by the contractor or his authorised representative and the Engineer***



Tenderer

Chief Engineer
APSS, Amaravati.

**3.6 Rejection of Materials/works**

**3.6.1** Any material brought to site which in the opinion of the Engineer is defective, sub-standard, damaged, contaminated, deteriorated or does not comply with the requirement of the specification shall be rejected. The contractor shall remove from site such materials within 4 hours of notice from site.

**3.6.2** If the work or portion of the work which in the opinion of the Engineer is found to be defective or unsound, the contractor shall pull it down and re-execute the same work at his own cost.

**3.7 Measurement Materials**

Materials requiring mixing should be measured separately in boxes of appropriate size before being mixed in the specified proportions.

**3.8 Storage of Materials**

**3.8.1** Adequate safe, dry storage shall be provided for all materials particularly cement.

**3.9 Codes**

3.9.1 Unless mentioned otherwise, current versions of all codes, specifications and standards issued by the Indian Standards Institution and Indian Roads Congress shall be fully applicable to these specifications. In the absence of appropriate publications by ISI or IRC, adoptable specification of the International Organisation for Standarization shall apply.

3.9.2 In case of any conflict in meaning between the specifications mentioned hear in and those of ISI or IRC, the provisions of these specifications shall prevail.

3.9.3 The following codes shall be applicable for the purpose. However the latest revision of these codes shall only be used.

i) **IS:383-1970** - Specification for coarse and fine aggregates from natural sources for concrete. (Second revision)

ii) **IS:2386-1963 (Part I to Part III)** - Methods of Tests for aggregates of concrete.

iii) **IS: 9103-1979** - Specification for admixtures for concrete.

iv) **IS:8112-1989** - Specification for 43 grade ordinary portland cement (first revision with amendment No.3.)

v) **IS:516-1959** - Methods of test for strength of concrete.

vi) **IS:1121-1974 (Part I to Part IV)** - Method of test for determination of strength properties of natural building stones.

vii) **IS:7779-1975 (Part I to Part III)** - Schedule of properties and availability of stones for construction purposes.

viii) **IS:1077-1976** - Specification for common burnt clay building bricks (Third revision)

ix) **IS:3495-1976 (Parts I to IV)** - Methods of test for burnt clay building bricks (Second revision)

x) **IS:1003-1977 (Parts I & II)**- Specification for timber panelled and glazed door, window and ventilator shutters.

xi) **IS:432-1982 (Parts I & II)** - Specification of mild steel and medium tensile steel bars and hard drawn steel wires for concrete reinforcement.

xii) **IS:1786-1985** - Specification of Cold - worked steel, high strength deformed bars for concrete reinforcement (revised).

xiii) **IS:226-1975** - Specification for structural steel (Standard quality) (Fifth revision).

xiv) **IS:14687-1999** - Guidelines for false work for concrete structures.

xv) **IS:1597-1967 (Parts 1 & 2)** - Code of practice for construction of stone masonry.

xvi) **IS:2212-1962** - Code of practice for brick work.

xvii) **IS:2572-1963** - Code of practice for construction of hollow concrete block masonry.

xviii) **IS:4326-1993** - Code of practice for earthquake resistant design and construction of buildings (Second revision).

xix) **IS:456-2000** - Code of practice for plain and reinforced concrete (Fourth revision).

xx) **IS:800-1984** - Code of practice for general construction in steel (Second revision).

xxi) **IS:3370-1967** (Parts I to IV) - Code of practice for concrete structures for the storage of liquids.

xxii) **IS:1893-1984** - Criteria for earthquake resistant design of structures.

xxiii) **IS:4082-1996** - Recommendations on stacking and storage of construction materials at site (Second revision).

xxiv) **IS:1642-1960** - Code of practice for general engineering drawings (Second revision).

xxv) **IS:696-1972** - Code of practice for general engineering drawings (Second revision).

xxvi) **IS:962-1967** - Code of practice for architectural and building drawings (First revision).

xxvii) **IS:1161-1979** - Specification for steel tubes for structural purposes (Third revision).

xxviii) **IS:816-1969** - Code of practice for use of metal arc welding for general construction in mild steel.

xxix) **IS:3414-1968** - Code of practice for design and insulation of joints in buildings.

xxx) **IS:1331-1992 (Part I & II)** - Methods of non-destructive testing of concrete soils : general requirement (third revision)

xxxi) I**S:1904-1986** - Code of practice for design and construction of foundations in soils : general requirement (Third revision)

xxxii) **IS:2502-1963** - Code of practice for bending and fixing of bars for concrete reinforcement.

## 3.10 PERFORMANCE OF WORK

### 3.10.1 Execution of Works

**3.10.1.1** All the works shall be executed in strict conformity with the provisions of the contract documents, explanatory detailed drawings and specifications.

**3.10.1.2** The site should be cleared of all obstructions, vegetation, loose stones and materials before start of work.

**3.10.1.3** The Engineer in charge, Supervisor will inspect the work on a Day-to-Day basis.

### 3.10.2 Work in Monsoon

3.10.2.1 The construction may entail working in monsoon also. The contractor must maintain a minimum labour force and execute the construction according to the prescribed schedule.

3.10.2.2 Contractor is responsible for keeping the construction work site free from water.

### 3.10.3 Plinth Levels

3.10.3.1 A proper level should be maintained, in terms of horizontal and vertical alignment. A minimum acceptable plinth level above road level shall be maintained. The plinth level shall be agreed with the Engineer's representative.

## 4.0 DETAILED SPECIFICATIONS OF MATERIALS

### 4.1 Water (APSS No. 129)

4.1.1 Water should be clean, fresh and free from all chemicals, salts and deleterious materials and vegetable growth. Water has to meet the requirements mentioned in Cl. 5.4 of IS:456-2000. Storage for water should be sufficient and adequate for the regular consumption of works and for the use of labour on site.

### 4.2 Earth (APSS No. 309 & 310)

4.2.1 For filling, the soil shall be free from all rubbish, organic or vegetable growth including roots, weeds etc. Black cotton soil should not be used for basement filling.

### 4.3 Sand (APSS No. 110)

4.3.1 Sand shall be clean river or pit sand of approved quality and free from salt, earth, dust or other impurities. Sand for plain and reinforced concrete shall confirm to IS: 383-1970. Sand for various purposes shall confirm grading as below.

Sand for Masonry --- table 110-A of APSS No.110

Sand for Plastering --- table 110-B & 110-C of APSS No. 110

Sand for Plain and Zone I to III of table 110-D of APSS No.110

Reinforced concrete

### 4.4 Stone for Masonry (APSS No. 107)

4.4.1 Stones used shall be strong, durable, dense, compact, close grained, homogeneous, fire resistant and shall be obtained from sources approved by Engineer. Stones shall additionally be hard, sound, free from cracks, decay and other flaws or weathering and shall be easily workable. Stones with round surfaces shall not be made use of.

4.4.2 Stones shall have a crushing strength of not less than 1000 Kg/cm$^2$. Stones with lesser crushing strength may be used in works with prior approval of the Engineer. Stones shall be non-porous and when tested in accordance with IS:1124-"Method of Test for Determination of Water Absorption" etc., shall show water absorption of less than 5% of its dry weight when soaked in water for 24 hours. Tests for durability and weathering shall be done in accordance with IS:1126 and IS:1125 respectively. The working of stones to required sizes and their dressing shall be as per IS:1127 "Recommendations for dimensions and workmanship of natural building stones for Masonry work" and IS:1129 "Dressing of Natural Building Stones". Stones especially

limestones and sand stones, shall be well seasoned by exposure to air before use in construction works.

**3.5 Cement (APSS No. 112)**

3.5.1 Cement should comply with the requirements of IS:8112-1989 and should be 43 grade ordinary Portland Cement, for making plain and reinforced concrete, mortar etc. The quality of cement shall be inconfirmity to the performance characteristics given in IS: 8112 - 1989. The contractor shall procure bulk cement required for the works only from reputed cement factories (main producers) acceptable to the Engineer and should obtain, furnish from suppliers of cement a test certificate for every consignment of cement. The cement bag shall bear the manufacturer's name or their registered trade mark. Cement shall be tested in accordance with IS: 4031-1988 and IS: 4032-1988.

4.5.2 When storing the bags, the floor should be raised 30 cms., above the ground and stacked in rows not exceeding 10 bags high, 60 cms clear from the walls in a closed water proof building protected from flood, rain and moisture and deterioration in such a manner that easy access and proper inspection and counting is possible.

4.5.3 The cement should be delivered to the site in sound dry bags and shall be stored properly. Cement packed in LDPE Bags may be preferred to ensure protection from moisture and dampness.

**4.6 Bricks (APSS No. 102)**

4.6.1 Bricks for masonry in foundations, walls and other locations shall be common burnt clay building bricks having minimum crushing strength of 40 Kg/cm². They shall be sound, hard and thoroughly well burnt, but not over-burnt, with uniform size having rectangular faces with parallel sides and sharp straight right angled edges and be of uniform colour with fine compact uniform texture. Bricks shall be of uniform deep red cherry or copper colour. They shall be free from flaws, cracks and nodules of free lime. Water absorption after 24 hours immersion in cold water shall be not more than 20% by weight. They shall not absorb more than 10% by weight of water after immersion for six hours. They shall emit a clear metallic ringing sound when struck by a mallet and shall not break when dropped on their face, from a height of 60 cm. Fractured surface shall show homogeneous, fine grained uniform texture, free from cracks, air holes, laminations, grits, lumps of lime, efflorescence or any other defect which may impair their strength, durability, appearance and usefulness for the purpose intended. Under-burnt or vitrified bricks shall not be used. Samples of bricks brought to the site shall be tested periodically for compression and other tests according to IS:3495, Parts-I, II & III - “Method of Test for Burnt Clay Building Bricks”.

**4.7 Coarse Aggregate (APSS No. 108)**

The coarse aggregate shall be from hard granite crushed stone conforming to IS: 383-1970. The pieces of aggregate shall be non porous, hard, strong durable clean and free from clay, rounded in shape and shall have granular or crystalline non powdery surfaces. The aggregate shall be well graded. Tests where required shall be carried out in accordance with IS : 2386 - 1963.

**4.8 Steel Reinforcement (APSS No. 126)**

4.8.1 Reinforcement shall be free from pitting due to corrosion and free from loose rust, mill scale, paint, oil, grease, adhering earth etc. The over laps in the

reinforcement shall be as per IS: 456-2000 wastage in steel will be at the cost of contractor.

4.8.2 The contractor shall procure MS and HYSD rods required for the works, only from the main manufacturing steel units to the prescribed specification of Bureau of Indian Standards or equivalent and licensed to affix to ISI or other equivalent certifications, marks and acceptable to the Engineer-in-charge. The contractor should obtain and furnish from suppliers of steel, necessary ISI test certificate for every consignment of steel, before use on work.

4.8.3 Mild steel bars shall conform to Grade I of IS:432.

4.8.4 High yield steel strength deformed bars shall conform to IS:1786. Binding wire shall conform to IS:280.

4.8.5 Erected and secured reinforcement after fabrication shall be inspected and approved by the Engineer prior to placement of concrete.

**4.90 GLAZED TILES (APSS No. 121)**

The tiles shall be covered by a glaze on the top and under side. The edges shall be free from glaze in order that the tiles may adhere properly to the base. The glaze shall be uniform in quality and free from welts, ships, craze, specks, crawlings, or other imperfections visible from a distance of one meter. The glazed tiles shall be white or color and size of 300mm x 200 mm or 200mm x 100 mm with a thickness of 6mm. The tiles shall be true to shape and conform to the performance requirements of IS 777 and supplier shall submit a certificate with respect to the quality of tiles and detailed there in.

## 5.0 DETAILED SPECIFICATION OF WORKS

### 5.1 Standard

A high standard of workmanship in all trades will be required. The Contractor shall ensure that only skilled and experienced workmen are employed.

### 5.2 Supervision

5.2.1 The Contractor's supervising staff shall be fully qualified and experienced in the types of work being carried out under the supervision and shall be capable of ensuring that they are done well and efficiently.

### 5.3 Temporary works

Where required, the Contractor shall furnish such details of his temporary works as may be called for by the Engineer and the Contractor shall satisfy the Engineer as to their safety and efficiency. The Engineer may direct that temporary works, which he considers unsafe or insufficient, shall be removed and replaced in a satisfactory manner.

### 5.4 Codes

5.4.1 Unless mentioned otherwise, current versions of all codes, specifications and standards issued by the Indian Standards Institution and Indian Roads Congress, wherever mentioned, shall be fully applicable to these specifications. Where standards are not yet published by the ISI or IRC, adaptable British Standards or Specifications of the International Organization for standardization shall apply.

5.4.2 In case of any conflict in meaning between the specifications mentioned herein and those of ISI or IRC, the provisions of these specifications shall be prevail.

**5.5 Base lines and bench marks**

5.5.1 The Contractor shall establish and maintain, to the satisfaction of Engineer, the base lines and bench marks, based on which the works are set out. Where such base lines and bench marks are provided by the Engineer, the Contractor shall maintain these throughout the period of construction without causing any disturbance to them.

**5.6 Setting out**

5.6.1 The Contractor shall set out all the works to be executed by him, in line with the standard base lines, position and bench marks and truly as per drawings within the accepted tolerance limits at no extra cost to Owner. The Contractor shall be solely responsible for the setting out of all the works, to be executed by him and the approval of such setting out by the Engineer shall in no way absolve the Contractor of his responsibility for carrying the work to the true lines, levels and positions as per drawings.

**5.7 Dewatering**

5.7.1 The Contractor shall carryout all the works, in dry and workable condition and maintain the same in dry condition till the final handing over of works at no extra cost to the Owner. For this the Contractor shall make at his cost all the necessary provisions of dewatering, wherever necessary, to the full satisfaction of the Engineer.

**5.8 Safety of existing work**

5.8.1 Before taking up any construction adjoining other property or existing work, the Contractor shall take all steps necessary for the safety and protection of such property or work.

**5.9 Protection of existing services**

5.9.1 The Contractor shall take all precautions necessary to prevent damage to or interference with under-ground or over-ground services such as cables, drains, piping or piles, whether shown on drawings or not. Equipment etc., mounted in position shall be protected against falling debris etc., by means of tarpaulin or such other material.

**5.10 Handing over of work site**

5.10.1 On completion of work, the Contractor shall remove all rubbish, debris, surplus materials, temporary work etc., from the site. The site shall be handed over in a tidy and workmanlike manner.

**5.11 CRS Masonry in CM (1:6) in 1st sort (APSS 107 & APSS 611)**

5.11.1 The work shall consist of a facing of selected stones hammer dressed at faces and joints with only a small proportion of smaller stones in the hearting.

5.11.2 The face stones shall be set in regular courses of uniform thickness from bottom to the top throughout. The height of the course should be uniform throughout by using stones of same height. The face stones shall be laid in headers and stretchers alternately so as to break joint by atleast 75mm and headers shall project atleast 100mm beyond stretchers. The stones shall be solidly bedded, set full in mortar with joints not exceeding 12mm in thickness and shall extend well back into the hearting.

5.11.3 Bond stones shall be placed in the wall @ interval of 2m in length and 600mm in height and shall run through the wall if the wall is not more than 600mm thick. If the wall is more than 600mm thick line of headers shall be laid from face to back each header overlapping the other by atleast 150mm.

5.11.4 The heart portion shall be filled with good flat bedded stones set as close as possible, well set in mortar.

5.11.5 The work on interior face shall be precisely the same as on the exterior face unless the work is to be plastered in which case the side joints need not be vertical.

**5.12 Coursed Rubble Masonry in CM (1:6) 2nd sort: (APSS NO. 612)**

5.12.1 This work shall be executed similar to the specifications for C.R.S. masonry 1st sort with the exception that the hearting and backing shall conform to the standard specification for random rubble masonry and bond with the face stones being carried up continuously with the face work.

**5.13 RRS Masonry in CM (1:6) (APSS 107 & APSS 615)**

5.13.1 The face stone be hammered dressed on the face, side and the beds to enable to come into close proximity with the neighboring stone. Face stone shall be of not less width in plan than 150mm for walls of 400mm thick, 200mm for walls of 450mm thick. The face stone shall be laid in headers and stretchers alternatively so as to break joints by at least 75mm. Care is to be taken to break joints vertically.

5.13.2 Bond stones should built in the wall at intervals of 2m in length and 600mm in height and shall run through the wall if the wall is not more than 600mm thick. The heart portion shall be filled with good flat bedded stone set as close as possible, well set in mortar.

**5.14 Brick Work: (APSS 102 APS 501 & 504)**

5.14.1 All bricks to be used in the work shall be thoroughly soaked in water before use to prevent absorption of water from the mortar.

5.14.2 The bricks shall be set in cement mortar of **1:6** proportions by adopting a proper bond (preferably either English bond or a Flemish bond) throughout the wall.

5.14.3 Every course shall be truly vertical. Vertical joints of consecutive courses shall not come directly over one another. Vertical joints, in alternate course shall come directly over one another. Joint's shall be fully filled with mortar and raked. Every brick shall be laid with full joints of cement mortar on its bed, ends and side in one operation. No feeding of mortar by using excess water shall be allowed.

**Reinforced Half Brick Partition Walls**

5.14.4 All bricks to be used in the work shall be thoroughly soaked in water before use to prevent absorption of water from the mortar.

5.14.5 The cement mortor used for reinforced brick work shall be in cm (1:4) and mortor used shall conform APSS No. 115. Reinforcement for half brick walls shall be in the form of MS Bars and shall be of specified qualities. The brick shall be constructed only in stretcher bond. The reinforcement shall be well embedded in cement mortor at every third course and half the joint thickness of mortor shall first be laid and the other half laid after the reinforcement is placed in the position. The free ends of the reinforcement where ever possible shall be pegged into the mortor joints of main brick walls.

**5.15 Reinforced cement concrete (A.P.S.S. 402 & 403)**

5.15.1 Reinforced cement concrete shall correspond to **M20** grade as per **IS 456** - 2000 equivalent to (1:1.5:3) nominal mix proportion and **M25** grade design mix.

5.15.2 All R.C.C. work shall be carried out in strict accordance with latest IS specification. No concrete work shall be cast in the absence of the works-in-charge/Engineer. All the materials used should be of good quality as mentioned in **Sec. 4.0** above.

5.15.3 Reinforcement shall be steel and shall be free from corrosion, oil, grease or paint. Bars shall be hooked or bent accurately and placed in position as per design and drawing, and bound together tight using M.S. binding wire properly annexed tying wire.

5.15.4 Binders, stirrups, links should be securely wired to the main ring. Reinforcement shall be lap jointed or spliced only if unavoidable. Not more than 33% of the bars as specified in drawing shall be lapped at one section.

5.15.5 Proper cover shall be maintained between the reinforcement and the shuttering.

5.15.6 All cement concrete shall be machine mixed and machine vibrated.

5.15.6.1 The proportions of cement concrete specified in the above schedule are nominal and are indication of approximate proportion of cement, fine aggregate and coarse aggregate which may have to be altered suitably at site to obtain desired strength and workability. However, the quantity of cement shall not be less than specified below.

| Nominal Mix | Cement in bags of 50 Kgs per one Cubic metre (net) of cement concrete |
|---|---|
| a. 1:1.5 : 3 | 8.84 bags of 50 Kgs. |
| b. 1:2:4 | 6.62 bags of 50 Kgs. |
| c. 1:2.5:5 | 5.30 bags of 50 Kgs. |
| d. 1:3:6 | 4 42 bags of 50 Kgs. |
| e. 1:4:8 | 3.31 bags of 50 Kgs. |
| f. 1:5:10 | 2.65 bags of 50 Kgs. |
| g. 1:6:12 | 2.21 bags of 50 Kgs. |
| h. 1:8:16 | 1.66 bags of 50 kgs. |

5.15.7 The quantity of water shall be varied to suit the moisture content of the aggregate and shall be just sufficient in produce a dense concrete with workability. Workability should be checked at frequent intervals as per **IS:1199.**

5.15.8 The Cement and aggregates shall be mixed thoroughly in the specified proportion in a mechanical mixer until the mixture is of uniform colour. Where machine mixing is done the concrete shall be mixed, until the mixture is of uniform colour and, in no case, for less than two minutes.

5.15.9 Transportation, placing, compaction and curing of concrete.

5.15.9.1 After mixing, the concrete shall be transported from the mixer to the position of placing as rapidly as possible by appropriate mean without causing separation or segregation of concrete, maintaining the required workability.

5.15.9.2. Concrete shall only be placed after the Engineer has inspected the shuttering and reinforcement. The concrete shall be placed and compacted before initial setting of concrete commences and should not be subsequently disturbed.

5.15.9.3 Concrete after depositing should be compacted thoroughly by means of a mechanical vibration. Over vibration and under vibration of concrete are harmful and should be avoided. Use of polythene sheet is recommended above the shuttering to arrest the slurry loss through the shuttering joints while placing and compacting the concrete.

5.15.9.4 Concreting shall be carried out continuously upto construction joints already planned. Joint shall be kept where shear force is minimum.

5.15.9.5 Rigid supervision shall be maintained for curing the concrete after lying. All exposed faces of concrete shall be kept moist for a minimum period of 21 days by spraying water or using gunny bags.

**5.16 CENTERING (FORM WORK) AND SHUTTERING**

5.16.1 Shall be substantially and rigidly constructed of steel and shall be true to the dimensions described.

5.16.2 All joints shall be sufficiently tight to prevent leakage of cement grout.

5.16.3 All faulty joints shall be adequately caulked.

5.16.4 Shuttering shall be erected true to line and braced and strutted to prevent deformation under the weight and pressured wet concrete and constructional loads, wind pressure and other forces.

5.16.5 All faces of shuttering and moulds in contact with wet concrete shall be treated with a coat of oil to prevent adherence to concrete. Release agent should be applied so as to provide thin uniform to the forms without coating the reinforcement.

5.16.6 Centering and shuttering shall be removed after maturity gradually without jerking. Before removal of the shuttering the concrete shall be examined properly. Form shall not be released until the concrete has achieved a strength of atleast twice the stress to which the concrete may be subjected at time of removal of form work. In normal circumstances where ordinary portland cement is used and adequate curing is done, minimum striking may be as follows:

| | **Type of Formwork Striking Formwork** | **Minimum Period Before** |
|---|---|---|
| a) | Vertical formwork to columns, walls, beams | 16-24 days |
| b) | Soffit formwork to slabs (Props to be refixed immediately after removal of formwork) | 3 days |
| c) | Soffit formwork to beams (Props to be refixed immediately after removal of formwork) | 7 days |
| d) | Props to slabs : | |
| | 1) Spanning up to 4.5 m | 7 days |
| | 2) Spanning over 4.5 m | 14 days |
| e) | Props to beams and arches : | |
| | 1) Spanning upto 6 m | 14 days |
| | 2) Spanning upto 6 m | 21 days |

The number of props left under, their sizes and dispositions shall be such as to be able to safely carry the full dead load and live load likely to occur during construction.

**5.17 Cement Plastering in two coats CM 1:6 & CM 1:4 (APSS 901, 903 & 904)**

5.17.1 The surface of the wall shall be kept wet for 2 hours before plastering.

5.17.2 The Mortar in 1:6 proportion shall be dashed and pressed over the surface and then brought to smooth and uniform surface by means of float and trowel. The plaster shall be well pressed into the joints. Plaster shall be started from the top and worked down towards plinth. The work shall be tested frequently with a plumb bob and straight edge.

5.17.3 After the first coat the surface is left rough to receive the second coat. The final coat shall be applied a day or two after the first coat put on has set, but

the first coat shall not be allowed to dry. The final coat shall consist of 1 part of cement to 4 parts of fine sieved sand and shall be applied as in the first coat and brought to a uniform surface and then finished with a sponge to give granular appearance.

The finished surface shall be watered for a period of atleast 10 days.

**5.18 Water proof plaster over the roof**

5.18.1 On the clean wet surface of the concrete slab, before it has set, a layer of cement plaster shall be laid to give an average depth of 20mm over the concrete.

5.18.2 The Mortar to be used shall be of CM 1:3 proportion mixed thoroughly with a standard water proofing material with water repelling properties to ensure non-absorption.

5.18.3 Gauges should be put on the floor about ten feet apart to ensure even thickness.

5.18.4 Plastering must be done in squares or strips to avoid cracks. After the floor has been completed, it shall be covered with two inches of grass; sand or saw-dust and kept wet for three weeks.

**5.19 Pointing: (APSS - 906)**

5.19.1 Cement mortar for pointing shall conform to SS:115 and shall be of 1:3 proportion. The joints in the masonry shall be raked out to a depth not less than the width of the joint, when the mortar is green. Joints are to be brushed clean of dust and loose particles with a stiff brush. The area shall then be washed and the joints thoroughly wetted before pointing is commenced.

5.19.2 The mortar shall be pressed into the raked out joints according to the type of joint required. The mortar shall not be spread over the corners, edges or surface of the masonry. The pointing shall then be finished with proper tool. The superfluous mortar shall be cut off from the edges of the line and the surface of the masonry shall be cleaned of all mortar.

5.19.3 Pointing could be either flush pointing, or groove pointing.

**5.20 Flooring: (APSS 701 & 702)**

5.20.1 Flooring shall be with 25.4mm thick Shabad**/** Cuddapah polished stones laid over a bed of 100mm thick plain cement concrete in 1:5:10 proportion/RCC slab.

5.20.2 All the stones in one room shall be preferably of same width. The width of all the slabs in one row must be uniform with longitudinal joints parallel to each other.

5.20.3 The joint width shall be kept minimum and the sides of the slab shall be chisel dressed to ensure a correct joint.

**Granolothic concrete flooring (APSS No. 701 & 710)**

5.20.4 The mix proportions for the Granolothic concrete floor topping shall be (1:1:2) (Cement : F.A. : C.A) by volume. The minimum amount of water which will give necessary workability for adequate compaction shall be added. The grading of the course aggregate for Ganolothic concrete shall be from 6mm to 12mm. The finished thickness of flooring shall be 20mm and the panels into which the floor is divided for laying the Granolothic concrete shall not have any panel dimensions in excess of 1.5m.

5.20.5 Flooring in kitchens shall be with Rough Shahabad Stones laid over a bed of 100mm thick P.C.C. (1:5:10) proportion and pointed with CM : 1:3.

**5.21 Joinery:**

5.21.1 Aluminium doors, windows and glazing.

5.21.1.1 **Design** - Main members shall be designed to withstand a wind load of 100 M.P.H. wind pressure at 10m height. Deflections of millions at this wind load shall not exceed 1/175 of the clear span. Evidence of compliance of this design load shall be submitted with mathematical proof as and when required by the Engineer-in-Charge

5.21.1.2 **General.**

i) The unit assemblies shall be as per drawings or as directed by the Engineer-in-Charge.

ii) The unit assemblies that shall be anodized finish. Anodizing shall be minimum 20 to 25 microns thick of mat texture non directional and non-specular. Anodized surface shall be suitably protected during transportation, storage and erection.

iii) Sub units shall be joined together by concealed screws jamb member shall be self mulling type obviating use of separate mullions, thus increasing clear light of each unit.

iv) Joints shall either be mitred or coped mitred joints shall have extruded corner reinforcements. All joints shall be neat, hair line, and sealed with epoxy resin to make them water proof.

v) Openable shutters shall have a single row continuous neoprene or PVC weather strips to prevent infiltration. Weather strips shall not be interrupted by any fittings.

vi) All windows shall be glazed from inside with PVC rubber shalimar putty glazing beads shall be snap fit and shall be fitted without use of screws. No screws other then those on some of the hardware shall be visible.

vii) Glass in windows shall be 6 mm. thick and for doors and 6mm thick of clear ordinary quality unless otherwise specified in the schedule of quantities.

viii) Where glass of full length is not available in fixed glazing, the butt joint shall be provided with "U foam" insulation or Nylon packing as directed by the Engineer –in-Charge.

5.21.1.3 **Fixing**

Fixing shall be done as for steel windows mentioned above.

5.21.1.4 **Measurement** -

Measurement shall be in Sq.M of net area fixed at site.

**5.21.2 Flush shutters for doors:**

5.21.2.1 **Flush shutters (Double/Single):** should be factory made ISI marked confirming to IS 2202-1991 (part-I), 35mm thick with bond wood solid block board type core having cross bonds and face veneers hot pressed bonded with water proof phenol formaldehyde synthetic resin, with lipping on all sides.

5.21.2.2 **Construction:** The block board core shall confirm to the requirements specified in clause 7.1.1. of **IS** 2202 (Part I) : 1991. The frame constructed of stiles and rails shall be provided for holding the core. The width of the frame including internal lipping shall not be less than 45 mm and not more than 75 mm.

5.21.2.3 **Plywood:** used in flush door shutter shall confirm to IS 710:1976 with surface requirements confirming to type AB of IS 303:1989.

5.21.2.4 Cross-bands used in flush door shutter shall confirm to the requirements laid down in IS 710:1976.

5.21.2.5 **Face Veneers:** used in flush door shutters shall confirm tot he requirements laid down for veneer for BWP grade plywood in IS 710:1976.

5.21.2.6 All Plywood, cross – boards and veneer used shall be treated in accordance with clause 6.1.5.1. of IS 2202 (Part I) : 1991.

5.21.2.7 Adhesive used for bonding plywood or cross bond and face veneer to core shall be phenol formal dehyde synthetic resin adhesive confirming to BWP grade specified in IS 949:1974.

5.21.2.8 Internal lipping shall be of Teak wood and shall have a total depth not less than 25mm. It may be provided separately, when it is of species different from that of backing or as one piece with the style, designated as frame-cum-lipping, when internal lipping and backing are of the same species.

5.21.2.9 External lipping shall be of teak wood and shall be solid and shall measure at least 6mm on the face of the door. It shall be provided all round the shutter in case of single shutter and on three sides in case of double shutter.

5.21.2.10 In case of double leaved shutters, the sheeting of the stiles shall be rebated by 8mm to 10mm. The rebating shall be either splayed or square type as per clause 7.7 of IS 2202 (Part – I):1991. The depth of lipping at the meeting of stiles shall not be less than 30mm.

5.21.2.11 Shutter shall be shop prepared for taking mortise locks or latches as may be ordered.

5.21.2.12 Workmanship and the finish of the face panels shall be in confirmity with those specified in IS 303:1989

5.21.2.13 **Tests:** Knife test, glue Adhesion test, End Immersion test, slamming test shall be carried out as per clause 10 of IS 2202 (Part – I) 1991. The sampling and criteria for conformity, making etc. shall also be as per I IS 2202 (Part – I): 1991.

**5.21.4 Q.C. Clearance: The doors & windows (both frames & shutters) and ventilators should be got cleared by the Engineer/ Quality Control agency authorised by the Engineer-in-Chief. The tests will be conducted at the manufacturer's place and Q.C. clearance certificate will be issued for the lot before supply to site for use in construction. All the arrangements for testing at the manufacturer's place should be made by the contractor at his cost. No door, window or ventilator should be fixed without clearance of Engineer/ Q.C. agency. The contractor should inform the Engineer/Q.C. agency for testing and clearing at least 7 days in advance.**

**5.22 NOTES ON MASONRY**

i) All stones, bricks etc., used in the masonry work shall be thoroughly soaked in water before use to prevent absorption of water from the mortar.

ii) Stones shall be laid on their broadest faces which gives better opportunity to fill the faces between stones.

iii) To give sufficient lateral bond a stone in any course shall overlap the stone in the course below i.e. joints parallel to the pressure in two

adjoining course shall not lie too closely in the same vertical line. A minimum overlap of 6" shall be maintained.

iv) To give sufficient transverse bond, prescribed no. of headers shall be used.

v) The practice of building two thin faces, tying width occasionally through stones and filling up the middle with small stones or dry packing shall be strictly guarded against.

vi) Jambs for door and window opening shall be formed with quions of the full height of the course. The quion shall be of breadth atleast one and a half times the depth for the course and in length atleast twice the depth.

vii) It is advisable to erect the door and window frames first and build the masonry around.

viii) Thickness of the joint should not be more than 12mm.

ix) Every course of the masonry shall be truly vertical. Use of plumb bob to check verticality by the mason shall be encouraged.

x) Care should be taken to keep all corners and sides including door and window opening truly vertical.

**5.23 Notes on Pointing**

i) Flush pointing with a groove or a line appears neat and does not spoil the look of the stone or brick masonry.

ii) As far as possible a minimum amount of mortar shall be used to avoid wastage.

iii) The edges shall be neatly trimmed with a trowel and a straight edge.

iv) While mortar is green a groove shall be formed by running a tool along the center lines of the joints. This operation shall be continued till a smooth and hard surface is obtained.

v) Even the vertical joints shall be finished in a similar fashion.

vi) Even when the job is done carefully, there is always an amount of superfluous mortar sticking to the masonry. This should be wiped off with a wet cloth.

vii) After the work is set and dry i.e., after one or two days the stones shall be cleaned with a strong acid so as to remove the cement stains.

viii) After cleaning with acid the stones shall be cleaned with soap water to ensure natural colour of the stones.

ix) If care is taken as shown above the pointing work will look attractive and neat, and the natural appearance of the stone masonry is retained.

**5.24 Cover Blocks**

a) A cover block is used to separate the reinforcement from the shuttering before concrete is laid so that when the concrete is set the reinforcement is well within the concrete section at a distance from the outer surface, with specified cover to reinforcement.

b) This assures a proper bond between concrete and steel and the steel is prevented from rusting.

c) Normally a bottom cover of 12mm to 15mm is sufficient for slabs. For columns the cover should be about 40mm, and for beams it is 25mm.

d) A cover block has to be reasonably good for using in R.C.C.

e) The mortar for preparing cover blocks shall atleast be of proportion 1:2.

f) The block shall be prepared on a clean and level platform by spreading the mortar in the moulds of required size and depth.

g) When the mortar is still green strands of tying wire shall be inserted into each block. This wire is useful for tying the block to the reinforcement.

h) After 24 hours the block shall be removed from the mould and cured for about seven days.

i) A properly made cover block does not get crushed when the reinforcement is tied over it and during the concreting work.

j) Use of 20mm stone chips as cover for the reinforcement will not be accepted.

k) Use of mortar cover blocks ensures long life to steel and to the R.C.C. construction.

**5.25 Reinforcement chairs**

a) When the reinforcement is tied there is a need to separate bottom steel from the top steel and to maintain correct effective depth.

b) This is achieved by using reinforcement spacers or chairs.

c) For example if the slab is 100 mm thick the reinforcement section shall be 75 mm after allowing for top and bottom cover.

d) This could be achieved by making a reinforcement chair of slightly lesser size so as to accommodate the chair underneath the top steel,

e) The chair shall be minimum 450mm long and should have legs bent in opposite directions to ensure stability,

f) The chairs shall be placed on a cover block so that the legs do not stick out once the shuttering is removed.

g) Apart from ensuring separation to top and bottom steel the chair also takes the load of the movement of workers when concrete is being laid and the reinforcement work does not get disturbed.

h) Use of chairs in the reinforcement work is a good construction practice.

i) Use of large sized stones or bricks to separate top and bottom steel will not be accepted.

**5.26 Bearings of R.C.C. Slabs & Beams**

a) Where supports are not monolithic with the beam or slab the bearing surface shall be plastered with cement mortar 1:3 with the craft paper laid over the plaster, before laying the concrete.

b) The vertical face of the masonry rebate at bearings shall be plastered smooth with CM 1:3. For beams the craft paper shall be continued to the sides by folding the paper neatly to the plastered vertical face of the masonry opening.

**6.0 ADDITIONAL SPECIFICATIONS:**

**6.1 Anti Termite Treatment**

If the site is infected with white ants, all the ant hills shall be dug out completely and queen ants destroyed. Antitermite treatment, before

construction in foundation and basement where required shall be done as per I.S. code 6313 Part II 1971.

Chemicals used, the relevant I.S. specifications for the same and their usual concentrations as water emulsions for soil treatment are given in table 201 A of S.S. 201 APSS as furnished below.

**6.2 Blasting Operations**

Blasting operations when considered necessary shall be resorted to only with written permission of the Engineer-in-charge. Prior inspection shall be carried out for the safety and stability of the public property. Blasting operations in the proximity of over head power lines, communication lines, or other structures shall not be carried until the operator or the owner of both of such lines have been notified and precautionary measures deemed necessary shall be taken as per the procedure laid down in S.S. No. 203 APSS and code 4081-1967 shall be followed.

**6.3 a) Expansion Joints**

Structures in which marked changes in plan dimension take place abruptly shall be provided with expansion joint at the section where such changes occur. Expansion joint shall be so provided that the necessary movement occurs with a minimum resistance at the joint. The structures adjacent should preferably supported on separate columns of walls but not necessarily on separate foundations reinforcement shall not extend across an expansion joint and the break between the sections shall be complete. The details as to the length of a structure where expansion joints have to be provided can be determined after taking into consideration various factors such as temperature exposure to weather etc. For the purpose of general guidance however it is recommended that structure exceeding 45M in level to shall be decided by one or more expansion joints (SS No. 403.8 & IS 456).

b) **Construction Joints**

Vertical joints in floor and roof slabs shall be provided in the case of long building of more than 30M in length specially when the width or depth of such buildings are less than 15M and when narrow corridors connect blocks of relatively greater width. The most suitable position for such vertical joints are where the corridors take off from inner blocks. On soils such as black cotton, such joints are more essential shall be invariably provided at the places shown in the drawing or as directed by the Engineer-in-charge. Construction joints when necessary shall be located as follows.

In the main beam over the centre of support. No vertical joint shall be permitted in case of main beams. In other cases they shall be provided if necessary in the following location.

i) In subsidiary beams at mid span.

ii) In the case of slabs the joints wherever possible shall be parallel to main reinforcement. In the case of one way reinforced slabs and over the centre of supporting beams or walls in other cases. In general the joints shall not be provided in locations of considerable shear or under concentrated loads.

Suitable water stops as specified shall be provided in the case of water retain structures (SS No. 403.7).

**6.4 Load testing of structures**

Load testing of structures shall conform to SS No. 403 APSS. Load tests on completed structures shall be made of required by the specifications or

condition of contract or by the Executive Engineer in the event of reasonable doubt as to the adequacy of the strength of the structure. Such tests shall be carried out after expiry of 56 days of effective hardening of the concrete test loading of structures, allowable deflections, recovery of deflection etc., shall be as per clause 17.6 of IS: 456-2000.

6.5 Theoretical requirement of cement should be as follows:-

| | | |
|---|---|---|
| a. | C.R.S. Masonry in C.M. (1:6) | 1.54 bags per Cum |
| b. | C.R.S. Masonry in C.M. (1:8) | 1.15 bags per Cum |
| c. | Brick Masonry in C.M. (1:4) | 1.44 bags per Cum |
| d. | Brick Masonry in C.M(1:6) | 0.96 bags per Cum |
| e. | Brick Masonry in C.M. (1:8) | 0.72 bags per Cum |
| f. | 12 mm plastering in C.M. (1:5)& C.M. (1:3) | 1.02 bags per 10 Sqm. |
| g. | 12 mm plastering in C.M. (1:6)& C.M. (1:4) | 0.82 bags per 10 Sqm. |
| h. | 20 mm plastering in C.M. (1:6)& C.M. (1:4) | 1.15 bags per 10 Sqm. |
| i. | 12 mm plastering in C.M. (1:4) | 1.08 bags per 10 Sqm. |
| j. | 12 mm plastering in C.M. (1:6) | 0.72 bags per 10 Sqm. |

## 7.0 ELECTRIFICATION WORKS (TECHNICAL SPECIFICATION)

## 7.1 INTERNAL ELECTRIFICATION

### 7.1.1 WIRING INSTALLATION :

### 7.1.1 Scope

The scope under this section covers wiring installation comprising of

a) Lighting/Fan/Exhaust Fan/Circuit bell points.

b) Power circuits and Air Condition circuits.

c) Circuit wiring.

### 3.06.1.2 Standards

The following IS standards are applicable.

a) IS:**732-1989**- Code of Practice for Electrical Wiring Installation

b) **IS:1646-1961**- Code of Practice for Fire Safety of Buildings (General Electrical Installation)

c) **IS:3206** Part I, - Code of practice principles Part II and Part III for good lighting and aspects of design, schedule for values of illumination and glare level and calculation of coefficient of utilization

d) **IS:1985**- National electrical code

e) **IS: 2509** - Specification for rigid non-metallic conduits for electrical installations.

f) **IS:3419-** Specification for fitting for rigid non-metallic conduits.

g) **IS:5133- Part-I-1969** - Specification for boxes for enclosure of electrical accessories.

h) **IS:694-1977**- Specifications for P.V.C. insulated cable for working voltage upto and including 1100 volts.

i) **IS:8130-1984-** Specifications for conductors for insulated electrical cables and flexible cords.

j) **IS:5561-** Electrical power connectors

k) **IS:3854-1966-** Flush type switches

l) **IS:1293-1967-3** pin plug sockets

m) **IS:6538-1978-** Plugs

**7.1.1.3 Recessed conduit wiring system with rigid conduits**

a) **Type and size of conduit:** All rigid non-metallic conduits shall conform to accepted standards and shall be used to corresponding accessories. Conduits shall provide adequate mechanical protection for the enclosed cables and the interior of the conduit shall be free from obstructions. No non-metallic conduit less than 20 mm in dia shall be used. The number of insulated cables that can be drawn into rigid non-metallic conduits are given in separate enclosure (Ref: Exhibit (1) page no. 161)

b) **Bunching of cables:** Unless otherwise specified, insulated conductors of AC supply shall be bunched in separate conduits. For lighting and small power outlet circuits, phase segregation in separate conduits is recommended.

c) **Conduit Joints:** Conduits (metallic) shall be joined by means of screwed couplers and screwed accessories only. In long distance straight runs of conduit, inspection type boxes at reasonable intervals shall be provided. Cut ends of conduit pipes shall have no sharp edges nor any burrs left to avoid damage to the insulation of conductors while pulling them through such conduits.

d) Inspection type conduit fittings such as inspection boxes, deep boxes, bends, elbows and tees shall be so installed that they remain accessible for such purposes as withdrawal of existing cables or installation of additional cables.

e) Metallic switch board boxes shall be fabricated from **1.6mm** thick sheet metal of 16 gauge GI sheet and wooden switch board boxes shall be of Teakwood. The switch boards should be flush mounting type. The MS switch boards should be painted with two coats of red oxide and two coats of synthetic enamel paint of approved grade and make before fixing in position.

The switch boxes should be covered with **3mm** thick plate switches.

f) The chase in the wall shall be neatly made and be of ample dimensions to permit the conduit to be fixed in the manner desired. In the case of building under construction chases shall be provided in the walls, ceiling etc., at the time of their construction and shall be filled up neatly after erection of conduit and brought to original finish of the walls.

g) The conduits shall be fixed in chases by means of staples or saddles not more than **60 cms** apart. Fixing of standard bends or elbows shall be avoided as far as practicable and all curves maintained by bending the conduit pipe itself with a long radius which will permit easy drawing in of conductors. All threaded joints of rigid steel conduit shall be treated with some approved preservative compound to secure protection against rust.

h) Suitable inspection boxes shall be provided to permit periodical inspection and to facilitate removal of wires if necessary. Minimum size of inspection boxes shall be **75** x **75 mm**.

i) The **PLATE** switch board boxes, junction boxes etc., should be efficiently earthed with conduit by a suitable means of earth attachment.

j) When crossing through expansion joints in Buildings, the conduit section across the joint may be through flexible conduits of same size as the rigid conduit.

k) **Wires:** Wires shall comply with the following features.

- Annealed copper conductor, multi strand, PVC insulated, **1100** volts grade cables.
- The following colour coding shall be followed :-

  Phases - R - Red

  - Y - Yellow

  - B - Blue

  Neutral - Black

  Earth - Green

l) On each (lighting/Ceiling Fan/Exhaust Fan) circuit not more than **10** points or **800-1000** watts load should be connected. Example:- If on one of the switch boards there are only **4** switches to control **4** lights, other switch board another **3** switches to control lights etc, then for arriving at the circuit length, the shortest distance from circuit breaker in the MCB distribution board to the nearest switch board shall be considered. Inter connections between such switch boards shall be allowed by providing same wires as are used for light points and no measurement in circuit wiring is allowed for such inter connections. A separate conduit pipe has to be provided for running circuit mains and the conduits for light points shall never be used for the same.

For **5** Amps **3** pin on separate board the circuit measurement to first nearest **5** Amps **3** pin socket is considered. No measurement will be separately considered for looping of switch boards in circuit wiring.

m) The mounting height of switch boards (bottom of MS Box) shall be **4'6"** from finished floor level.

The **5** Amps/**15** Amps **3** pin sockets with shutter protection shall be at 1 mtr. level.

Wiring for power circuits i.e., **15** Amps **3** pin and AC points shall be provided in separate conduit pipes.

n) Neat holes shall be punched on MS Switch Board Boxes for conduit pipe entries. Rough, burred holes with chisel shall be avoided. Conduit pipes to be fixed to MS Switch Board Boxes, MCB Distribution Boards etc., by providing check nut arrangement. Before drawing of PVC insulated cables inside the conduits, ebonite/ nylon bushes to be provided at conduit ends in order to avoid damage to cables during drawing.

o) **3mm** thick **PLATE TYPE** covers provided on Switch Boards should be fixed to switch boards by providing brass coated screws and plated cup washers.

p) Any loose holes on Switch Board boxes and Distribution Boards shall be properly closed, so as to prevent entry of lizards etc.

q) Whenever cables of size **6.0** Sq.mm and above are connected inside switch, socket or MCB, metallic plug point etc, proper type and size of lugs to be crimped to cable leads before making the permanent connection in switches etc.

r) Telephone/intercom cables shall not be laid in the same conduit where electric lighting/power cables are drawn. Separate conduit pipes to be used for drawing of telephone/intercom cables.

s) Loose joints with PVC insulation shall be avoided. Wherever possible joints of cables shall be avoided. If found necessary proper type and size of connectors shall be used.

t) The drop of voltage between the main switch/distant. Terminals and the farthest current consuming apparatus shall not exceed **2%** with all devices switched on.

**7.1.1.4 Testing**

The entire installation shall be tested for

a) Insulation Resistance

b) Earth continuity

c) Polarity of single pole switches

Tests shall be conducted in the presence of Site Engineer. Test results to be tabulated and submitted to the site engineer.

**7.1.2 MCB DISTRIBUTION BOARDS, MCB's AND ELCB's:**

**7.1.2.1 Scope**

The scope under this section covers installation comprising.

a) Low voltage distribution boards

**7.1.2.2 Standards**

The following IS Standards are applicable

a) IS:**8623**- Factory built assemblies

b) IS:**2147-IP42**- Protection against ingress

c) IS:**8828**-BS-**3871**- Specification for Miniature **CEEP-19 CIRCUIT** breakers

d) IP **20**- Protection category

e) IS:**12640-1980**- Residual current operated circuit breakers

**7.1.2.3 Details**

a) The Distribution Boards shall have vermin, dust, rust proof painting done by powder coating process.

b) The cables entering the D. Board should be properly bunched and dressed before making connection in MCB's.

c) Cable glands shall be provided wherever armoured cables are connected to switches.

d) Cable leads shall be provided with proper type and size of lugs crimped to leads before making permanent connection inside MCB's.

e) Permanent circuit identification shall be provided on the distribution boards.

f) The mounting height of MCB distribution boards etc., (bottom line) shall be **6'-6"** from finished floor level.

**7.1.3 CABLES:**

**7.1.3.1 Scope**

The scope under this section covers.

a) Power cables

### 7.1.3.2 Standards

a) IS:**8130** - Specification for conductors for insulated electric cables

b) IS:**1554**-Part I - Specification for Armoured/unarmoured power cables

c) IS:**3961** - Recommended current ratings for cables

d) IS:**5831-1984** - Specifications for PVC insulation and sheating of electric cables

### 7.1.3.3 General requirements for cables

a) Cables should be stranded aluminium conductors for **6mm** thick and above.

b) L.V. cables shall be **1100** Volts grade.

c) Cables shall have colour coded insulation.

d) PVC inner and outer sheathing shall be applied by extrusion.

e) Steel armouring between inner and outer sheathing.

f) The PVC insulation and sheathing shall confirm to IS:**5831-1984.**

g) The armouring for cables upto **16mm²** shall be of round steel wire and that above **16mm²** shall be of galvanised steel strings.

### 7.1.3.4 Laying of Cables

a) Cables if laid underground shall be at a depth of not less than **60** Cms, in a trench. Sand filling shall be provided at the bottom of trench before laying the cable. Bricks shall be provided on either side of the laid cable. Sand filling shall be done to cover the cable laid. Bricks shall be provided on the top. Earth filling shall be done.

M.S. cable identification tags, route indicators embedded in C.C. are to be provided at every 8 meters length of cable laid.

b) Hume pipe, trenches/tunnels with proper precast slabs to withstand wear and tear of vehicular traffic shall be provided at road crossings.

c) Cables if laid in the air shall be laid on cable trays and shall be properly clamped to the trays by plated MS. saddles at proper intervals. Cables shall be properly dressed before fixing on the cable trays.

d) Extra cable loops of minimum **500 mm** shall be provided at each end of cables laid.

e) Cables shall be bent to a radius of **20** times the diameter of the cable with a minimum of **10** times diameter at restricted space.

f) Control/Telephone cables shall be laid away from power cables on separate cable trays.

### 7.1.3.5 Testing

Manufacturers test report shall be submitted for tests on cables in accordance with Indian standards specifications.

Cables shall be tested after installation before commissioning by using 1000 Volts Megger and the following readings shall be obtained and tabulated.

- Continuity on all conductors

- Insulation Resistance

a) between conductors

b) all conductors and ground

The tests shall be conducted in the presence of Site Engineer and results submitted.

**7.1.4 CABLE TRAYS AND ACCESSORIES:**

**7.1.4.1 Scope**

The scope covers MS cable trays and cable tray accessories.

**7.1.4.2 Standards**

(IS. specifications shall be adhered to)

**7.1.4.3 Specifications**

Material: Hot rolled plain sheets of tested quality "O" grade as per IS **1079.**

Thickness of material: **2.0 mm**

Cable loading on tray: **50** Kg/MTR

Span between cable tray supports: **1.5** meters to **2.0** Mtrs.

Surface finish: Hot dip galvanising iron as per IS **2629,** minimum **70** microns thickness

Length of cable trays: **2.5** Meters

Width of Cable trays: (outside to outside width to be taken)

a) Ladder type - Bolted/welded construction **300** mm/450mm/600mm (depending on number of cables to be laid)

b) Perforated cable trays (Same as above)

**7.1.4.4 Sizes of Cable Trays:**

a) Ladder type - Bolted/welded construction

Side rail

* Flange width **15** mm

* Depth **70** mm

* Two coupler holes of **10**mm diameter required on each side of side rail

* Rungs

* Channel section: **20** x **40** x **20** mm

* Slot size on rungs: **20** x **10** mm (oblong holes)

* Interval between rungs not more than **250m**m

b) Perforated type construction

* Flange width : **30**mm

* Slot size : **20** x **10** mm (oblong)

**7.1.4.5 Sizes of Coupler Plates:**

a) Ladder type - Bolted/welded construction

Size: **90** x **45** mm

Thickness of material: **2/2.5** mm

Slot size: **20** x **10**mm oblong holes - Two numbers

Round holes **10**mm diameter Two numbers

Finish: Hot dip galvanised as per IS:**2629**

b) Perforated type construction:

Size: **210**mm x **25**mm

Thickness of material: **3**mm

Slot size: Oblong holes **20** x **10** mm - **2** numbers

Round holes **10**mm diameter - Two numbers

Finish: Hot dip galvanised as per IS:**2629**

**7.1.4.6 Hardware for coupler plate:** (Electro galvanised)

a) Hexagonal Head Bolts - **4** Nos.

b) Plain washers **- 8** Nos.

c) Hexagonal nuts - **4** Nos.

Number of coupler plates per cable tray - Two numbers.

**7.1.4.7 Cable tray Accessories:**

Material: Hot rolled plain sheets of tested quality "O" grade as per IS: **1079**

Finish: Hot dip galvanised as per IS:**2629**

Minimum bending Radius - **450**mm

Tees, Horizontal/vertical elbows, cross and reducers for both ladder type - welded/bolted and perforated construction shall be as per standard manufacturers drawings.

**7.1.4.8 Erection**

Cable trays shall be erected on walls, trenches (if necessary) by drilling holes in the wall by power drilling machine. Cable Tray shall be fixed to wall by providing proper size Anchor expandable type bolt and nut arrangement.

Proper type of cable tray accessories shall be selected depending on the site condition.

**7.1.5 CABLE TERMINATION:**

Cable gland body shall be made of brass castings and machined to final size. The general construction of the glands should be as per standard manufacturer's drawings. It mainly consists

a) Compression Nut - Brass - **1** No.

b) Gland body with Hexagonal head - Brass - **1** No.

c) Rubber Ring - Rubber - **1** No.

d) Brass washers - Brass - **3** Nos.

e) Check nuts - Brass - **1** No.

Metal parts of the gland shall be free from blow holes and surface shall be machined smoothly.

All edges shall be deborred and then nickle plated wherever necessary. The cable glands shall be of single compression type.

## 7.1.6 L.T. PANEL BOARDS

**Scope**

The Scope covers the requirement of designs, construction, assembly, testing, Supply and installation of Panel Boards.

### 7.1.6.2 Standards:

IS:**4064** - Code of practice for switch of use units.

IS**:4237** - General requirements for switch gear and control gear for voltages not exceeding **1000** Volts.

IS:**2147** - Degree of protection provided by enclosures for low voltage switch gear. for cables

IS:**9224** - Fuse class category

IS:**2705** - Current Transformers

IS:**1248** - Indicating Instruments

IS:**375** - Marking and arrangement of bus bars

IS:**3156** - Voltage Transformers

IS:**3231** - Relays

IS:**722** - Integrating Information

IS:**6975** - Control Switches and push buttons

IS:**2959** - Auxiliary contactors

IS:**2516** - Air circuit Breakers

IS:**2208** - HRC fuse links

IS:**8623** - Factory Built Assemblies of switch gear and control gear.

### 7.1.6.3 CONSTRUCTION:

The panel board shall be:

i) of the metal enclosed, indoor, floor mounted, free standing type.

ii) be made up of the requisite vertical sections, which, when coupled together shall form continuous dead front switchboards.

iii) provide dust and damp protection, the degree of protection being no less than IP, **51** to IS. **2147**.

iv) be readily extensible on both sides by the addition of vertical sections after removal of the end covers.

**7.1.6.3.1** The panel boards shall be constructed only of materials capable of withstanding the mechanical, electrical and thermal stresses, as well as the effects of humidity, which are likely to be encountered in normal service.

**7.1.6.3.2** Each vertical section shall comprise:

i) A front framed structure of rolled/folded sheet steel channel section, of minimum **2** mm thickness, rigidly bolted together. This structure shall house the components contributing on the major weight of the equipment, such as circuit breaker fuse switch units, main horizontal busbars, vertical risers and other front mounted accessories.

The structure shall be mounted on a rigid base frame of folded sheet steel of minimum **2mm** thickness and **100mm** height. The design shall ensure that the weight of the components is adequately supported

without deformation or loss of alignment during transit or during operation.

ii) A cable chamber housing the cable and connections, and power/control cable terminations. The design shall ensure generous availability of space for ease of installation and maintenance of cabling, and adequate safety for working in one vertical section without coming into accidental contact with live parts in an adjacent section.

iii) Front and rear doors fitted with dust excluding neoprene gaskets with fasteners designed to ensure proper compression of the gaskets. When covers are provided in place of doors, generous overlap shall be assured between sheet steel surface with closely spaced fasteners to prelude the entry of dust.

**7.1.6.3.3** The height of the panel should not be more than **2400** mm. The total depth of the panel should be adequate to cater for proper cabling space.

**7.1.6.3.4** Doors and covers shall be minimum **1.5**mm thick sheet steel. Sheet steel shrouds and partitions shall be of minimum **1.5**mm thickness. All sheet steel work forming the exterior of switch boards shall be smoothly finished, levelled and free from flaws. The corners should be rounded.

**7.1.6.3.5** The apparatus and circuits in the panel boards shall be so arranged as to facilitate their operation and maintenance and at the same time to ensure the necessary degree of safety.

**7.1.6.3.6** Apparatus forming part of the panel boards shall have the following minimum clearances:

i) Between phases - **25** mm

ii) Between phases and earth - **25** mm

iii) Between phases and earth - **25** mm

iv) Between neutral and earth - **19** mm

When, for any reason, the above clearances are not available, suitable insulation shall be provided. Clearances shall be maintained during normal service conditions.

Creepage distances shall comply to those specified in relevant standards.

**7.1.6.3.7** All insulating material used in the construction of the equipment shall be of non-hygroscopic material, duly treated to withstand the effects of high humidity, high temperature tropical ambient service conditions.

**7.1.6.3.8** Functional units such as circuit breakers and fuse switches shall be arranged in multi-tier formation, except that not more than two air circuit breakers shall be housed in a single vertical section.

**7.1.6.3.9** Metallic/insulated barriers shall be provided within vertical sections and between adjacent sections to ensure prevention of accidental contact with:

i) Main busbars and vertical risers during operation, inspection or maintenance of functional units and front mounted accessories.

ii) Cable terminations of one functional unit, when working on those of adjacent unit/units.

**7.1.6.3.10** All doors/covers providing access to live power equipment/circuits shall be provided with tool operated fasteners to prevent unauthorised access.

**7.1.6.3.11** Provision shall be made for permanently earthing the frames and other metal parts of the switch gears by two independent connections.

**7.1.6.4 METAL TREATMENT AND FINISH:**

**7.1.6.4.1** All steelwork used in the construction of the switchboards, should have undergone a rigorous metal treatment process as follows: (Seven tank process.)

i) Effective cleaning by hot alkaline degreasing solution followed by cold water rinsing to remove traces of alkaline solution.

ii) Pickling in dilute sulphuric acid to remove oxide scales and rust formation, if any, followed by cold water rinsing to remove traces of acidic solution.

iii) A recognised phosphating process to facilitate durable coating of the paint on the metal surface and also to prevent the spread of rusting in the event of the paint film being mechanically damaged. This again, shall be followed by hot water rinsing to remove traces of phosphate solution.

iv) Passivating in de-oxalite solution to retain and augment the effects of phosphating.

v) Drying with compressed air in a dust free atmosphere.

vi) Primer coating with two coats of a highly corrosion resistant primer, applied wet on wet & stove dried under strictly controlled conditions of temperature and time.

vii) A finishing coat of stoving synthetic enamel paint to the specified shade of IS.**5.** The total thickness of paint should not be less than **15** to **20** microns.

**7.1.6.5 BUS BARS:**

**7.1.6.5.1** The busbars shall be air insulated and made of high conductivity, high strength aluminium alloy complying with the requirements of grade E91 of IS 5082.

**7.1.6.5.2** The busbars shall be suitably braced with non-hygroscopic SMC supports to provide as through fault withstand capacity of **50** KA RMS symmetrical for one second and a peak short circuit withstand of **105** KA minimum. The neutral as well as the earth bar should also be capable of withstanding the above fault level. Ridges shall be provided on the SMC supports to prevent trackling between adjacent bus bars.

**7.1.6.5.3** Large clearances and creepage distance shall be provided on the busbars system to minimise the possibility of a fault.

**7.1.6.5.4** High tensile bolts and spring washers shall be provided at all busbar joints.

**7.1.6.5.5** The cross sections of the busbars risers for various ratings shall have been decided on the basis of temperature rise tests carried out under conditions closely similar to actual service conditions. For a total operating temperature of **110** deg. C. at an ambient of **40** deg. at the standard current ratings and corresponding cross sections of the main busbars should be such that the bus bar shall carry **1** Amp. per Sq.mm.

**7.1.6.5.6** The main phase busbars shall have continuous current rating throughout the length and the neutral busbars shall have a continuous rating of atleast 50% of the phase busbars.

**7.1.6.5.7** Connections from the main busbars to functional circuit shall be arranged and supported so as to withstand without any damage or deformation the thermal and dynamic stresses due to short circuit currents.

**7.1.6.5.8** Busbars shall be colour coded for easy identification of individual phases and neutral.

**7.1.6.5.9** The busbars shall be suitably supported with epoxy resin mould insulators.

**7.1.6.6 Tests:**

The panel Board shall be inspected as per relevant standards in presence of the Site Engineer and shall include.

a) High voltage test

b) Insulation test

c) Constructional and safety features

**7.1.6.7 Name Plates:**

Main name plates shall be fixed at the top centre. Name plate giving feeder detail shall be provided and are to be fixed by screws.

**7.1.7 Earth Electrodes:**

**7.1.7.1 Scope:**

The scope included both pipe earth electrodes and plate earth electrodes.

**7.1.7..2 Standards:**

IS:**3043** - Code of practice for earthing

**7.1.7.3 Construction:**

**Pipe Earth Electrode** : G.I pipe shall not be less than **38**mm diameter and **2½**meter long. It shall be buried vertically into the earth pit with the top not less than **1.25** meters below ground level. The G.I pipe should be “C” / "B" class type.

**Plate Earth Electrode**: Copper plate of **600** x **600** x **3.15** mm. shall be burried in the earth with faces vertical and top shall not be less than **1.5** metres below ground level.

The electrodes shall be surmounted by alternate layers of charcoal or coke and salt. Watering arrangement with **½”** G.I pipe with a funnel shall be provided, the later being housed with chamber (masonry) of inner size **300** x **300** mm CI hinged cover with CI frame to be provided on the top.

**7.1.7.4 Tests:**

The resistance of earth electrode shall be less than 5 Ohms.

**7.1.1.3 (a) Exhibit (I)**

**MAXIMUM PERMISSIBLE NUMBER OF 250 V GRADE SINGLE-CORE CABLES THAT CAN BE DRAWN INTO RIGID NON-METALLIC CONDUITS**

**(Ref.IS:732-1989)**

| Size of Cable | | Size of Conduit, mm | | | | | |
|---|---|---|---|---|---|---|---|
| Nominal Cross-Sectional Area mm² | Number and Diameter of Wires, mm | 16 | 20 | 25 | 32 | 40 | 50 |
| | | Number of Cables, Max | | | | | |
| 1.0 | 1/1.12* | 5 | 7 | 13 | 20 | - | - |
| 1.5 | 1/1.40 | 4 | 6 | 10 | 14 | - | - |
| 2.5 | ¦ 1/1.80 ¦ ¦ 3/1.06* ¦ | 3 | 5 | 10 | 14 | - | - |



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| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 4 | ¦ 1/2.24 ¦<br>¦ 7/0.85* ¦ | 2 | 3 | 6 | 10 | 14 | - |
| 6 | ¦ 1/2.80 ¦<br>¦ 7/1.06* ¦ | - | 2 | 5 | 9 | 11 | - |
| 10 | ¦11/3.55+¦<br>¦ 7/1.40* ¦ | - | - | 4 | 7 | 9 | - |
| 16 | 7/1.70 | - | - | 2 | 4 | 5 | 12 |
| 25 | 7/2.24 | - | - | - | 2 | 2 | 6 |
| 35 | 7/2.50 | - | - | - | - | 2 | 5 |
| 50 | ¦ 7/3.00+¦<br>¦19/1.80 ¦ | - | - | - | - | 2 | 3 |

* For copper conductors only.
\+ For aluminium conductors only.

**7.3 Contractor**

The contractor for electrical works executing substations, earthing, transformers, lighting shall hold valid class 'A' licence issued by Andhra Pradesh electrical licensing board and he should submit his license copy and his previous experiences along with the tenders and also before commencement of work The contractor shall quote and employ the number of electrical engineers, supervisors, wiremen in his position for such type of works, who are processing necessary permits/certificates/licences.

Liquidated damages will be levied in case of failure to complete the job in time as per standard clauses.

The contractor shall be able to read the drawings and prepare the drawings as per site conditions and any modifications necessary for submissions to the electrical inspector.

The contractor shall have laison with APSEB officials and electrical inspectors, get the drawings and installations approved and also getting power supply released from APTRANSCO.

It is complete responsibility of the contractor to get the electrical inspector's approval, including and getting power supply. He should have good laison with APTRANSCO.

The contractor can strictly follow the drawings and specifications for carrying out the works and he can get clarifications from site Engineer or Consultant in case of doubt. The contractor shall submit three copies of drawings for conduit routing inside the building, which he is going to carry out and submit one copy to consultant and two copies to client. This also includes the point wiring, telephone point conduit and power plug wiring conduit for each building before starting of the civil works atleast one week in advance for verification by client and consultant. The contractor shall also give the total quantity of various conduit to be used in each building before starting work as well the quantity of various wires to be used for each building.

The contractor should plan properly for all electrical material and works entrusted to him 15 days in advance and in form the Engineer in charge and

consultant about his work progress. He should co- ordinate with civil persons for recessed conduict laying and also switch boards recessed fixing and any other electrical work associated with civil works as well as electrical Engineer at site and electrical consultant for his works.

**7.4 Standards for Electrical Equipment**

7.4.1 Unless otherwise stipulated in this specification, all equipment or material covered under this specifications shall be designed, manufactured and tested in accordance with the latest standards of Indian Standard's specifications.

7.4.2 All equipment shall conform to latest Indian electricity Rules, Indian electricity act and Indian Insurance rules as regard safety, earthing and other essential provisions specified in for installation and operation of electrical equipments.

7.4.3 Extreme care shall be taken to make enclosures for switch gears proof against rodents, lizards and other creeping vermin.

7.4.4 Continuity of power supply is to be given maximum consideration and the design of the equipment shall be such as to simplify inspection maintenance and testing at site. The design shall include all reasonable precautions and provisions for safety of operating personnel and maintenance personnel.

**7.5 COMPLETETION CERTIFICATE:**

7.5.1 After completion of the work the contractor should submit completion certificate in the following format:

I/We certify that the installation detailed below has been installed by me/us and tested and that to the best of my/our knowledge and belief, it complies with Indian Electricity Rules 1956.

Electrical installation at .....

Voltage and system of supply .....

Particulars of Works :

a) Internal Electrical Installation.

| | No. | Total Load | Type of system of wiring |
|---|---|---|---|
| i) | Light point | | |
| ii) | Fan point | | |
| iii) | Plug point | | |
| | 3-pin 5 A | | |
| | 3-pin 15 A. | | |

| b) Others | Description | hp/kW | Type of starting |
|---|---|---|---|
| 1) Motors : | | | |
| i) | | | |
| ii) | | | |
| iii) | | | |
| 2) Other plants: | | | |

c) If the work involves installations of over head line and/or underground cable.



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1) i) Type and description of overhead line

ii) Total length and No. of spans

iii) No. of street lights and its description.

2) i) Total length of underground cable and its size

ii) No. of joints :

End joint :

Tee Joint :

Straight through joint :

Earthing:

i) Description of earthing electrode

ii) No. of earth electrodes

iii) Size of main earth lead

Test Results:

a) Insulation Resistance

i) Insulation resistance of the whole system of conductors to earth ..... Megaohms.

ii) Insulation resistance between the phase conductor and neutral.

Between phase R and neutral ..... Megaohms

Between phase Y and neutral ..... Megaohms.

Between phase B and neutral ..... Megaohms

iii) Insulation resistance between the phase conductors

Between phase R and neutral ..... Megaohms.

Between phase Y and neutral ..... Megaohms

Between phase B and neutral ..... Megaohms

b) Polarity test :

Polarity of non-linked single pole branch switches.

c) Earth continuity test :

Maximum resistance between any point in the earth continuity conductor including metal conduits and main earthing lead ..... Ohms.

d) Earth electrode resistance :

Resistance of each earth electrode.

i) ..... Ohms.

ii) ..... Ohms.

iii) ..... Ohms.

iv) ..... Ohms.

e) Lightning protective system.



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Resistance of the whole of lightning protective system of earth before any bonding is effected with earth electrode and metal in/on the structure ..... Ohms.

| **Signature of the Supervisor** | **Signature of Contractor** |
|---|---|
| Name and Address | Name and Address |
| ..... | ..... |
| ..... | ..... |
| ..... | ..... |

* * *



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**7.4.7 LIST OF MAKES FOR WATER SUPPLY AND SANITARY ITEMS TO BE USED.**

| Sl.No. | Item | Makes Suggested |
|---|---|---|
| 1. | CI pipes | **BPL/ INDO/ BIC/ BIF/** or any other equivalent ISI marked makes. |
| 2. | GI pipes | **Zindal / Tata / Zenith /** or any other equivalent ISI marked makes. |
| 3. | GM Valves | **Zolota / Zenith / Anchor** or any other ISI marked makes. |
| 4. | Cast Iron Valves | **IVC/ Kirloskar / Durga** or any other ISI marked makes |
| 5. | HDPE Pipes | **Superflow / Godavari / Premier Plastics** or any other equivalent ISI accredited / ISI marked makes. |
| 6. | PVC pipes | **Phenelex / Sudhakar / Nandi** or any other ISO accredited / ISI marked makes. |
| 7. | Wash Basins | **Parryware / Neycer / HSW** or any other ISI marked makes. |
| 8. | IWC | **Parryware / Neycer / HSW** or any other ISI marked makes. |
| 9. | EWC | **Parryware / Neycer / HSW** or any other ISI marked makes. |
| 10. | Porcelain Flush Tanks | **Parryware / Neycer / HSW** or any other ISI marked makes. |
| 11. | Urinals | **Parryware / Neycer / HSW** or any other ISI marked makes. |
| 12. | Pillar / Bib / Stop cocks | **Seiko / Reico / Jaquar** or any other equivalent makes. |
| 13. | EWC | **Parryware / Neycer / HSW** or any other ISI marked makes. |
| 14. | Paints | |
| | a) Synthetic enamel paint, Oil Bound distemper | **ASIAN / NEROLAC** or any other ISI marked makes |
| | b) White primer coat external walls | **JK / BIRLA** or any other ISI marked makes |

**7.4.8 LIST OF MAKES FOR CIVIL ITEMS TO BE USED.**

| Sl. No. | Item | Makes Suggested |
|---|---|---|
| 1. | Cement | **Ultra Tech/ Ramco/ Priya/ Rajashree/ Birla/ Kesoram** or any other ISI marked or ISO accredited makes. |
| 2. | Steel | **Visakha Steel Plant/ Tata Steel/ SAIL** |
| 3. | Ceramic Tiles | |
| | (a) Dadooing Tiles | **Jhonson / Somani / Kajaria** or any equivalent ISO accredited makes. |
| | (b) Flooring Tiles | **Jhonson/ Somani/ Kajaria/ Spartek/ Regency** or any equivalent ISO accredited makes. |
| 4. | Flush Door Shutters | **Kutti/ Anand/ Raveela/ Subhdwar** or any equivalent ISI marked makes. |
| 5. | Aluminium Fixtures for wood and iron works | **Jyothi / Classic / Oxford** or any other ISI marked makes. |
| 6. | Aluminium Sections | **INDAL / HM Dalco** or any other equivalent makes. |
| 7. | M S Tubes | **TATA / Khandelwal / Zenith** or any other equivalent makes. |



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## DRAWINGS



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**1.0 DRAWINGS:**

**1.1 The plans enclosed with the tender are only for tender purpose and liable to be altered during execution of work as per necessity of site conditions. The premium quoted by the contractor for various items shall hold good for execution of work even with altered plans.**

1.2 One set of drawings, on the basis of which actual execution of the work is to proceed shall be furnished free of cost to the contractor by the Chief Engineer / Executive Engineer progressively according to the work program submitted by the contractor and accepted by the Chief Engineer / Executive Engineer. Drawings for any particular activity shall be issued to the contractor in advance of the scheduled date of the start of the activity. However, no extra claims by the contractor toward any delay in issue of drawing or issue of any revision / change to the drawings issued earlier shall be admissible. The Chief Engineer shall intimate the contractor in advance regarding any delay to issue of drawings, for any particular stage of works. If work gets effected due to delay in issue of drawings, for any particular stage of work the contractor shall be granted extension of time as per clause No. 27 of General Conditions of Contract.

1.3 Signed drawings above shall not be deemed to be an order for work unless they entered in the agreement and schedule of drawings under proper alterations has been sent to the contractor by the Engineer-in-charge with a covering letter confirming that the drawing is authority for work in contract.

**2.0 DISCREPANCIES:**

2.1 In case of discrepancies between documents the following order of procedure shall apply:-

2.1.1 Between the written description of written dimensions in the drawings and the corresponding one in the specifications, the latter shall apply.

2.1.2 Figured dimensions shall supercede scaled dimensions. The drawings on a larger scale shall take precedence over those on a smaller scale.

2.1.3 Drawings issued as construction drawings from time to time shall supercede tender drawings and also the correspondence drawings previously issued.

**Note: The contractor should not execute any component of work without obtaining the working drawings. Any work done without drawings shall be at the contractors responsibility only in the absence of any written order from the Engineer-in-Charge. Acceptance for such work will be at the discretion of the Engineer-in-charge.**

**3.0 SECRECY CLAUSE:**

The drawings and specifications made available to the tenderer shall exclusively be used on the work and they are retained from passing on each plan to any unauthorised hand either in parts or in full under the provisions of Section-3 and 5 of the official secrets Act 1923. Any violation in this regard will entail suitable action under appropriate clause or official secret Act 1923.

**BILL OF QUANTITIES**
**AND**
**PRICE BID**
**(Schedule – A)**



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**NAME OF WORK: Construction of two rooms/ one room and renovation of existing rooms for SG MPL HS ANKASANI CHERUVU for the establishment of Autism centres in Konaseema District**

**Est. Cost. Rs. 40.00 Lakhs.**

## BILL OF QUANTITIES

## PREAMBLE

1. The Bill of Quantities shall be read in conjunction with the instructions to Tenderers, General and Special conditions of Contract Technical Specifications and Drawings.

2. The quantities given in the Bill of Quantities are estimated and provisional and are given to provide common basis for tendering. *The quantities here given are those upon which the lumpsum tender cost of the work is based but they are subject to alterations, omissions, deductions or additions as provided for in the conditions of this contract and do not necessarily show the actual quantities of work to be done*. The basis of payment will be actual quantities of work ordered and carried out as measured by the Contractor and verified by the Engineer and valued at the estimate rate plus or minus tender percentage quoted in the Bill of Quantities where applicable, and otherwise at such rates and prices as the Engineer-in-Charge may fix within the terms of Contract.

3. The estimate rates in the Bill of Quantities shall, except in so-far as it is otherwise provided under the Contract include cost of all constructional material, labour, machinery, transportation, erection, maintenance, profit, taxes and duties together with all general risks, liabilities and obligations set out or implied in the Contract.

4. The plans enclosed with the tender are liable to be altered during execution of work as per necessity of site conditions. The Tender percentage quoted by the tenderer shall hold good for execution of work even with altered plans.

5. The whole cost of complying with the provisions of the Contract shall be included in the estimated rates for items provided in the Bill of Quantities and where no items are provided in the Bill of Quantities, their cost shall be deemed to be distributed among the estimate rates entered for the related items of work.

6. General directions and descriptions of work and materials are not necessarily repeated nor summarised in the Bill of Quantities. References to the relevant sections of the Contract documentation shall be made before entering estimate rate against each item in the Bill of Quantities.

7. The method of measurements of completed work for payment shall be in accordance with the relevant B.I.S. Codes & A. P. S. Specifications.

8. All items of work are to be executed as per the drawings / specifications supplied with the contract documents.

   If there is any contradiction between the drawings and the text of the specifications, the later shall prevail.

9. The Tenderer should inspect and select the quarries of his choice before he quotes the tender percentage in the Schedule of Bill of Quantities and satisfy himself about the availability of required quantum of materials.

10. Diversion drains should be excavated before completion of the embankments and the useful soils should be used in the nearby embankments.



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11. The actual mix proportion by weight to be adopted during execution will be got designed in the laboratories to suit the grade of concrete and mortar to be used. It will be the responsibility of the contractor to manufacture concrete and mortar of required strength.

12. The quantum of measurement for all items of earthwork involving conveyance manually or by machinery shall be as assessed by level measurement. The measurements for the embankment will be for the consolidated banks only.

13. Wherever bailing out of water is involved either for excavation or for foundations or for constructions, the percentage quoted shall take into account the dewatering charges necessary. No separate payment will be made for dewatering.

14. Wherever embankment work is involved, useful soils approved by the Engineer-in-Charge from the cutting reaches and diversion drains shall be taken and used for forming nearby embankments soils used for constructions will be at free of cost.

15. The quoted tender percentage shall also include the work of any kind necessary for the due and satisfactory construction, completion and maintenance of the works according to the drawings and these specifications and further drawings and orders that may be issued by the Engineer-in-Charge from time to time. The quoted tender percentage shall include compliance by the Contractor with all the general conditions of contract, whether specifically mentioned or not in the various clauses of these specifications, all materials, machinery, plant, equipment, tools, fuel, water, strutting, timbering, transport, offices, stores, workshop staff, labour and the provision of proper and sufficient protective works, diversions, temporary fencing and lighting. It shall also include safety of workers, first aid equipments suitable accommodation for the staff and workmen, with adequate sanitary arrangements, the effecting and maintenance of all insurances, the payment of all wages, salaries, fees, royalties / Taxes, duties or other charges arising out of the execution of works and the regular clearance of rubbish, reinstatement and clearing-up of the site as may be required on completion of works safety of the public and protection of the works and adjoining land. The work of Building in quality control / assurance shall be deemed to be covered in the quoted percentage.

16. The Contractor shall ensure that, the quoted tender percentage shall cover all stages of work such as setting out, selection of materials, selection of construction methods, selection of equipment and plant, deployment of personnel and supervisory staff, quality control testing etc. The work quality assurance shall be deemed to be covered in the tender percentage.

17. a) The special attention of the tenderer is drawn to the conditions in the tender notices wherein reference has been made to the Andhra Pradesh Standard Specifications [APSS] and the Standard preliminary specifications containing therein. These preliminary specifications shall apply to the agreement to be entered into between the contractor and the Government of Andhra Pradesh and shall form an in-separable condition of the contract along with the estimate. All these documents taken together shall be deemed to form one contract and shall be complimentary to another.

    b) The tenderer shall carefully study the drawings and additional specifications and all the documents, which form part of the agreement to be entered into by the successful tenderer. The APSS and other documents connected with contract such as estimate plans, specifications, can be seen on all working days in the office of the Chief Engineer, Samagra Shiksha, Amaravati.

18. The tenderers attention is directed to requirements for materials under the clause ‘materials and workmanship’ in the preliminary specifications of APSS. Materials

conforming to the Bureau of Indian Standards specifications, APSS etc., shall be used on the work and the tenderers shall quote his overall tender percentage accordingly.

19. The tenderer has to do his own testing of materials and satisfy himself that they conform to the specifications of respective I.S.I. Codes before tendering.

20. The contractor shall himself procure the required construction materials of approved quality including the earth for formation of embankment and water from quarries/ sources of his choice. All such quarries / sources of materials required for the work shall be got approved by the Engineer-in-Charge in writing well before their use of the work.

21. The contractor shall himself procure the steel, cement, Bitumen, Blasting materials, sand, metal, soils, etc., and such other materials required for the work well in advance. The contractor has to bear the cost of materials for conveyance. The department will not take any responsibility for fluctuations in market in cost of the materials, transportation and for loss of materials etc.

22. Inspection of site and quarries by the tenderer: Every tenderer is expected before quoting his overall tender percentage, to inspect the site of proposed work. He should also inspect the quarries and satisfy himself about the quality, and availability of materials. The best class of materials to be obtained from quarries, or other sources shall be used on the work. In every case the materials must comply with the relevant standard specifications. Samples of materials as called for in the standard specifications or in this tender notice, or as required by the Engineer-in-charge, in any case, shall be submitted for the Engineer-in-charge's approval before the supply to site of work is begun.

23. The tenderer's particular attention is drawn to the sections and clauses in the A.P. standard specification dealing with

    a) Test, inspection and rejection of defective materials and work.
    b) Carriage
    c) Construction plant
    d) Water and lighting
    e) Cleaning up during the progress and for delivery.
    f) Accidents
    g) Delays
    h) Particulars of payments.

    The contractor should closely peruse all the specification clauses, which govern the overall tender percentage he is tendering.

24. The defect liability period of contract is twenty four months plus defects correction period.

25. The estimate rates for items shown in the Schedule "A" include all construction materials. No escalation in rates will be paid unless specified in the tender document. The tenderer has to quote an overall tender percentage considering all the aspects of the tender to complete the finished item of work as per the APSS / B.I.S. specifications, the special specifications appended, Drawings etc.

26. If there is any contradiction between APSS and B.I.S. specifications, listed and detailed technical specifications, the latter shall prevail.



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27. In case of a job for which specifications are not available with the Schedule or in APSS/ B.I.S. code and are required to be prescribed, such work shall be carried out in accordance with the written instructions of the Engineer-in-charge.

28. The contractor should use the excavated useful soils and stone for construction purpose. Soils used for construction either for homogeneous section in hearting or in casing zone based on the suitability will be at free of cost and the cost of stone used for construction purpose will be recovered from the contractors bill.

    The contractor should quote his tender percentage keeping in view of the above aspects.

29. Additions and alternations by the Tenderer in the Schedule of quantities will disqualify the tender.

    In the case of discrepancies between the written description of the item in the bill of quantities (Schedule “A”) and the detailed description in the specification of the same item, the later shall be adopted

30. The Unit rates noted below are those governing payment of extras or deductions for omissions according to the conditions of the contract as set-forth in the preliminary specifications of the A.P. standard specifications and other conditions of specification of this contract.

31. It is to be expressly understood that the measured work is to be taken according to the actual quantities when in place and finished according to the drawings or as may be ordered from time to time by the Executive Engineer and the cost calculated by measurement or weight at their respective rates without any additional charge for any necessary or contingent works connected works connected herewith. The Percentage Excess or less on ECV quoted are for works in situ and complete in every respect.

32. For all items of work in excess of the quantities indicated the rates payable for such excess quantities will be tendered rates i.e., estimate rates plus or minus tender percentage.

33. For all items of work, intermediate payment will be made provisionally as per relevant clause. Full-accepted agreement rates will be paid only after all the items of works are completed.

34. The contractor is bound to execute all supplemental works that are found essential incidental and inevitable during execution of main work.

35. The payment of rates for supplement items of work will be regulated as under.

    a) Supplemental items directly deductible from similar items in the original agreement.

    b) The rates shall be derived by adding to or subtracting from the agreement rate of such similar item the cost of the difference in the quantity of materials labour between the new items and similar items in the agreement worked out with reference to the schedule of rates adopted in the sanctioned estimate with which the tenders are accepted plus or minus overall tender percentage.

    c) Similar items but the rates of which cannot be directly deducted from the original agreement.

    d) Purely new items which do not correspond to any item in the agreement.

    e) The rate of all such items shall be estimated rates plus or minus overall tender percentage.



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**36 Extra Items:**

36.1 Extra items of work shall not vitiate the contract. The contractor shall be bound to execute extra items of work as directed by the Engineer-in-Charge. The rates for extra items shall be worked out by the Engineer-in-charge as per the conditions of the Contract and the same are binding on the Contractor.

36.2 The contractor shall before the 15th day of each month, submit in writing to the Engineer-in-charge a statement of extra items if any that they have executed during the preceding month failing which the contractor shall not be entitled to claim any.

37 Entrustment of additional items:

37.1 Where ever additional items not contingent on the main work and outside the scope of original agreement are to be entrusted to the original contractor dispensing with bids and if the value of such items exceeds the limits upto which the officer is empowered to entrust works initially to contractor without calling for tenders, approval of **competent authority** shall be obtained. Entrustment of such items on nomination shall be at rates not exceeding the estimated rates.

37.2 Entrustment of the additional items contingent on the main work will be authorised by the officers upto the monetary limits upto which they themselves are competent to accept items in the original agreement so long as the total amounts upto which they are competent to accept in an original agreement rates for such items shall be worked out in accordance with the procedure - For all items of work in excess of the quantities shown in the Bill of Quantities of the Tenders, the rate payable for such items shall be estimate rates for the items (+) or (-) over all tender percentage accepted by the competent authority.

37.3 Entrustment of either the additional or supplemental items shall be subject to the provisions under para 176 (B) of APWD Code, the items shall not be ordered by an officer on his own responsibility if the revised estimate or deviation statement providing for the same requires the sanction of competent authority. The Chief Engineer being the authority next higher to Chief Engineer, who entered into the agreement approves the rate for the items / variation in quantity in the current agreement.

**Note:** It may be noted that the term estimate rate used above means the rate in the sanctioned estimate with which the tender's compared or if no such rate is available in the estimate the rate derived will be with reference to the schedule of rates adopted in the sanctioned estimate with which tenders are compared.

## BILL OF QUANTITIES

## PART- I

**NAME OF WORK: Construction of two rooms/ one room and renovation of existing rooms for SG MPL HS ANKASANI CHERUVU for the establishment of Autism centres in Konaseema District**

**Est. Cost. Rs. 40.00 Lakhs.**

| S. No | Approximate Quantity In figures/ words | Description of work | Unit In figures/ words | Estimate Rate In figures / words | Amount in Rs. |
|---|---|---|---|---|---|
| | | Schedule - A Enclosed | | | |



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Schedule - B
List of Drawings

| S.No. | Drawing No. | Description |
|---|---|---|

- Enclosed -

DECLARATION

I/We hereby declare that I / We have inspected and satisfied myself / ourselves thoroughly and I / We am / are conversant with the local conditions, regarding all materials and about required for the work on which I / We have based my / our rates for the work. The Specifications, Plans, Designs and conditions of contract on which the offer has been based completely studied by me / us before submitting the tender.

**TENDERER**



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## SCHEDULE -C

## LIST OF SPECIFICATIONS FOR THE VARIOUS ITEMS OF WORKS SUPPLEMENTING THOSE DESCRIBED IN SCHEDULE 'A' BY S.S. NUMBERS

| Sl.No. | Short Title | I. S. Number |
|---|---|---|
| I. | **CEMENT:** | |
| | 1. 43 Grade ordinary port portland cement | 8112-1989 |
| | 2. Methods of physical tests for hydraulic comments | 4031 (Part 1 to 15) - 1988 |
| II. | **AGGREGATES :** | |
| | 1. Aggregates (Coarse and fine) from natural source for concrete | 383 - 1970 |
| | 2. Specification for sand for masonry: | 2116-1980 |
| | 3. Method of tests for aggregates for concrete: | 2386 (Part-I to IV) - 1963 |
| III. | **BUILDING STONES :** | |
| | 1. Method of Tests for determination of strength properties of natural building stones<br>Part - I : Compressive strength<br>Part - II : Transverse strength<br>Part - III : Tensile strength<br>Part - IV : Shear strength | 1121 (Part I to IV) - 1974 |
| | 2. Quarrying stones for construction purposes, recommended practice | 83831-1977 |
| | 3. Measurement of buildings and Civil Engineering Works | 1200 (Part-IV) - 1976 |
| | 4. Stone Masonry: Specifications for dressing natural building stones | 1129-1972 (Part – IV) |
| | 5. Drilling and permeability tests | 5529 (Part III) - 1973 |
| | 6. Code of practice for permeability tests (during and after construction) | 11216 - 1985 |
| IV. | **STEEL** | |
| | 1. Code of practice for bending and fixing of bars for concrete reinforcement | 2502-1963 |
| | 2. Specifications for High Strength Deformed Steel bars and Wires for concrete reinforcement : | 1781-1985 |
| | 3. Recommendation for detailing of reinforcement in reinforced Cement concrete works | 5525-1969 |
| | 4. Mild steel and Medium Tensile steel bars for concrete reinforcement | 432 (Part – VIII) – 1982 |
| | 5. Measurement of Building and Civil Engineering works Part - VIII Steel works and iron work) | 1200 (Part – VIII) - 1993 |
| V. | **MASONRY** | |
| | 1. Code of practice for construction of stone masonry Rubble stone masonry | 1597 (Part-1) - 1992 |
| | 2. Measurement of building and Civil Engineering Works Plastering and pointing. | 1200 (Part-XII) - 1976 |



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VI. **CONCRETE**:

| | | |
|---|---|---|
| 1. | Measurement of building and Civil Engineering Works method of concrete works | 1200 (Part-II) - 1974 |
| 2. | Concrete works: Code of practice for plain and reinforced concrete | 456 - 1978 |
| 3. | Precast concrete coping blocks | 5751 - 1984 |
| 4. | Method of Testing for strength of concrete | 516 - 1959 |
| 5. | Specification for Admixtures for concrete | 9103 - 1979 |
| 6. | Method of sampling and analysis of concrete | 1199 - 1976 |
| 7. | Concrete mixer - batch type | 1791 - 1968 |
| 8. | Concrete Vibrators - Immersible type | 2505 - 1980 |

VII. **EARTH WORK:**

| | | |
|---|---|---|
| 1. | Measurement of building and Civil Engineers Works method for earth work | 1200 (Part – 1) - 1974 |
| 2. | Safety Code for filling and other deep foundations | 5121 - 1969 |
| 3. | Safety Code for excavation works | 3701 - 1966 |
| 4. | Code of practice for earth work on canals | 4701 - 1982 |
| 5. | Method of testing for soils Determinations of Water content | 2720 (Part – II) - 1973 |
| 6. | Determination of water content dry density relation using light compaction | 2720 (Part-VIII) - 1980 |
| 7. | Determination of Dry density of soils in place by sand replacement method (first revision) | 2720 (Part – XXVII) - 1974 |
| 8. | Determination of dry density of soils in – situ | 270 (Part – XXIX) - 1975 |

VIII. **OTHER SUBJECTS**:

| | | |
|---|---|---|
| 1. | Safety code for scaffolds | 3698 (Part – I) - 1968 |
| 2. | Safety Code for ladders | 3696 (Part – II) - 1966 |
| 3. | Recommendation of stacking and storage of construction materials at site | 4082 – 1977 |



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## SCHEDULE - D
## LIST OF I.S. CODES FOR ELECTRICAL INSTALLATIONS

### INSTALLATIONS

| | | |
|---|---|---|
| 1. | Electrical wiring installation (system voltage not exceeding 650 V) | IS 732-1989 |
| 2. | Graphical symbols used in Electro-technology art-XI-Electrical Installation buildings. | IS 2032-1969 |
| 3. | Fire safety of building (General) Electrical Installation | IS 1646-1961 |
| 4. | PVC Insulated cables | IS 694-1977 |
| 5. | PVC Insulated cables (heavy duty) | IS 1554-1964 |
| 6. | 5 A tumbler switches | IS 1087 |
| 7. | 15A tumbler switches | IS 2120 |
| 8. | 3 Pin plugs and sockets | IS 1293 |
| 9. | Aluminum conductors for insulated cables | IS 1753-1967 |
| 10. | Recommended current ratings for cables | IS 1961-1967 |
| 11. | Installation and maintenance of paper insulated power cables | IS 1155-1967 |
| 12. | Earthing | IS 3043-1966 |
| 13. | Rigid steel conduits for electrical wiring | IS 9537-PII-1989 |
| 14. | Fittings for electrical wiring | IS 2667-1964 |
| 15. | Flexible steel conduits electrical wiring | IS 3430-1966 |
| 16. | Accessories for rigid steel conduit insulated cables | IS 3837-1966 |
| 17. | A C circuit breakers | IS 2516-1977 |
| 18. | Heavy duty airbreak switches and fuses for voltage not exceeding 1000V | IS 4047-1967 |
| 19. | 11 KV XLPE cables | IS 7089-part-II |
| 20. | Fuses | IS 9224-p-I & II |
| 21. | Current Transformers | IS 2705 |
| 22. | Voltage Transformers | IS 3516 |
| 23. | Indicating Instruments | IS 1248 |
| 24. | Relays | IS IS - 722 |
| 25. | Integrating instruments | IS IS - 3231 |
| 26. | Auxillary contracts | IS 2959 |
| 27. | Auxillary switches and push buttons | IS 6875 |
| 28. | A C isolators and earthing switches | IS 1818-1972 |
| 29. | Central gear for voltage not exceeding 1000 V | IS 4337-1967 |
| 30. | Installation and maintenance of switchgear | IS 3072-1965 |

| | | |
|---|---|---|
| 31. | Enclosed distribution fuse board and cutouts for voltage 100 V | IS 2675-1964 |
| 32. | Airbreak isolators for voltage 1000 V | IS 2607-1976 |
| 33. | Marking alignment or switchgear bushers main Connections and auxillary wiring | IS 375-1963 |
| 34. | Selection installation and maintenance of fuses (Voltages not exceeding 650 V) | IS 3106-1966 |
| 35. | General and safety requirements for electric Lighting fittings | IS 1913-1969 |
| 36. | Lighting public thorough fares | IS 1944-1970 |
| 37. | Waterproof electric lighting fitting | IS 3528-1966 |
| 38. | Water tight electric lighting fittings | IS 3553-1966 |
| 39. | Luminaries for street lighting | IS 2149-1970 |
| 40. | Mild steel tubulars and other wrought steel pipe fittings | IS 1239-1968 |
| 41. | Ceiling fans | IS 374-1951 |
| 42. | Transformers | IS 2026-1962 |
| 43. | Installation and commissioning of transformers | IS 1886-1967 |
| 44. | Protecting of buildings and allied structures against lighting | IS 2309-1967 |
| 45. | Interior Illumination | IS 3636-1966 |
| 46. | Busbar ratings | IS 8084-1976 |
| 47. | Onload change over switches | IS 4064-1978 |
| 48. | Alluminium alloy for bus bars | IS 5082 |
| 49. | Factory built assemblies of switchgear and control gear | IS 8623 |
| 50. | General requirements of switchgear and control gear for voltages not exceeding 1000 V. | IS 4237 |
| 51. | Degree of protection provided by enclosing for low voltage switchgear | IS 2147 |

## LIST OF APPROVED MAKES OF ELECTRICAL MATERIALS

| | Descriptions | Make |
|---|---|---|
| 1. | Annealed copper conductor, PVC insulated, 1100 volts grade flexible copper cables manufactured to IS-694 | Q-Flex, Nicco, Finolex, Pawancab Anchor, Havells, RPG, with FR & ISI marking. |
| 2. | PVC insulated aluminium conductor, PVC sheathed, Armored Underground cables to IS 1554-Part-I-1976 or latest revision. | Universal, ICC, CCI, Nicco, Pawan cab, Havells, Suncab with ISI marking. |
| 3. | M.S. screw type black enameled conduit pipes manufactured as per IS 9538-part-II 1981 or latest revision. | Gupta Brothers, Supreme, Sunce, Adarsh, BCG, Bharat |
| 4. | M.S. Boxes | Make out of 16 SWG sheet steel. |
| 5. | Flush type deluxe piano marked switches & ISI marked. | Anchor, Leader, Maru with I.S.I marked |
| 6. | 3 Plate ceiling Roses ISI marked. | Anchor, Leader, Maru with I.S.I. marked |
| 7. | 5 A 3 pin / 2 pin socket & 15 A/5A 3 pin power plug sockets of flush type switches. | Anchor, Leader, Maru with ISI marked |
| 8. | Bell pushes | Anchor, Leader, Maru with I.S.I. marked |
| 9. | MCB, MCCBs, MCB distribution Boards | Standard/Havells/MDS/ Indo-Copp/L&T. |
| 10. | Switch Fuse Unit | L&T /English Electric/ Siemens/Havells/HPL /standard. |
| 11. | Distribution Fuse Board Type 5N 100 TNG E.E. or equipment. | English Electric/Larsen Toubro/ Siemens/Havells/HPL |
| 12. | Energy saving model 28Watts with Electronic Choke & Ordinary 40W tube. | Unit: Trinic Tube: Philips/Wipro/Crompton. |
| 13. | Ceiling Fans | Crompton Greaves/Polar/ Orient/Bajaj with ISI marked & DGS & D approved model |
| 14. | Exhaust fans : Heavy & Light duty | Crompton Greaves/Orient Polar/Bajaj |



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## SCHEDULE OF TECHNICAL PARTICULARS
## FOR FIRE FIGHTING

**MANUAL CALL POINTS:**

| | | |
|---|---|---|
| Make | : | Globe / Safeway |
| Material of body | : | M.S. |
| Thickness of body | : | 1.5 mm |
| Material of frangible element | : | Safex make Safety glass (if broken it will be of diamond bits) |
| Contact making/breaking for alarm | : | Breaking |
| Overall dimensions | : | 100mm x 100mm |
| Size of frangible element | : | 70mm x 70mm |



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# BILL OF QUANTITIES
## Part-II

## Details of Maximum amount reimbursable to the Contractor.

**1) GST : As per government norms**

**a) "The rates mentioned in "BOQ" (Schedule – "A" are including overhead charges and contractors profit but excluding GST charges. The overhead charges include engaging technical persons by the contractor and work insurance. Hence, the contractor quoted rates is inclusive of engaging technical persons and work insurance and no reimbursement for these will be made separately".**

**b) The rates mentioned in Bill of Quantities (Part –I) are inclusive of overhead charges and Contractors profit but exclusive of GST and seigniorage charges. The items covered under overhead charges are as per the MoRT&H standard data book for analysis of rates (First Revision).**

**c) In each bill 0.10% of the value of the gross bill will be added to the bill amount and equal amount will be deducted from the bill amount creditable to Corpus fund of NAC.**

**2) Seignorage Charges : As per government norms**

**3) Corpus fund towards NAC : As per government norms**

0.1% of the Value of the Gross bill will be added to the bill amount and equal amount will be deducted from the bill amount creditable to corpus fund of NAC.



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## PRICE BID

**NAME OF WORK:** **Construction of two rooms/ one room and renovation of existing rooms for SG MPL HS ANKASANI CHERUVU for the establishment of Autism centres in Konaseema District**

**Est. Cost. Rs. 40.00 Lakhs.**

Estimated contract value (in figures & words): **Rs. 32,12,152.00/-ps**

I Sri / Smt./M/s. ……... do hereby express my willingness to execute the aforesaid work as per the conditions, standards, specifications, rules, regulations, etc., stipulated in the tender documents.

a) at an overall tender percentage of …….. (in figures)…….
(in words) **less** over estimated value.

***OR***

b) at estimate value.

SIGNATURE, NAME OF THE TENDERER / AUTHORISED SIGNATORY.



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## FORMATS OF SECURITIES



Tenderer

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**PROFORMA**

**BANK GUARANTEE *FOR* EARNEST MONEY DEPOSIT**

(At the time of submission of tender)

WHEREAS …….. (Name of the Contractor) (here in after called "the Tenderer") has submitted his tender response to NIT No……. dated:…… for the work " ……..." (Name of work) (hereinafter called "the tender").

KNOWN ALL MEN by these present that we …… ……. (Name and Address of Bank) ……. (hereinafter called "the Bank" are bound unto …… / (name of the designated PAO) in the sum of * ……. for which payment will and truly to be made to the said Department, the Bank binds itself, his successors and assigns by these presents.

SEALED with the Common Seal of the Bank this …….. day of ……..200….

THE CONDITIONS of this obligation are:-

(1) If after Tender opening the tenderer withdraws or modifies his Bid during the period of bid validity specified in the Form of Tender.

(2) If the Tenderer having been notified of the acceptance of his bid by the Department during the period of validity.

(a) fails or refuses to execute the Form of Agreement in accordance with the Instructions to Tenderers, if required; or

(b) fails or refuses to furnish the balance EMD and additional performance Security in accordance with the instructions of Tenderers.

We undertake to pay to the Department up to the above amount upon receipt of his first written demand, without the Department having to substantiate his demand, provided that in his demand the Department will note the amount claimed by him is due to him owning to the occurrence of one or both of the two conditions, specifying the occurred condition or conditions.

This Guarantee will remain in force up to and including the date** …… after the dead line for submission of Tenders as such deadline is stated in the Instructions to Tenders or as it may be extended by the Department, notice of which extension(s) to the Bank is hereby waived. Any demand in respect of this Guarantee should reach the Bank not later than the above date.

DATE……. SIGNATURE OF THE BANK ……..

WITNESS…….. SEAL……..

---

(Signature, Name and Address)

---



Tenderer

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APSS, Amaravati.

**PROFORMA**

**BANK GUARANTEE FOR BALANCE "E.M.D."**

(At the time of Agreement)

_____(name & address of Department)

_____

WHEREAS_____

_____

_____

_____ __ _____(name and address of Contractor ) (hereinafter called "the Contractor") has undertaken, in pursuance of Contract No. _____ dated: _____ to execute the work of _____ [name of work];

AND WHEREAS it has been stipulated by you in the said Contract that the Contractor shall furnish you with a Bank Guarantee by a Schedule bank for the sum specified therein as balance EMD / EMD for compliance with his obligations in accordance with the Contract;

AND WHEREAS we have agreed to give the Contractor such a Bank Guarantee;

NOW THEREFORE we hereby affirm that we are the Guarantor and responsible to you, on behalf of the Contractor, up to a total of _____ [amount of guarantee] _____ [in words], such sum being payable and we undertake to pay you, upon your first written demand and without cavil or argument, any sum or sums within the limits of _____ [amount of guarantee] as aforesaid without your needing to prove or to show grounds or reasons for your demand for the sum specified therein.

We hereby waive the necessity of your demanding the said debt from the Contractor before presenting us with the demand.

We further agree that no change or addition to or other modification of the terms of the Contract or of the Works to be performed thereunder or of any of the contract documents which may be made between you and the Contractor shall in any way release us from any liability under this guarantee, and we hereby waive notice of any such change, addition or modification.

This guarantee shall be valid upto …….. i.e., until 28 days from the date of expiry of the Defects Liability period.

Signature & seal of the Guarantor_____

Name of Bank_____

Address_____

Date_____



Tenderer

Chief Engineer
APSS, Amaravati.

**PROFORMA**

**BANK GUARANTEE FOR ADDITIONAL SECURITY**

_____(name and address of Department)

_____

WHEREAS _____ (name and address of Contractor ) (hereinafter called "the Contractor") has undertaken, in pursuance of Contract No. _____ dated: _____ to execute _____ [name of Contract and brief description of works] (hereinafter called "the Contractor");

AND WHEREAS it has been stipulated by you in the said Contract that the Contractor shall furnish you with a Bank Guarantee by a Schedule bank for the sum specified therein as Additional further security bank guarantee for compliance with his obligations in accordance with the Contract;

AND WHEREAS we have agreed to give the Contractor such a Bank Guarantee;

NOW THEREFORE we hereby affirm that we are the Guarantor and responsible to you, on behalf of the Contractor, up to a total of *Rs.*_____ [amount of guarantee] _____ [in words], such sum being payable and we undertake to pay you, upon your first written demand and without cavil or argument, any sum or sums within the limits of _____ [amount of guarantee] as aforesaid without your needing to prove or to show grounds or reasons for your demand for the sum specified therein.

We hereby waive the necessity of your demanding the said debt from the Contractor before presenting us with the demand.

We further agree that no change or addition to or other modification of the terms of the Contract or of the Works to be performed thereunder or of any of the contract documents which may be made between you and the Contractor shall in any way release us from any liability under this guarantee, and we hereby waive notice of any such change, addition or modification.

This guarantee shall be valid upto and until 28 days from the date completion.

Signature & seal of the Guarantor_____

Name of Bank_____

Address_____

Date_____

**ANNEXURE – I (B).**

**FORM OF SOLVENCY CERTIFICATE BY BANKS**

**(From Nationalized Banks/ Scheduled Banks)**

**I, _____ Managing Director / Manager / General Manager / Agent of _____ Bank Limited do hereby certify that a _____ _____ *[here the Names and addresses of the contractor]* to be solvent to the extent of Rs._____ [Rupees _____ _____] as disclosed by the information and record which are available with the aforesaid bank.**

**For the _____ Bank**

**Date:**

**Place:**

**Signature of Bank Manager**

**[Authorised to Sign]**

**FORMAT FOR EVIDENCE OF ACCESS TO OR**

**AVAILABILITY OF CREDIT FACILITIES**

**(CLAUSE 8 (iii) OF TENDER NOTICE)**

**(From Nationalized Banks / Scheduled Banks)**

***BANK CERTIFICATE***

**This is to certify that M/s. …….....……. is a reputed company with a good financial standing. If the contract for the work namely …… ……...…….. is awarded to the above firm, we shall be able to provide over draft/credit facilities to the extent of Rs.…… to meet their working capital requirements for executing the above contract.**

**Signature of Sr. Bank Manager**

**Name of Bank**

**Address of Bank**